How We Found Out

Evidence, Error, and the Making of Medical Knowledge

Preface

This book follows medical questions through time. Each chapter asks not only what physicians eventually came to believe, but how they found out: which observations challenged clinical intuition, which experiments separated competing explanations, and why some evidence was accepted while other evidence was ignored. Medical knowledge is not produced by a single dramatic discovery. It is built through repeated observation, measurement, comparison, criticism, and practical consequences. Throughout, the focus is on what changed for patients and populations: whether an advance prevented death, restored function, reduced suffering, or gave people more control over daily life. Benefits were rarely uniform; access, risk, and cost shaped who could share them.

Across the chapters, the book asks how investigators established that microorganisms cause disease, mosquitoes transmit malaria, HIV causes AIDS, smoking causes lung cancer, high blood pressure raises the risk of heart disease and stroke, bacteria cause stomach ulcers, vitamin deficiency causes scurvy, insulin controls blood glucose, blood comes in different groups, and DNA carries genetic information. It also follows how discoveries became prevention, treatment, and care.

The book also follows the diseases that made these advances urgent. It traces how clean water, vaccines, antibiotics, rehydration, and modern diagnosis helped people survive infections that had once killed millions, while showing how access to those breakthroughs remained uneven.

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Chapter 1

Germ Theory of Disease

1.1 The question behind germ theory

For most of medical history, doctors could observe that illness passed between people without being able to identify what carried it. The word "contagion" described a pattern of spread; it did not yet name a mechanism. The stronger claim—that particular living organisms could enter a body and produce a particular disease—required instruments, laboratory methods, and standards of proof that did not yet exist. Germ theory did not mean that every illness had a microbial cause. It proposed that specific microorganisms could cause specific infectious diseases, a claim that investigators had to test one disease and one organism at a time.

This distinction explains why clinical intuition alone was insufficient. A physician might notice that a fever followed contact with a sick person, but the same fever could also follow bad air, seasonal change, diet, crowding, or a hidden environmental poison. A useful theory had to predict what would happen when exposure was interrupted and had to survive attempts to disprove it.

1.2 Miasma, contagion, and inherited assumptions

The miasma theory attributed epidemics to poisonous or corrupted air. It was not irrational: foul smells accompanied crowded housing, sewage, decomposing organic matter, and many outbreaks. Sanitary reforms based on this theory could still improve health. But miasma did not explain why certain illnesses spread along particular chains of contact or why some individuals remained well after the same exposure.

Older traditions also contained versions of contagion. Girolamo Fracastoro proposed in the sixteenth century that tiny transferable particles could spread disease, and Antonie

van Leeuwenhoek later observed microscopic organisms. Yet observation was not causation. Seeing small "animalcules" in water did not show that they caused human disease, and the available microscopes could not reliably connect a particular organism to a particular illness.

1.3 Semmelweis and the evidence of an intervention

Ignaz Semmelweis supplied one of the clearest early demonstrations that medical practice could interrupt disease transmission. At the Vienna General Hospital, he compared two obstetric clinics. Women treated by physicians and medical students died of puerperal fever far more often than women treated by midwives. The difference was not explained by the patients' bodies or by the calendar; it followed the organization of care.

Semmelweis noticed that doctors moved from autopsies to the maternity ward. He required them to wash their hands in chlorinated lime before examining women. Mortality fell dramatically. The intervention worked before its mechanism was accepted, but Semmelweis could not yet identify the infectious material or provide a laboratory demonstration. His evidence was therefore powerful in practice but vulnerable in a culture that expected a theory to be anatomically and chemically complete. Semmelweis showed that an infectious material could be carried on physicians' hands and removed by washing, even though he could not identify it as a microorganism [13]. Later historians also emphasize that professional conflict, institutional resistance, and Semmelweis's increasingly polemical relations with colleagues helped bury an effective observation.

The episode is important because it separates two questions: "Does an intervention work?" and "Do we understand why it works?" Clinical results can precede mechanism, but the mechanism is often needed to generalize the result to new settings.

1.4 John Snow and the epidemiology of cholera

John Snow approached cholera through patterns of cases rather than through a microscope. During the 1854 outbreak in London's Soho district, he mapped cases around public water pumps and compared households using different water supplies. The concentration of cases around the Broad Street pump was difficult to reconcile with a simple theory of poisonous air. When the pump handle was removed, the local outbreak subsided, although the epidemic had already begun to decline.

Snow's argument was not a perfect experiment. People moved, exposures overlapped, and the organism responsible for cholera had not been isolated. The pump handle was removed while the outbreak was already declining, so its removal alone could not establish why the

outbreak ended. Snow's case rested on several kinds of comparison: the geography of deaths, households supplied by different water companies, and people who avoided the pump despite living nearby. His 1855 account made an important methodological point: disease causation could be tested at the level of populations, and epidemiology could establish a route of transmission before microbiology identified the agent [14].

1.5 Pasteur turns fermentation into a medical argument

Louis Pasteur's work on fermentation changed the debate. Fermentation had often been treated as a chemical transformation that occurred spontaneously. Pasteur showed that different fermentations were associated with different living microorganisms, and that heating could prevent unwanted microbial growth. His swan-neck flask experiments showed how boiled broth could remain free of growth when airborne particles were kept out, even though air could enter. This challenged spontaneous generation under the conditions tested. His work did not begin as a program to explain human disease, but it supplied a model: if microorganisms could produce changes in wine or milk, they might also produce changes in living tissues [15].

Pasteur then studied diseases of silkworms and demonstrated that microscopic agents could be associated with contagious disease in animals. He linked laboratory observation to practical prevention through selective breeding, separation of diseased stock, and, in later work, vaccination against anthrax and rabies. His conclusions met strong opposition because they challenged spontaneous generation and because many organisms were difficult to see or cultivate. The decisive change came not from one observation but from the accumulation of reproducible results across fermentation, animal disease, and vaccine experiments.

1.6 Koch and the search for specific causes

Robert Koch made the germ theory more exact. Studying anthrax, he used staining, microscopy, animal transmission, and culture methods to connect a specific bacillus with a specific disease. Improvements in solid culture media, staining, and photography allowed researchers to separate organisms and preserve evidence that could be examined by others.

In 1882, Koch presented the tubercle bacillus as the cause of tuberculosis. He later set out criteria, now known as Koch's postulates, for linking a microorganism to disease: the organism should be found in diseased hosts, be isolated and grown, produce the disease in a susceptible host, and be recovered again. Koch's work on tuberculosis supplied a striking example of this chain of evidence: he stained the bacillus in diseased tissue, cultivated it,

and reproduced tuberculosis in experimental animals. He announced the discovery on 24 March 1882 and published his account soon afterward [16]. These criteria gave bacteriology a disciplined causal grammar and helped make microbiology cumulative rather than dependent on a charismatic observer. Koch received the 1905 Nobel Prize for his investigations of tuberculosis [12].

The postulates were never a universal machine for proving every disease. Some microbes cannot be grown in artificial culture; some people carry pathogens without symptoms; and viruses require living cells. Even bacteria can cause different outcomes depending on dose, immunity, genetics, and environment. Their value was methodological rather than absolute: they made investigators state what evidence they had and what evidence was still missing.

1.7 Antisepsis, vaccines, and public health

Once microorganisms became plausible causes, medicine acquired new ways to test the theory. Joseph Lister applied carbolic acid to wounds, instruments, and dressings in an antiseptic system intended to prevent contamination during surgery [17]. Pasteur's work supported heat treatment of liquids and the development of vaccines. Public-health authorities expanded clean water, sewage control, isolation, sterilization, and safer handling of food and milk.

These measures also exposed a tension in the new theory. Germ theory could encourage narrow focus on the microbe while obscuring poverty, housing, labor, nutrition, and access to care. A bacterium may be necessary for disease without being sufficient to explain who is exposed, who becomes ill, and who survives. The modern infectious-disease model therefore combines agent, host, and environment rather than treating the organism as the whole story.

1.8 Why acceptance took time

The victory of germ theory is often narrated as if Pasteur and Koch simply defeated an irrational past. The actual process was slower. Many physicians accepted useful antiseptic practices before accepting a universal germ theory; others rejected individual claims because early cultures were contaminated or because proposed organisms could not be reproduced. Different diseases required different kinds of proof. Cholera, tuberculosis, wound infection, and viral illnesses did not all fit the same experimental design.

The theory became persuasive when it generated successful predictions. If a microbe caused disease, interrupting its route should prevent cases; killing it with heat or disinfectant should reduce transmission; and immunizing the host should change susceptibility. Each

success made the theory more than an explanation of past observations. It became a tool for finding interventions.

1.9 The legacy of the discovery

The discovery that microorganisms cause many diseases did not make clinical judgment obsolete. Doctors still have to recognize atypical presentations, interpret imperfect tests, and account for the patient's circumstances. What changed was the evidentiary standard. A plausible clinical impression could now be challenged by culture, microscopy, epidemiological comparison, controlled intervention, and later molecular methods.

The enduring lesson is not that laboratory science replaced bedside medicine. It is that medicine became more reliable when observation and experiment were made to correct one another. Clinical intuition suggested questions; population patterns revealed routes; laboratory methods identified agents; and practical interventions tested whether the explanation mattered. Germ theory succeeded because it joined all four forms of evidence.

Timeline

1546 Fracastoro describes transferable "seeds" of contagion.

1670s Leeuwenhoek observes microscopic organisms.

1847 Semmelweis introduces chlorinated handwashing in a maternity ward.

1854 John Snow uses mapping and water-supply comparisons during a cholera outbreak.

1860s–1870s Pasteur links fermentation and animal disease to microorganisms.

1867 Lister publishes his antiseptic principles for surgery.

1876 Koch links a specific bacterium to anthrax.

1882 Koch presents the tubercle bacillus as the cause of tuberculosis.

1905 Koch receives the Nobel Prize for his tuberculosis investigations.

Chapter 2

Antibiotics

Penicillin is often used as shorthand for the antibiotic revolution, but the change came from a growing family of medicines. Antibiotics made it possible to treat many bacterial infections from within the body, rather than relying only on wound care, surgery, isolation, or the patient's own defenses. Their history includes synthetic chemistry, the study of soil microbes, industrial production, and the continuing struggle against bacterial resistance.

2.1 The search for selective toxicity

The central problem was finding a substance that harmed a pathogen more than the person infected. In the early twentieth century, Paul Ehrlich's work on arsphenamine, also known as Salvarsan, suggested that chemicals could be designed to attack particular microbes. It was used to treat syphilis, though it was toxic and difficult to administer. Researchers continued to look for safer drugs that could selectively interfere with bacterial life [93].

In the 1930s, Gerhard Domagk and colleagues tested synthetic dyes for antibacterial effects. Their work showed that Prontosil could protect infected animals and patients from some bacterial diseases. The active component was sulfanilamide, which gave rise to a group of related medicines called sulfonamides or sulfa drugs. They were among the first widely useful systemic antibacterial medicines and were used for infections such as streptococcal disease before penicillin could be produced at scale [94, 93].

2.2 Antibiotics from the natural world

Penicillin's development into a medicine showed that microbes could make substances that inhibited other microbes. Researchers began screening fungi and bacteria, especially organisms

living in soil. Selman Waksman and his laboratory at Rutgers developed a systematic screening program. In 1943, graduate student Albert Schatz isolated streptomycin from *Streptomyces griseus*; Elizabeth Bugie also contributed to the collaborative research and coauthored its initial report. Streptomycin was the first antibiotic effective against tuberculosis, a disease for which sanatorium care had been the mainstay. Clinical studies established its value. Waksman received the 1952 Nobel Prize for the soil-microbe research program that led to its discovery [96, 95, 93].

Penicillin and streptomycin were followed by chloramphenicol, tetracyclines, macrolides, and other families. Each worked in a different way: some interfered with bacterial cell-wall construction, while others disrupted protein production or other essential processes. This variety mattered because bacteria differ, infections occur in different tissues, and a drug effective against one organism may fail against another. Laboratory testing and clinical judgment helped match an antibiotic to the likely cause of infection [93].

2.3 A transformation in medicine

Antibiotics changed the likely outcomes of pneumonia, wound infections, bloodstream infections, and many other bacterial diseases. They also made other parts of modern medicine safer to attempt. Operations, cancer treatment, intensive care, and organ transplantation can leave patients vulnerable to infection; effective antibiotics help clinicians prevent or treat some of these complications. They are not a cure for every infection: they do not work against viruses, and some bacterial illnesses resolve without them. Their benefit depends on using a drug that is active against the organism and can reach the infected site [97, 93].

2.4 Resistance and the limits of a miracle drug

Antibiotic resistance is an evolutionary response. Within a bacterial population, some organisms may carry traits that let them survive a drug. When susceptible bacteria are killed, resistant ones can multiply and spread. Resistance is not a person becoming resistant to a medicine; it is the bacteria changing in ways that make treatment less effective. Resistant infections can be harder to treat and may require medicines with greater risks or more complex care [93, 97].

These medicines remain essential, but each use creates some selection pressure. Using antibiotics when they are not useful exposes people to side effects without treating the cause, and unnecessary use can accelerate resistance. Stewardship means choosing an appropriate drug when treatment is needed and using it according to clinical guidance. Infection prevention,

vaccination, accurate diagnosis, and surveillance also help preserve the effectiveness of existing drugs [97].

The antibiotic era did not eliminate infectious disease. It changed the balance between humans and bacteria, making many once-dangerous infections treatable while giving microbes a powerful new pressure to adapt. The story is therefore both an account of therapeutic discovery and a reminder that medical tools must be matched to evidence and continually renewed.

Timeline

1910 Arsphenamine, developed by Paul Ehrlich and Sahachiro Hata, enters use for syphilis.

1932 Gerhard Domagk's group demonstrates the antibacterial effects of Prontosil in animals.

1935 Sulfonamide treatment enters clinical practice after the active antibacterial component of Prontosil is identified.

1943 Albert Schatz isolates streptomycin in Selman Waksman's Rutgers laboratory.

1940s–1960s Multiple antibiotic families enter clinical use, greatly expanding treatment options for bacterial disease.

Today Antibiotics remain essential, while stewardship and resistance surveillance help protect their effectiveness.

Chapter 3

Penicillin Therapy

3.1 The mystery of bacterial infection

Before antibiotics, physicians could clean wounds, drain abscesses, remove infected tissue, and sometimes use antiseptics, but they had few medicines that could selectively kill bacteria inside the body. Sulfonamides offered an early answer in the 1930s, yet many serious infections remained difficult to treat. The challenge was not simply to find something that killed bacteria in a dish. The substance had to reach infected tissue, remain active in the body, and do less harm to the patient than to the microorganism.

This problem made the history of penicillin a story of two discoveries. Alexander Fleming noticed an antibacterial phenomenon in 1928. More than a decade later, Howard Florey, Ernst Chain, Norman Heatley, and their colleagues converted that phenomenon into a purified, testable, and manufacturable drug. The famous laboratory accident was therefore the beginning of a research program, not its completion.

3.2 Fleming notices a clear zone

In August 1928, Fleming returned to his laboratory at St Mary's Hospital after a holiday. One culture plate containing staphylococci had been contaminated by a mold. Around the mold was a clear zone where bacterial growth had been inhibited. Contamination usually meant that an experiment had been spoiled; Fleming's attention turned the spoiled plate into evidence.

He isolated the mold and showed that it released a substance that inhibited several bacteria. Fleming called the substance penicillin, after the genus *Penicillium*, and published his findings in 1929 [22]. He also recognized an important therapeutic property: penicillin

appeared more toxic to bacteria than to human tissues. That selectivity made it promising, but promise was not proof of clinical usefulness.

3.3 Why the discovery went quiet

Fleming could not easily purify enough penicillin. The substance was unstable, present in small quantities, and difficult to extract from the culture fluid. His experiments showed inhibition but did not supply a reliable dose for a patient. Other researchers were occupied with sulfa drugs, vaccines, and the many unsolved problems of infectious disease. The penicillin observation was real, but the technical path from mold to medicine was unclear.

This period corrects the popular version of the story. Serendipity made the first observation possible, but serendipity did not solve purification, pharmacology, toxicity, clinical testing, or manufacturing. A useful discovery can remain dormant when the surrounding methods and institutions are not ready to develop it.

3.4 The Oxford team asks a new question

In 1939, Florey and Chain at Oxford began a systematic study of natural antibacterial substances. Chain, a biochemist, examined Fleming's reports and worked on extracting penicillin. Heatley developed methods for measuring its activity and recovering it from liquid broth. Their question was no longer whether mold inhibited bacteria; it was whether enough active material could be made, purified, and tested in living animals.

The team used a biological assay: a sample of extract was placed on a culture of susceptible bacteria, and the clear zone was measured. This assay was essential because chemical analysis could not yet identify the unstable substance directly. Activity itself became the guide through each purification step. A process that produced a chemically cleaner fraction but lost activity was a failure, even if it looked better in a flask.

3.5 Animal experiments and the first patients

In 1940, the Oxford researchers showed that purified penicillin could protect mice infected with lethal bacteria. Treated mice survived while control mice died. The experiment linked three parts of the causal argument: penicillin inhibited the organism in culture, it reached an infected animal, and it could alter the outcome of disease without obvious toxicity at the tested dose [93].

Human treatment followed cautiously. In February 1941, an Oxford policeman with a severe staphylococcal and streptococcal infection received penicillin and improved rapidly. The supply was too small to complete his treatment, and he eventually died. Later patients showed more striking recoveries, but the early cases also exposed the danger of confusing a dramatic response with a complete solution. A drug cannot change practice until it can be supplied reliably and used in an adequate course.

3.6 War, scale, and the moldy cantaloupe

World War II created both urgency and resources. In Britain, the war made large-scale pharmaceutical production difficult. Florey and Heatley travelled to the United States in 1941 seeking industrial and government support. At the US Department of Agriculture laboratory in Peoria, Illinois, researchers searched for higher-yielding molds and improved the growth medium.

A strain of *Penicillium chrysogenum* from a moldy cantaloupe produced far more penicillin than Fleming's original strain. Corn steep liquor, a plentiful by-product of starch production, improved yields. Engineers and pharmaceutical companies then adapted surface culture to deep-tank fermentation, allowing oxygenated broth to support enormous quantities of mold. The drug's history therefore belongs partly to industrial microbiology and process engineering, not only to clinical medicine.

By 1943, production had expanded enough to supply Allied forces. Penicillin reduced deaths from wound infections and bacterial illnesses among soldiers and civilians. The wartime setting accelerated cooperation between academic laboratories, government agencies, and industry, but it also shaped whose needs were prioritized and how access was organized.

3.7 From a treatment to an antibiotic era

Penicillin's success changed the expectations of medicine. Researchers began searching soil, fungi, and bacteria for other antimicrobial compounds. New drug classes followed, and infections that had once required prolonged hospitalization could often be treated at home. The 1945 Nobel Prize was awarded jointly to Fleming, Chain, and Florey "for the discovery of penicillin and its curative effect in various infectious diseases" [23].

The shared award is historically revealing. Fleming supplied the original observation and early characterization; Chain and Florey led the work that demonstrated therapeutic potential; Heatley and many other researchers solved practical problems; and industrial teams created the supply. The cure was not the property of one heroic moment.

3.8 Resistance appears in the success story

Fleming warned that bacteria exposed to insufficient concentrations of penicillin could become resistant. His warning reflected evolutionary logic: variation already present in a bacterial population could be selected when a drug killed susceptible organisms but allowed resistant ones to multiply. Resistance was observed even before penicillin became widely available, and it expanded with use.

This did not invalidate the drug. It changed the conditions of its usefulness. The right medicine, dose, duration, diagnosis, and infection-control practice all became part of antibiotic stewardship. The history of penicillin thus contains both a triumph and a warning: a treatment can be extraordinarily effective while still being temporary in evolutionary time.

3.9 What the story teaches about discovery

The penicillin story is engaging because each stage corrects a tempting but incomplete explanation. Fleming's plate shows that accidental observation can open a question. The Oxford experiments show that a clear zone is not yet a drug. Animal studies show that activity in culture can become treatment in a living host. Wartime scale-up shows that clinical impact depends on production. Resistance shows that practice must respond to biology after the discovery.

Medical knowledge advanced because different kinds of expertise were joined: clinical observation, bacteriology, biochemistry, pharmacology, animal studies, engineering, and manufacturing. No biography alone explains the transition. The central historical question is not who noticed the mold, but how a fragile laboratory signal became a reproducible intervention that changed what doctors could promise to patients.

Timeline

1928 Fleming observes that a Penicillium mold inhibits staphylococci.

1929 Fleming publishes his report on penicillin.

1939–1940
Florey, Chain, Heatley, and colleagues purify and test penicillin.

1941 The first human patients receive Oxford-produced penicillin.

1941–1943 US–British cooperation develops high-yield strains and deep-tank fermentation.

1943 Penicillin supplies become sufficient for Allied military demands.

1945 Fleming, Chain, and Florey receive the Nobel Prize.

Chapter 4

Mass Production of Medicines

A medicine discovered in a laboratory helps few people until it can be made reliably in quantities that clinics and hospitals can use. Pharmaceutical manufacturing joined chemistry, microbiology, engineering, machinery, and quality testing to turn experimental treatments into repeatable products. Penicillin's wartime scale-up, described in the preceding chapter, showed that production itself could determine whether a medical discovery reached patients. Later advances in automated processing expanded that possibility across tablets, vaccines, insulin, and many other medicines.

4.1 Scaling up production

Making a medicine at industrial scale required more than building a larger version of a laboratory. Penicillin had to be grown in sterile conditions, first in smaller cultures and then in deep fermentation tanks, before it could be separated from the broth and purified. Scientists and engineers improved the mold strain, nutrients, oxygen supply, mixing, and extraction. Industrial equipment made it possible to produce amounts that small flasks could never supply [337].

Other medicines depended on different production methods. Chemical drugs could be synthesized in batches and formed into tablets or capsules with mechanized equipment. Injectable medicines and vaccines required controlled formulation and filling into sterile containers. Insulin production expanded from extraction from animal pancreases to recombinant human insulin made by engineered microorganisms. These methods differ, but all rely on repeatable processes that deliver the intended amount of active medicine in a usable form.

4.2 Automation, consistency, and quality

Automated equipment can measure ingredients, control temperature and mixing, move materials between steps, fill containers, and record process conditions. Sensors and computer controls help operators detect when a process is drifting. Instead of relying only on final testing, manufacturers monitor important steps and test samples from production batches. Good manufacturing practice rules require controls and records so that medicines meet standards for identity, strength, quality, and purity [338].

Traditional production often proceeds in separate batches, with pauses between steps. Continuous manufacturing links stages so that raw materials enter a controlled process and finished product emerges steadily. It can reduce some interruptions and support closer process monitoring, but it requires careful validation and regulatory oversight. It is an emerging approach rather than a universal replacement for batch manufacturing [339].

4.3 More treatment, with limits

Large-scale production helped make antibiotics, insulin, vaccines, and many routine medicines available beyond a small number of hospitals. Consistent dosing and dependable supply supported treatment in clinics, vaccination campaigns, emergency care, and long-term management of chronic diseases. Manufacturing capacity alone does not guarantee that patients can obtain those products: cost, distribution, health-system capacity, and shortages still shape access.

Scale also increases the consequences of errors. Contamination or a process failure can affect an entire batch, as the history of vaccine production has shown. Automated equipment does not remove the need for trained staff, validated processes, independent quality oversight, and reporting of defects. The breakthrough was the ability to make large quantities while preserving reliable quality, then connect that production to systems that deliver medicines to patients.

Timeline

1940s Deep-tank fermentation makes wartime mass production of penicillin possible.

1920s onward
Commercial insulin production expands access to treatment for diabetes.

1970s onward

Recombinant DNA methods create new ways to manufacture human insulin and other biologic medicines.

2010s onward

Regulators begin approving medicines made through continuous manufacturing.

Today Automated monitoring, batch records, and quality systems support large-scale production of many medicines.

Chapter 5

Pneumonia and Respiratory Infections

At the start of the twentieth century, pneumonia was among the leading causes of death in the United States, alongside tuberculosis and diarrheal disease. The word described inflammation in the lungs, not one single illness: bacteria, influenza, and other infections could all produce it. Before effective antimicrobial treatment, severe bacterial pneumonia often progressed quickly, and doctors had limited ways to alter its course. Better identification of causes, targeted treatment, vaccines, oxygen, and organized supportive care gradually changed the odds [279, 280].

5.1 Identifying the cause made treatment more specific

Physicians had long recognized pneumococci as an important cause, but laboratory typing showed that pneumococci were not a single uniform target. In the early twentieth century, Rockefeller Institute researchers including Rufus Cole and Oswald Avery developed type-specific serum treatment. When patients received serum matched to their pneumococcal type early enough, the risk of death fell. This work helped establish that laboratory diagnosis and treatment matched to the infecting organism could matter in a disease previously viewed as nearly untreatable [280].

Sulfonamide medicines in the 1930s and penicillin in the 1940s made effective antibacterial treatment more widely available. Antibiotics turned many bacterial pneumonias from rapidly fatal infections into treatable illnesses. They could not treat viral pneumonia, and they did not eliminate deaths from bacterial infection; oxygen, careful fluid management, nursing, and later intensive care remained important for severe disease [280].

5.2 Prevention and care address different risks

Vaccines against influenza and pneumococcal disease prevent some of the infections that can cause pneumonia. Pneumococcal vaccines developed from early serotype research, and conjugate vaccines introduced around 2000 made routine infant immunization more effective while also reducing spread in communities. Vaccines do not prevent every pneumonia because many organisms and causes can inflame the lungs. Their contribution is to lower the risk of particular severe infections [280].

The survival gains came from a chain of advances: recognizing different causes, identifying bacterial types, using matched serum and then antibiotics, preventing disease with vaccines, and supporting breathing and circulation when infection became severe. Unequal access to timely diagnosis, antibiotics, oxygen, and vaccination still shapes who benefits.

Timeline

1890s Antipneumococcal serum treatment is attempted.

1910s–1920s
Serotyping and controlled studies improve type-specific serum treatment.

1930s Sulfonamides reduce deaths from susceptible bacterial pneumonia.

1940s Penicillin expands effective treatment for bacterial pneumonia.

2000 onward
Pneumococcal conjugate vaccines protect infants and reduce community transmission.

Chapter 6

Modern Vaccination

6.1 From variolation to vaccination

Modern vaccination grew out of an older attempt to prevent smallpox. In variolation, material from a smallpox lesion was deliberately introduced into the body. Survivors were often protected from later infection, but the procedure could cause severe disease and spread smallpox to others. The problem was how to produce protection without deliberately exposing people to the full danger of the disease.

In 1796, the English physician Edward Jenner tested the observation that people who had caught cowpox seemed protected from smallpox. He inoculated eight-year- old James Phipps with material from a cowpox lesion and later exposed him to smallpox. Phipps did not develop the disease. Jenner's work helped establish vaccination as a preventive method, though his experiment would not meet modern standards for research ethics or informed consent. It also built on knowledge and practices that existed before Jenner, rather than appearing from a single insight [114].

6.2 From a protective observation to laboratory vaccines

Jenner's vaccine used material from a related disease. In the late nineteenth century, Louis Pasteur and other investigators showed that vaccines could be developed through laboratory work on disease-causing organisms. By weakening microbes or otherwise altering them, researchers could sometimes stimulate protection without causing the full disease. Pasteur's work on anthrax and rabies helped make vaccination a program of experimental biology as well as an empirical practice [15].

Later methods expanded the possibilities. Vaccines could contain killed organisms, weakened live organisms, inactivated toxins, or selected parts of a microbe. Each approach posed different questions: would it stimulate lasting protection, could it be made consistently, and could the production process ensure that it did not cause the disease it was meant to prevent? The answer had to be demonstrated for each product, not assumed from the word "vaccine."

6.3 The polio trials and the need for controlled evidence

Polio made vaccine research a public priority. Epidemics frightened families because infection could paralyze children or impair breathing. In the early 1950s, Jonas Salk and his colleagues developed an inactivated poliovirus vaccine. There was disagreement over whether a killed virus could provide adequate protection, so the question required a large and carefully organized field trial rather than confidence in a theory alone.

The 1954 trial compared vaccinated children with control groups across many communities. Thomas Francis Jr. and the University of Michigan evaluation team collected and analyzed results using common procedures. In April 1955, the trial results showed that the vaccine protected against paralytic polio, and the vaccine was licensed soon afterward [36].

The rollout also exposed how manufacturing failures could undermine a sound scientific result. Some vaccine batches made by Cutter Laboratories contained live poliovirus that had not been fully inactivated. The resulting Cutter incident caused cases of polio, including paralysis and deaths. Vaccination was temporarily suspended while authorities investigated; the response strengthened oversight and made quality control and batch testing central to the vaccination system [37].

This episode altered the meaning of vaccine safety. A successful trial could show that a vaccine worked under study conditions, but it could not guarantee that every manufacturing batch was safe. Vaccines therefore required consistent production, independent review, and monitoring after they reached the public.

6.4 Smallpox eradication depended on more than a vaccine

Smallpox became the clearest demonstration that vaccination could change the fate of a disease. The World Health Organization intensified its eradication program in 1967. The work combined vaccination with case-finding, surveillance, rapid local response, improved

vaccine production, and coordination among national health services. In areas where mass vaccination alone could not reach everyone, teams increasingly focused on finding cases and vaccinating their contacts.

The last naturally occurring case was reported in 1977, and the World Health Assembly declared smallpox eradicated in 1980. This achievement depended on the vaccine, but also on durable field programs, trained local workers, international cooperation, and methods for identifying outbreaks before they spread [38].

Eradication showed that vaccines can protect individuals and, when a disease has no continuing animal reservoir and transmission can be interrupted, help remove the disease from human populations. Most vaccines do not eradicate their target diseases, but widespread immunization can reduce illness, disability, and transmission even when a pathogen continues to circulate.

6.5 Safety monitoring and public trust

Modern vaccine development uses laboratory studies, clinical trials, review of manufacturing, and surveillance after approval. Clinical trials can identify common reactions and measure protection, but even very large trials may not reveal a very rare adverse event. Ongoing systems therefore compare health outcomes in large populations and investigate safety signals. The US Vaccine Safety Datalink, established in 1990, links health records across participating health systems to study possible rare or serious events [39].

Safety systems do not mean that a vaccine has no risks. They make it possible to detect, investigate, and communicate risks in proportion to the evidence. The history of the Cutter incident, as well as later monitoring systems, shows why transparency matters: public confidence depends not only on a product's demonstrated benefits, but also on honest reporting when something goes wrong and visible action to correct it.

6.6 New platforms and old scientific questions

Vaccines today use several platforms, including weakened or inactivated organisms, purified proteins, viral vectors, and nucleic acids. Messenger RNA (mRNA) vaccines deliver instructions that let cells briefly produce a microbial protein for the immune system to recognize. The first mRNA vaccine products used widely in people emerged during the COVID-19 pandemic, but the technology rested on decades of basic research into RNA, delivery systems, and immune responses. Researchers were able to design candidates quickly and evaluate

them in clinical trials; speed came from accumulated science, parallel work, and substantial resources, not from skipping the need for evidence [40].

New platforms do not remove the central questions. Investigators still ask whether a vaccine prevents disease, how long protection lasts, what adverse effects occur, how consistently it can be manufactured, and who can obtain it. During the COVID-19 pandemic, gaps in access between countries made clear that scientific innovation and public-health benefit are not the same thing. A vaccine can change outcomes only when reliable supplies, delivery systems, clear communication, and public trust allow people to receive it.

6.7 What modern vaccination has made possible

Vaccination transformed prevention from an individual act of exposure into a field that joins immunology, microbiology, clinical trials, manufacturing, regulation, surveillance, and public health. Jenner's early work showed that one infection could protect against another. Later investigators learned to design and test vaccines for specific pathogens, while eradication campaigns demonstrated what coordinated delivery could accomplish.

The history is not a straight line from discovery to acceptance. Failures in production, gaps in access, shifting disease patterns, and distrust have all shaped vaccination programs. Modern vaccination is therefore both a biological technology and a social system: its success depends on evidence that a product works, quality that makes each dose reliable, and institutions capable of delivering it fairly and responding openly to harm.

Timeline

1796 Jenner tests cowpox inoculation as protection against smallpox.

1880s Pasteur develops experimental vaccines, including rabies and anthrax vaccines.

1954 The large field trial of Salk's inactivated polio vaccine begins.

1955 Trial results support the vaccine's effectiveness; the Cutter incident exposes serious manufacturing failures.

1967 WHO intensifies the global smallpox eradication program.

1977 The last naturally occurring case of smallpox is reported.

1980 WHO declares smallpox eradicated.

1990 The US Vaccine Safety Datalink is established.

2020–2021

mRNA COVID-19 vaccines move from trials to broad public use.

Chapter 7

Sanitation and Clean Water

Many medical advances act on disease after a person becomes ill. Sanitation and safe water work earlier: they make it harder for infectious organisms to pass from human waste into drinking water, food, and homes. Their impact depends on public works as much as clinical knowledge: sewers, drains, protected water sources, treatment plants, and the institutions that build and maintain them.

7.1 The urban sanitation problem

Rapidly growing nineteenth-century cities struggled to remove human waste and refuse. Drinking water might come from a river or shallow well polluted by nearby cesspools, privies, and drains. Repeated outbreaks of cholera, typhoid, and other intestinal diseases made the danger visible, although many people still explained illness through miasma, or poisonous air.

In Britain, Edwin Chadwick's 1842 report on the sanitary conditions of the labouring population documented the relationship between poverty, crowded housing, waste, and disease. The 1848 Public Health Act enabled local boards to organize drainage, water supply, and refuse removal. Early reformers did not yet understand every pathogen or route of infection, but investments in cleaner surroundings could still interrupt exposure and improve health [98, 99].

7.2 John Snow follows the water

During London's 1854 cholera outbreak, physician John Snow investigated the cluster of cases around the Broad Street pump. He mapped deaths, interviewed residents, and compared water sources. His evidence pointed to contaminated pump water as the likely route of

infection; local authorities removed the pump handle. Snow's work was not the sole origin of modern epidemiology, nor did the map alone settle the scientific debate. It showed how observations about place, water supply, and illness could guide preventive action even before the responsible organism was established [100].

The evidence for waterborne cholera also challenged the idea that foul air was the main explanation. Later research identified specific infectious organisms and clarified how sewage could carry them into water supplies. This made sanitation more than a matter of cleanliness or civic pride: it became a way to block defined pathways of transmission.

7.3 Engineering clean water at scale

Cities gradually built systems that separated drinking-water sources from sewage and carried wastewater away from homes. Filtration reduced particles and many contaminants. Disinfection added another barrier by inactivating pathogens that remained. In the United States, Jersey City began routine disinfection of its community water supply in 1908, a step followed by many other municipalities. Water treatment, wastewater management, and household hygiene together reduced exposure to cholera, typhoid, dysentery, and other water-related infections [101, 102].

The transformation required continuing public investment. Pipes could leak, sewers overflow, and untreated waste could contaminate rivers downstream. Growing cities needed trained engineers, reliable monitoring, enforceable standards, and maintenance budgets. These systems also revealed an equity problem: communities with fewer resources often have less reliable access to safe water and sanitation. Safe supply is therefore both a technical achievement and a public responsibility [102].

7.4 A foundation of modern public health

Clean water and sanitation prevent infections without requiring each person to recognize an exposure or obtain treatment. They have supported longer, healthier lives and made urban growth safer. Yet they do not prevent every infection: pathogens may spread through food, direct contact, or other routes, and water systems require protection from new contamination.

The history of sanitation shows how public health combines observation, engineering, and collective action. Snow's investigation helped make a waterborne route visible; later microbiology explained the organisms; and civil engineers and local governments built systems that translated that knowledge into routine protection. The result is often quiet and easy to overlook: disease is prevented before it reaches the clinic.

The strongest historical lesson is that these gains depended on public systems, not a single household technology. Filtering and treating municipal water, separating sewage from drinking supplies, and maintaining pipes required local government, sustained funding, and public trust. The resulting protection reached people before they needed a diagnosis or a hospital, but it remained uneven wherever infrastructure was absent or unreliable [101, 102].

Timeline

1842 Edwin Chadwick publishes his report on the sanitary conditions of Britain's labouring population.

1848 Britain's Public Health Act enables local authorities to address drainage, water supply, and refuse.

1854 John Snow investigates London's Broad Street cholera outbreak and identifies contaminated pump water as the likely source.

Late nineteenth century Growing cities expand sewerage, filtration, and protected water supplies.

1908 Jersey City begins routine disinfection of its community drinking water supply.

Today Water treatment, sanitation, surveillance, and maintenance remain core public-health infrastructure.

Chapter 8

Antisepsis and Aseptic Surgery

For much of the nineteenth century, an operation could repair an injury only to be followed by a deadly wound infection. Surgeons knew that pus, fever, and tissue decay often followed surgery, but they lacked a reliable way to prevent it. Antisepsis and aseptic surgery transformed the operating room by changing how surgeons understood contamination and how they organized their work.

8.1 Lister's antiseptic system

Ignaz Semmelweis had shown that handwashing could reduce childbed fever, and Louis Pasteur's research strengthened the case that microorganisms caused fermentation and decay. In Glasgow, surgeon Joseph Lister applied these ideas to wound infection. After treating a compound fracture with carbolic acid in 1865, he developed a system for cleaning wounds, dressings, hands, and instruments with the chemical. In 1867, he published a series of cases and explained how antiseptic treatment could prevent suppuration and infection [13, 17, 103].

Lister's method was more than a single disinfectant. It asked surgeons to change their routines: cover wounds with treated dressings, clean tools, wash hands, and take steps to reduce contamination during an operation. His carbolic spray, intended to destroy germs in the air around the wound, became a conspicuous part of early Listerism. The method was difficult to master, and many surgeons doubted its value. Lister revised his practice as experience accumulated; historical accounts emphasize that antiseptic surgery developed through repeated adjustments, not as a finished system introduced all at once [103, 104].

8.2 From killing germs to excluding them

As bacteriology and sterilization methods advanced, surgeons increasingly tried to keep microorganisms from entering the wound in the first place. This approach, called asepsis, differed in emphasis from antisepsis: rather than relying mainly on chemicals to kill contaminants, it aimed to create and preserve a sterile field. German-speaking surgeons helped develop practices that included boiling instruments, using steam to sterilize dressings and equipment, and wearing clean clothing. Autoclaves and sterilization containers made these routines more dependable [105].

The shift required coordinated work by the whole surgical team. Instruments and dressings had to remain sterile until use; hands, gowns, and the operating surface had to be prepared; and staff needed to avoid contaminating the wound during the procedure. William Halsted introduced rubber gloves at Johns Hopkins around 1890 after surgical nurse Caroline Hampton developed a skin reaction to the antiseptic solutions used at work. Gloves first solved a practical problem for Hampton, then became part of efforts to reduce contamination during surgery [106].

Antiseptic and aseptic methods were not mutually exclusive stages in a simple sequence. Modern operating rooms combine careful hand hygiene, sterile instruments, protective barriers, skin antisepsis, and appropriate disinfection. Each measure blocks a different route by which infection can reach a patient. Hospitals still rely on validated cleaning and sterilization processes to keep reusable instruments from transmitting pathogens [107].

8.3 A new kind of surgery

Safer wound care changed what surgeons could attempt. As deaths from postoperative infection fell, operations could become more deliberate and complex rather than being limited to procedures that could be completed quickly. Antisepsis and asepsis did not eliminate surgical-site infections, and they did not make every operation safe. They made infection prevention a planned part of surgery and a shared responsibility of hospitals and surgical teams.

The operating room became a system, not merely a place where a skilled surgeon worked. Its safety depended on the preparation of instruments, materials, people, and space. That change helped make modern surgery possible, from joint replacement to transplantation, by reducing a danger that had once made even a technically successful operation perilous.

The change was visible in the operating room: cleaning hands and wounds, sterilizing instruments, and maintaining a clean field became linked practices. Their adoption was

gradual, and improvements in survival cannot be attributed to one measure alone; anesthesia, better surgery, and hospital organization also mattered. The lasting breakthrough was a reproducible system for preventing contamination during care [105, 107].

Timeline

1847 Ignaz Semmelweis introduces chlorinated handwashing in a maternity clinic.

1865 Joseph Lister begins applying carbolic acid treatment to compound fractures.

1867 Lister publishes his antiseptic principles for surgical wounds.

1880s Aseptic methods using boiling and steam sterilization spread in European surgical clinics.

1890 Rubber gloves are introduced in the Johns Hopkins operating room.

Today Sterile instruments, barriers, hand hygiene, and validated reprocessing are routine parts of infection prevention in surgery.

Chapter 9

Insulin Therapy

9.1 Before insulin: controlling a disease by starvation

Diabetes had been recognized for centuries by the sweet taste of urine and the progressive thirst, urination, and wasting it caused. By the late nineteenth century, experiments had connected the pancreas to the disease. In 1889, Oskar Minkowski and Joseph von Mering removed a dog's pancreas and observed that the animal developed diabetes. Researchers then suspected that the pancreas released an internal secretion needed to regulate the body's use of sugar, but they could not extract a safe, effective preparation for patients.

Before insulin, physicians could restrict food to slow the rise of glucose in the urine. Frederick Allen's severe "starvation treatment" sometimes delayed death, but only by keeping patients dangerously undernourished. For children with the rapidly progressive form now called type 1 diabetes, there was no lasting treatment. Extracts of pancreas had been tried, but digestive enzymes and impurities made them toxic or ineffective [24].

9.2 From pancreatic experiments to a usable extract

In 1921, Frederick Banting proposed that a pancreatic extract might work if the pancreas's digestive tissue was allowed to degenerate while its islet cells were preserved. At the University of Toronto, he and medical student Charles Best tested extracts in dogs whose pancreases had been removed. John Macleod provided laboratory space, equipment, and scientific supervision; James Collip, a biochemist who joined the group later that year, developed a more purified extract suitable for human use. The work drew on decades of physiological research and depended on several distinct contributions, not a single moment of discovery [25, 26].

The team tested its extracts by measuring their effect on diabetic dogs. When an extract

lowered blood sugar and reduced sugar in the urine, it offered evidence that the pancreas supplied a missing substance. The result was promising, but a preparation that worked in a dog was not yet a medicine. It had to be purified, standardized, produced in sufficient quantity, and tested carefully in people.

9.3 Leonard Thompson and the first successful treatment

On 11 January 1922, fourteen-year-old Leonard Thompson received an early pancreatic extract at Toronto General Hospital. It produced little benefit and caused an abscess at the injection site. Collip continued refining the preparation. On 23 January, Thompson received the improved extract; his blood and urine glucose fell, and his clinical condition improved. The result provided compelling evidence that a purified pancreatic substance could reverse the immediate metabolic crisis of severe diabetes [29, 25].

More patients were treated soon afterward, and the response was dramatic. Some children who had been near death regained strength. The extract did not eliminate diabetes or restore the pancreas, however. It replaced a hormone the body could not supply in adequate amounts. Treatment required repeated injections, careful adjustment to meals and activity, and close attention to dangerously low blood sugar as well as high blood sugar.

The Nobel Prize in Physiology or Medicine for 1923 went to Banting and Macleod for the discovery of insulin. The award did not settle disputes about credit: Banting shared his prize money with Best, and Macleod shared his with Collip. Their actions reflected a central fact about the work: the clinical treatment depended on a team that included experimental researchers, a supervising physiologist, a medical student, a biochemist, and hospital clinicians [27, 26].

9.4 Making insulin at scale

The first insulin came from animal pancreases, principally those of cattle and pigs. Demand quickly exceeded what one university laboratory could supply. The University of Toronto licensed manufacturers and required batches to be tested and standardized, because the strength and purity of early preparations could vary. By 1923, commercial production had begun. Wider supply turned a laboratory success into a treatment that physicians could prescribe beyond Toronto [24, 25].

Animal insulin saved lives but had practical limits. It was difficult to purify, could cause

reactions in some patients, and its duration of action made dosing difficult. Researchers developed longer-acting formulations, first by combining insulin with protamine and zinc. Glucose monitoring also changed slowly: urine tests were indirect and could not show a patient's blood glucose at that moment. For decades, people and clinicians had to make treatment decisions with incomplete information [25, 24].

9.5 Human insulin made with recombinant DNA

In the late 1970s, researchers used recombinant DNA techniques to produce human insulin in bacteria. The method offered a more reliable supply that did not depend on slaughterhouse pancreases and made highly purified human insulin possible. In 1982, the US Food and Drug Administration approved Humulin, the first biosynthetic human insulin and the first approved medical product made using recombinant DNA technology [24].

Later insulin analogs altered the molecule slightly to change how quickly it was absorbed or how long it acted. Fast-acting preparations could better match the rise in glucose after meals, while longer-acting preparations could cover periods between meals and overnight. Home blood glucose meters, insulin pens, pumps, and eventually continuous glucose monitors gave patients and clinicians more information and more flexible ways to adjust treatment. Each advance addressed a limit of earlier therapy, while leaving the daily work of managing diabetes in place [25, 24].

9.6 Testing what better control could achieve

The Diabetes Control and Complications Trial (DCCT), begun in 1983, tested whether intensive insulin treatment could reduce long-term complications of type 1 diabetes. It compared a regimen of frequent glucose checks and at least three daily injections or an insulin pump with the conventional practice of one or two injections and less frequent testing. The trial showed that more intensive glucose control reduced the risk of eye, kidney, and nerve damage. It also made the trade-off visible: intensive treatment increased the risk of severe hypoglycemia and required substantially more work from participants [28].

This evidence changed the goal of insulin therapy. The aim was no longer simply to prevent immediate collapse; it was also to keep glucose near a safer range over years and thereby reduce complications. Follow-up studies found that the benefits of improved control persisted long after the trial ended [28].

9.7 What insulin therapy changed

Insulin transformed type 1 diabetes from a rapidly fatal illness into a condition that many people could live with for decades. It also helped explain the role of the pancreas and established hormone replacement as a powerful medical treatment. Yet the discovery did not end the disease. Insulin must be matched to food, activity, illness, and the body's changing needs; too little can lead to dangerous hyperglycemia, and too much can cause hypoglycemia.

The history is therefore both a story of rescue and one of continuing experimentation. Animal studies established the principle, purification made human treatment possible, manufacturing expanded access, and clinical trials showed how treatment could reduce long-term harm. Later devices improved monitoring and delivery, but the benefits of insulin still depend on patients being able to obtain it and use it safely. Its discovery did not produce a finished cure. It created a durable therapy whose results have improved as evidence, technology, and clinical practice have advanced.

Timeline

1889 Minkowski and von Mering show that removing a dog's pancreas causes diabetes.

1921 Banting, Best, and Macleod's Toronto team demonstrates the effect of pancreatic extracts in diabetic dogs; Collip works to purify the extract.

January 1922 Leonard Thompson receives the first successful insulin treatment.

1923 Commercial insulin production begins; Banting and Macleod receive the Nobel Prize.

1930s Protamine and zinc formulations extend insulin's action.

1982 FDA approves Humulin, the first recombinant human insulin product.

1983–1993 The DCCT establishes that intensive glucose control reduces complications of type 1 diabetes.

Chapter 10

X-ray Imaging

Before X-ray imaging, a physician could suspect a broken bone or a hidden foreign object from examination, but seeing its exact position often required exploration or guesswork. X-rays created a new kind of evidence: a record of structures inside the body made without an incision. The discovery connected experimental physics to clinical diagnosis and eventually made radiography a routine part of medicine.

10.1 Röntgen discovers a new kind of ray

In November 1895, German physicist Wilhelm Conrad Röntgen was studying electrical discharges in a vacuum tube. He noticed that a screen coated with a fluorescent material glowed even when the tube was covered. Something invisible was passing through the covering and exciting the screen. Röntgen investigated the rays and found that different materials absorbed them by different amounts. Bone and metal blocked more of the radiation than soft tissue, creating contrasting shadows on photographic plates.

In December, Röntgen made an image of his wife Bertha's hand, in which the bones and her ring were visible. He called the radiation "X-rays" because its nature was unknown. The images captured public attention, and in 1901 Röntgen received the first Nobel Prize in Physics for discovering the rays [108, 109].

10.2 From striking image to clinical tool

Physicians and surgeons began experimenting with X-rays soon after the discovery. They could locate fractures, bullets, and other foreign objects without cutting into tissue. But an image did not interpret itself: the equipment needed to be operated, exposure had to be

controlled, and doctors had to learn what a shadow could establish. Historical studies show that early enthusiasm did not make X-rays instantly routine in hospitals; staffing, hospital organization, and clinical practice shaped how the technology was used [110].

In a conventional radiograph, an X-ray beam passes through the body to a detector. Dense structures absorb more radiation and appear differently from less dense tissues, producing a two-dimensional projection in which internal structures can overlap. Film-based imaging later gave way to digital detectors, making images easier to store, share, and adjust. Fluoroscopy extends the same principle to moving images, allowing clinicians to guide some procedures in real time. Computed tomography, developed later, uses many X-ray measurements to reconstruct cross-sectional images [111].

10.3 Radiation and responsibility

X-rays are a form of ionizing radiation. Their power to pass through the body also means that exposure must be justified and kept as low as reasonably achievable while still answering the clinical question. Early users did not understand all long-term effects, and repeated exposures injured some radiographers and patients. Protective practices, dose control, and better equipment developed as the risks became clearer. For a medically appropriate examination, the diagnostic benefit generally outweighs the small radiation risk, but unnecessary exposure should be avoided [111, 110].

The deeper change was epistemic as well as technical. An X-ray image made parts of the living body visible to clinicians who had previously relied on symptoms, touch, and surgery. It did not replace those forms of evidence: a radiograph could be misleading or incomplete without a clinical question and skilled interpretation. X-ray imaging helped create modern diagnostic radiology by turning invisible anatomy into a record that could be examined, compared, taught, and used to plan treatment.

As equipment and technique improved, X-rays moved from a scientific demonstration to routine diagnosis of fractures, chest disease, and other conditions that had previously required inference or exploratory surgery. Their value came with a new responsibility: repeated or poorly controlled exposure can cause harm, so dose and clinical need became part of safe imaging practice [110, 111].

Timeline

November 1895

Wilhelm Conrad Röntgen discovers a previously unknown penetrating radiation.

December 1895
: Röntgen makes an X-ray image of Bertha Röntgen's hand.

1896
: Physicians begin experimenting with radiographs to locate fractures and foreign objects.

1901
: Röntgen receives the first Nobel Prize in Physics.

Twentieth century
: Improved tubes, detectors, fluoroscopy, and radiation protection expand clinical use.

Today
: Radiography remains widely used, alongside newer modalities such as CT, MRI, and ultrasound.

Chapter 11

Blood Groups and Transfusion

For centuries, physicians tried to replace blood lost in childbirth, injury, and surgery. Some transfusions appeared to help; others caused fever, shock, jaundice, or death. Before blood groups were understood, doctors had no reliable way to tell whether one person's blood could safely be given to another.

The discovery of blood groups changed transfusion from a gamble into a procedure that could be tested. Later advances in storage, donor services, component therapy, and infectious-disease screening made transfusion a routine part of modern medicine. Safety depends on all of these steps, not just on matching two letters on a blood-type card.

11.1 Why some transfusions caused a reaction

Blood is made up of cells suspended in plasma. Red blood cells carry oxygen; their surfaces display molecules called antigens. Plasma contains antibodies that can recognize particular antigens. If a recipient's antibodies bind to antigens on transfused red cells, the cells can clump and break apart. This immune reaction can obstruct small vessels and release substances that injure the body.

In 1901, the Austrian physician Karl Landsteiner reported that people differed in how their blood reacted when mixed. He found that some combinations clumped while others did not, revealing that human blood was not all the same. By 1909, the familiar A, B, AB, and O groups had been classified. Landsteiner received the 1930 Nobel Prize for the discovery of human blood groups [41].

In the ABO system, group A red cells carry A antigen and the plasma usually contains anti-B antibodies. Group B has B antigen and anti-A antibodies. Group AB has both antigens and neither of these antibodies, while group O has neither antigen and usually has both antibodies. The danger is reciprocal: the donor's red cells must not be attacked by the

recipient's antibodies. These descriptions concern the ABO system; other antigens can also matter.

11.2 Typing and crossmatching

Knowing a patient's ABO group was a major advance, but it was not a complete compatibility test. In 1907, physician Ludvig Hektoen proposed comparing donor and recipient blood before transfusion. That year, Reuben Ottenberg performed transfusions using blood typing and crossmatching. In a crossmatch, the laboratory tests whether the recipient's plasma reacts with the donor's red cells. This direct check can reveal incompatibilities that a basic blood-type label misses [42].

The Rh blood-group system added another important layer. In 1939–1940, Karl Landsteiner, Alexander Wiener, Philip Levine, and R. E. Stetson described the system now called Rh. The most clinically important Rh antigen is D. People whose red cells carry it are commonly called Rh-positive; those whose cells do not are Rh-negative. An Rh-negative person can develop anti-D after exposure to Rh-positive red cells, including through transfusion or pregnancy. That antibody can complicate later transfusions and, in pregnancy, may harm an Rh-positive fetus. Rh testing and appropriate care greatly reduced these risks [42].

For red-cell transfusion, a simplified compatibility guide is shown below. It includes ABO and RhD only; it leaves out many other red-cell antigens and does not replace antibody screening and a laboratory crossmatch.

Recipient type	Compatible red-cell donor types
O negative	O negative
O positive	O negative, O positive
A negative	O negative, A negative
A positive	O negative, O positive, A negative, A positive
B negative	O negative, B negative
B positive	O negative, O positive, B negative, B positive
AB negative	O negative, A negative, B negative, AB negative
AB positive	All eight ABO/RhD types

This chart applies to red cells, not to plasma, whose compatibility rules are different. O-negative red cells can be useful when there is no time to establish a patient's type, but they are not compatible with every antigen system and are not a universal blood product. In planned care, clinicians type and screen the recipient and select the safest available component [43].

11.3 From fresh blood to a blood bank

Even compatible blood was difficult to use if it clotted before it reached a patient. Sodium citrate, an anticoagulant, made it possible to keep donated blood fluid for longer. During the First World War, military physicians developed systems for collecting and storing citrated blood near the front. Oswald Robertson's work with stored blood helped show that transfusion need not depend on finding a donor at the bedside at the moment of crisis [42].

In 1937, Bernard Fantus established a hospital blood bank at Cook County Hospital in Chicago. The bank stored donations so they could be matched to patients when needed. During and after the Second World War, organized donor programs and improved preservation expanded the supply. Laboratories also learned to separate donated blood into red cells, plasma, and platelets, so a donation could be used to treat different needs rather than giving every patient whole blood [44].

Compatibility testing addresses immune reactions; it does not detect every infection that might be carried in a donation. Screening donors and testing donated blood became another essential part of safety. Today, donations are tested for multiple infectious-disease markers, while hospitals still perform patient-specific typing and compatibility checks before transfusion [45].

11.4 A system of checks

Blood groups made transfusion safer because they explained why certain combinations were dangerous and offered a way to avoid them. But a modern transfusion is not simply a matter of finding the same ABO group. Staff must identify the patient correctly, select an appropriate component, check for antibodies, perform compatibility testing, and monitor the transfusion. Donor recruitment, careful storage, and infection screening matter too.

This layered system grew from a basic observation about clumping blood. It also illustrates a broader change in medicine: a treatment becomes reliable when biological knowledge is combined with repeatable laboratory tests, trained staff, quality-controlled supplies, and procedures that catch human error.

Timeline

1818 James Blundell reports human-to-human transfusions for severe bleeding after childbirth.

1901 Karl Landsteiner identifies distinct human blood groups.

1907 Hektoen proposes crossmatching; Ottenberg uses typing and crossmatching in transfusion.

1914–1918
Citrate anticoagulation and stored blood support wartime transfusions.

1937 Bernard Fantus opens a hospital blood bank in Chicago.

1939–1940
The Rh blood-group system is described.

Today Blood typing, antibody screening, crossmatching, component preparation, and infection testing work together to reduce risk.

Chapter 12

Automated Laboratory Testing

Doctors rely on laboratory tests to detect disease, guide urgent treatment, and monitor chronic conditions. Before automation, many tests required a technologist to measure, mix, time, and read each specimen by hand. As demand grew, clinical laboratories needed methods that could produce reliable results for many patients in less time. Automated analyzers made routine blood and urine testing faster and more consistent, extending the reach of diagnostic medicine.

12.1 From manual assays to analyzers

In the 1950s, two different innovations began changing routine laboratory work. Leonard Skeggs developed the AutoAnalyzer, which moved samples and reagents through tubing and measured chemical reactions continuously. The same decade, Wallace Coulter's electronic cell counter used changes in electrical resistance as blood cells passed through a small opening. It replaced slow manual counting with rapid, reproducible measurements and helped make the complete blood count a routine test [335, 336].

As analyzers improved, laboratories could test more specimens and measure more substances from each one. Later AutoAnalyzer designs could measure more than 20 analytes while processing about 150 samples per hour. This means many laboratory results could be generated quickly; the exact number of patients served depends on the tests ordered, instrument capacity, staffing, and how quickly specimens arrive [335].

12.2 The laboratory becomes a connected system

Modern laboratory automation can link specimen identification, sorting, centrifugation, analyzer loading, result checking, and storage. Hematology analyzers count and classify blood cells; chemistry analyzers measure substances such as glucose, electrolytes, and kidney or liver markers; immunoassay systems measure hormones, proteins, and markers of infection or injury. Laboratory information systems connect orders and results to patient records. Automation therefore supports emergency diagnosis, routine health checks, blood-bank testing, cancer care, and the monitoring of long-term conditions [335, 336].

The machines do not replace laboratory professionals. Staff maintain instruments, use quality-control samples, investigate unexpected results, and review specimens flagged as abnormal. Some tests remain manual or take hours or days, especially when they require microbial growth or specialized analysis. A fast result is useful only if the specimen is correctly identified, the method is accurate, and a clinician interprets it in context. Automation can also make unnecessary testing easier, so speed and volume must be balanced with careful decisions about which tests a patient needs [335].

Large-scale testing is a quiet but essential breakthrough. It helps health systems identify anemia, infection, diabetes, kidney injury, and many other conditions across large patient populations, then repeat measurements to see whether treatment is working. Its benefits depend on trained staff, quality assurance, reliable specimen transport, and access to laboratories.

Timeline

1953 Wallace Coulter receives a patent for electronic particle counting, a principle later used in blood-cell analyzers.

1950s Coulter cell counters and Skeggs's AutoAnalyzer begin automating routine hematology and clinical chemistry.

1970s Computerized analyzers expand the number of tests that can be measured from patient specimens.

1980s onward Robotics, conveyor systems, and laboratory information systems connect more stages of testing.

Today High-volume laboratories process large numbers of specimens, with automation supporting but not replacing professional review.

Chapter 13

Modern Anesthesia

13.1 The problem anesthesia had to solve

Before the nineteenth century, surgery was constrained by the pain and terror of the operation. Surgeons worked quickly, and many procedures that are now routine were avoided or performed only when disease or injury left no alternative. Alcohol, opium, and other remedies could dull distress, but they could not reliably prevent pain or keep a patient still and unconscious.

Anesthesia changed what surgery could attempt. It is not one effect produced by one drug: depending on the procedure, clinicians may need unconsciousness, amnesia, pain relief, muscle relaxation, or only numbness in a limited area. The central challenge became how to produce the needed effects while keeping breathing, circulation, temperature, and other vital functions safe.

13.2 Ether and the public demonstration

Several people contributed to the arrival of surgical anesthesia. In 1842, Georgia physician Crawford Long used ether during surgery but did not publish his work until 1849. Dentist Horace Wells explored nitrous oxide for dental extractions; a public demonstration in 1845 did not convince its audience. On 16 October 1846, dentist William T. G. Morton gave a successful public demonstration of ether at Massachusetts General Hospital while surgeon John Collins Warren removed a neck tumor from Edward Gilbert Abbott. The operation showed other surgeons that a patient could be made insensible to pain during a major procedure, and news of the demonstration spread quickly [30, 31].

The event became known as “Ether Day,” but it did not settle who deserved credit for discovering anesthesia. Long’s earlier use, Wells’s work with nitrous oxide, Morton’s public

demonstration, and the contributions of chemist Charles Jackson and others all became part of a prolonged priority dispute. The distinction matters: a method can be used privately before it is published, demonstrated publicly, repeated elsewhere, and adopted as routine practice at different times. The 1846 operation was a turning point because it made surgical anesthesia visible and reproducible to the medical community, not because it erased the work that came before it [30].

Ether was soon joined by chloroform, which Scottish obstetrician James Young Simpson introduced in 1847. Both drugs could suppress pain, but neither was easy to control, and both could endanger life if the depth of anesthesia or the patient's breathing and circulation were neglected. Anesthesia reduced one major hazard of surgery while creating a new need: someone had to attend continuously to the patient's condition.

13.3 Different ways to block pain

Inhaled ether and chloroform produced general anesthesia, affecting the whole body. A different path opened in 1884, when ophthalmologist Carl Koller demonstrated that cocaine could numb the eye for surgery. Local anesthetics made it possible to block sensation in a limited area, and later regional techniques could numb a larger part of the body while a patient remained awake or lightly sedated [32].

These approaches widened the choices available to surgeons and patients. General anesthesia remained valuable for long or complex procedures, but it was no longer the only way to make an operation tolerable. Modern anesthesia grew as a set of techniques, with the method selected according to the operation and the patient's needs. The anesthetic plan also had to account for the possibility that a local or regional block might be incomplete or that a patient might need airway support.

13.4 Airways, muscle relaxation, and controlled breathing

By the early twentieth century, anesthesia was becoming a distinct medical specialty. Anesthetists developed methods to manage the airway and deliver anesthetic gases more predictably. Endotracheal tubes allowed clinicians to maintain an open airway and control ventilation during operations. In 1942, Harold Griffith and Enid Johnson introduced curare as a muscle relaxant during general anesthesia. Relaxed muscles made some operations easier, but paralysis also meant that a patient could not breathe independently; the technique therefore

depended on the ability to support and control breathing [33].

The result was a shift from simply rendering a patient unconscious to coordinating several effects. An anesthetic might combine a drug for unconsciousness, one for pain, and one for muscle relaxation, while the anesthesia team protected the airway and supported circulation. Newer intravenous and inhaled medicines made it easier to adjust the depth and duration of anesthesia. Equipment such as calibrated vaporizers helped deliver more predictable concentrations than early open-drop methods.

13.5 Monitoring turns observation into a safety system

For much of anesthesia's history, clinicians judged a patient's condition by watching skin color and chest movement, feeling a pulse, and listening to breathing. These observations remained essential, but they could miss early changes in oxygenation or ventilation. Electronic monitors began extending what an anesthetist could detect while keeping continuous observation at the center of care.

In 1986, the American Society of Anesthesiologists adopted basic monitoring standards requiring continual evaluation of oxygenation, ventilation, circulation, and temperature during anesthesia. The standards encouraged quantitative measures such as pulse oximetry for blood oxygenation and measurement of exhaled carbon dioxide to assess ventilation. They also specified continuous presence of qualified anesthesia personnel [34]. Monitoring did not replace clinical judgment; it gave the team earlier warning of problems and a chance to respond before they caused serious injury.

Safety improved through systems as well as devices: training, checklists, standardized equipment, recovery rooms, and review of adverse events. The Anesthesia Patient Safety Foundation, founded in 1985, helped make preventable injury a shared professional concern and supported wider adoption of monitoring technologies [35].

13.6 Modern anesthesia as a coordinated practice

Today, anesthesia may involve general anesthesia, regional or local anesthesia, sedation, or combinations of these approaches. Before an operation, the anesthesia professional reviews the planned procedure and the patient's health; during it, the team adjusts medication and supports vital functions; afterward, care continues through recovery and pain management. Anesthesia is therefore a practice that spans before, during, and after surgery, rather than a single injection or moment of unconsciousness.

Modern techniques have made many operations safer and more feasible, but they have not made risk disappear. Anesthetic drugs can affect breathing and blood pressure, and procedures still require careful preparation and continuous attention. The history of anesthesia is a story of expanding surgical possibility matched by growing responsibility: as clinicians learned to control pain and consciousness, they also learned to monitor and protect the whole patient.

Timeline

1842 Crawford Long uses ether for surgery in Georgia; he publishes the work in 1849.

1844–1845 Horace Wells explores nitrous oxide for dental anesthesia; his public demonstration fails to persuade.

1846 Morton publicly demonstrates ether anesthesia at Massachusetts General Hospital.

1847 James Young Simpson introduces chloroform in obstetrics.

1884 Carl Koller demonstrates cocaine as a local anesthetic for eye surgery.

1942 Curare is introduced as a muscle relaxant during general anesthesia.

1986 The American Society of Anesthesiologists adopts basic monitoring standards.

Chapter 14

Modern Operating Theatres

Modern surgery takes place in an operating theatre designed around a team and a sequence of safety checks. Surgeons, anesthesia professionals, nurses, and technicians coordinate the patient, procedure, sterile instruments, medicines, equipment, and recovery plan. Earlier chapters describe anesthesia and infection prevention; this chapter follows how research and teamwork helped bring these protections together in routine surgical practice.

14.1 The operating theatre as a safety system

As operations became more complex, preventing harm required attention to the whole process, not only the surgeon's technique. A patient could be harmed by an incorrect procedure or site, an unrecognized airway problem, a missed antibiotic dose, unavailable equipment, or a failed handoff after surgery. Monitoring, sterilization, blood services, and recovery care each address different risks. Reliable practice depends on the team coordinating them at the right time.

14.2 Research tests a surgical checklist

In 2007, the World Health Organization launched the Safe Surgery Saves Lives program. Its checklist organized evidence-based actions around three pauses: before anesthesia, before the incision, and before the patient leaves the operating room. Teams confirm the patient's identity and procedure, discuss anticipated problems, check anesthesia and essential equipment, verify infection-prevention steps, and plan postoperative care [325, 326].

A prospective study introduced the checklist in eight hospitals across different regions and levels of resources. Among the patients studied, inpatient complications fell from 11.0 percent

before implementation to 7.0 percent afterward; deaths fell from 1.5 percent to 0.8 percent. This was a multicentre before-and-after study, not a randomized trial, so the findings show an association and cannot establish that the checklist alone caused the entire improvement. The study nevertheless demonstrated that a shared safety routine could be evaluated across varied operating settings [327].

14.3 Making safety routine

A checklist works when staff use it to communicate, raise concerns, and catch problems before they reach the patient. The WHO implementation guidance emphasizes staff involvement, training, local adaptation, leadership, and feedback. A form completed without discussion cannot replace skilled clinicians, adequate equipment, sterile processing, or sound judgment [326].

Modern operating theatres combine physical design, technology, trained teams, and tested routines. Research helped identify preventable hazards and measure whether safety practices changed outcomes. The operating theatre became safer through this coordinated system, while the gains still depend on staffing, resources, and consistent practice.

Timeline

2007 WHO launches the Safe Surgery Saves Lives program.

2008 Eight hospitals pilot the surgical safety checklist and collect outcome data.

2009 The multicentre checklist study and WHO implementation guidance are published.

Today Surgical teams adapt safety checks to local workflows and review their results.

Chapter 15

Smallpox Eradication

Smallpox killed and disfigured millions across centuries, but it had features that made eradication possible: people were its only natural host, and infection usually caused a recognizable illness. A vaccine existed, yet vaccinating everyone at once proved difficult. The successful campaign combined immunization with surveillance, rapid investigation of outbreaks, and the containment of new cases.

15.1 From vaccination to a global goal

Edward Jenner's work in 1796 helped establish vaccination against smallpox, and national vaccination programs later reduced the disease in many places. But coverage was uneven, outbreaks continued, and global eradication required more than the existence of an effective vaccine. In 1958, the World Health Assembly called for eradication. The campaign gained momentum after the Assembly and WHO launched an intensified program in 1966–1967, with international coordination, vaccine support, and field teams [112, 113].

Early plans leaned heavily on mass vaccination. In many areas, however, health teams could not reach every person, and vaccine supplies and local resources were limited. The campaign therefore adapted. Workers searched actively for cases, investigated reports, and vaccinated people who had contact with an infected person and those in nearby communities. This surveillance-and-containment strategy focused effort where transmission was occurring rather than depending only on uniform coverage [114, 113].

15.2 Tools, teams, and the last cases

Practical innovations helped workers act quickly. The bifurcated needle carried a small amount of vaccine and made administration simple to teach. Freeze-dried vaccine could withstand difficult conditions better than some earlier preparations. More important still were trained local health workers who knew the communities, followed leads, traced contacts, and returned to check whether outbreaks had stopped. Eradication depended on logistics, trust, reporting, and political commitment as much as on vaccine biology [113, 114].

The last naturally occurring case was recorded in Somalia in 1977. After reviewing evidence from countries around the world, the WHO's Global Commission for the Certification of Smallpox Eradication concluded that transmission had ended. On 8 May 1980, the World Health Assembly formally endorsed the declaration that smallpox had been eradicated [112, 114].

15.3 What eradication required

Smallpox eradication was not a victory of technology acting alone. Vaccine made prevention possible; surveillance showed where disease remained; containment stopped chains of transmission; and international cooperation coordinated people, supplies, and evidence. The strategy also depended on the disease's biology: smallpox had no animal reservoir, and cases were usually visible enough to find and isolate. Other infections may not share these conditions.

The campaign changed public health's ambitions. It demonstrated that a coordinated international program could move beyond controlling a disease to ending its natural transmission. It also left an operational lesson: mass vaccination can establish broad protection, while careful case finding and targeted response can finish the work where transmission persists.

The campaign succeeded because vaccination was paired with surveillance and rapid containment. Teams searched for cases, traced contacts, and focused vaccination around outbreaks, making it possible to interrupt transmission even where routine coverage was incomplete. Smallpox's lack of an animal reservoir also made eradication achievable in a way that is not true for every infection [113, 114].

Timeline

1796 Edward Jenner conducts his early vaccination experiment against smallpox.

1958 The World Health Assembly calls for global smallpox eradication.

1966–1967

WHO organizes an intensified eradication program with vaccine support and field operations.

1970s Surveillance and containment help stop outbreaks in remaining endemic areas.

1977 The last naturally occurring smallpox case is recorded in Somalia.

8 May 1980

The World Health Assembly endorses the declaration that smallpox has been eradicated.

Chapter 16

Polio Vaccines

Poliomyelitis could infect many people without symptoms, but in a small fraction it damaged nerve cells and caused permanent paralysis or death. Outbreaks frightened families because healthy children could suddenly lose the ability to walk or breathe independently. Two different vaccines changed that history: Jonas Salk's inactivated vaccine, given by injection, and Albert Sabin's live-attenuated oral vaccine.

16.1 Growing the virus made vaccination possible

Polio research advanced when John Enders, Thomas Weller, and Frederick Robbins showed that poliovirus could be grown in non-nervous human tissue cultures. This made it easier to study the virus and produce material for vaccine development; the three researchers received the 1954 Nobel Prize for this work [115, 116].

Salk's team used virus that had been grown and then inactivated with formaldehyde, so it could stimulate immunity without reproducing in the recipient. A large 1954 field trial involved about 1.8 million children in the United States, Canada, and Finland. The results were announced on 12 April 1955, and the inactivated polio vaccine (IPV) was licensed that day. Its rapid introduction was clouded by the Cutter incident, in which improperly inactivated vaccine caused cases of polio. The event led to stricter oversight of vaccine production and testing [115, 116, 118].

16.2 The oral vaccine and mass campaigns

Sabin and other researchers pursued a different method: weakening live virus so it would stimulate immunity without causing the usual disease. The oral polio vaccine (OPV) was

easier to administer in large campaigns, and it created strong intestinal immunity that helped reduce transmission. Beginning in 1961, OPV became an important tool in many national and global programs [115, 116].

IPV and OPV have different advantages and limits. IPV cannot cause vaccine-associated paralytic polio, while the live vaccine very rarely can; OPV can also regain the ability to spread and cause disease in populations with low immunity. Countries have therefore used different vaccine strategies over time. These trade-offs are not a simple contest between two inventors: the best tool for protecting an individual may differ from the tool best suited to interrupt transmission in a community [116, 117].

16.3 A success still in progress

Polio vaccination drove down disease dramatically and eliminated wild poliovirus from most of the world. Global eradication remains incomplete, so vaccination and surveillance continue. The history illustrates the importance of both vaccine design and delivery: a potent vaccine protects only when it reaches people, and high coverage is needed to prevent the virus from finding new hosts [117].

The polio story also shows why vaccine programs must adapt. The injected and oral vaccines were created with different scientific methods and public health goals. Together with surveillance, laboratory testing, and sustained campaigns, they brought the world close to eliminating a feared cause of childhood paralysis.

Timeline

1949 Enders, Weller, and Robbins demonstrate growth of poliovirus in tissue culture.

1954 A large field trial evaluates Salk's inactivated polio vaccine.

12 April 1955
Results of the Salk vaccine trial are announced; IPV is licensed.

1955 The Cutter incident prompts stronger controls over vaccine production.

1961 Sabin's oral polio vaccine enters broad use.

1988 The World Health Assembly launches the Global Polio Eradication Initiative.

Today Vaccination, surveillance, and outbreak response continue toward global eradication.

Chapter 17

DNA Structure and Molecular Genetics

For much of the early twentieth century, scientists knew that chromosomes carried hereditary traits but did not agree about which molecules stored the instructions. Proteins seemed chemically diverse enough to encode complexity; DNA appeared too simple to do the job. Experiments that tracked how traits passed between cells, followed by the solution of DNA's structure, changed that view and helped create molecular genetics.

17.1 Experiments identify DNA as hereditary material

In 1928, Frederick Griffith found that a substance from heat-killed disease-causing pneumococci could transform living harmless bacteria into a virulent form. The responsible "transforming principle" was unknown. In 1944, Oswald Avery, Colin MacLeod, and Maclyn McCarty purified the active material and showed that it was DNA: destroying DNA removed its ability to transform the bacteria, while destroying proteins did not [119].

Some scientists remained cautious about extending this result from bacteria to all life. In 1952, Alfred Hershey and Martha Chase used radioactive labels to follow the protein and DNA of bacteriophages as they infected bacteria. Most of the viral DNA entered the bacterial cells and directed the production of new viruses, while most protein remained outside. The experiment provided independent evidence that DNA carried genetic information [120].

17.2 The double helix takes shape

To understand how DNA stored and copied information, researchers needed its three-dimensional structure. Erwin Chargaff's measurements showed patterns in the amounts of DNA bases, while X-ray diffraction experiments revealed a regular structure. At King's College London, Rosalind Franklin and her graduate student Raymond Gosling produced important diffraction images, including the image known as Photo 51. Maurice Wilkins and his colleagues also contributed X-ray work.

On 25 April 1953, James Watson and Francis Crick published a model in *Nature*: two strands wound around each other, with complementary base pairs joining them. Franklin and Gosling's and Wilkins and colleagues' papers appeared in the same issue, presenting supporting experimental evidence. Franklin's careful measurements and images were essential to the structure's solution, even though her role was long under-recognized in popular retellings [121, 122].

The paired structure suggested a way the molecule could be copied. If the two strands separated, each could serve as a pattern for a new partner. Complementary pairing also explained how a sequence of bases could preserve and transmit biological instructions. Matthew Meselson and Franklin Stahl's 1958 experiment later showed that DNA replication follows the predicted semi-conservative pattern, with each new molecule containing one old strand and one newly made strand [121, 123].

17.3 From molecular structure to genetics

The double helix connected chemistry to heredity. Mutations could be understood as changes in DNA sequence; gene expression as the use of that sequence to make RNA and proteins; and inheritance as the copying and transmission of DNA. The model did not by itself explain every gene or disease. It gave researchers a framework for asking precise questions about how biological information is stored, copied, and used.

DNA's discovery was not a solitary flash of insight. It depended on microbiology, biochemical purification, radioactive tracing, X-ray diffraction, model building, and the exchange of evidence between laboratories. Molecular genetics emerged when those different methods converged on a molecule whose structure made heredity experimentally tractable.

Timeline

1928 Frederick Griffith reports bacterial transformation.

1944 Avery, MacLeod, and McCarty identify DNA as the transforming principle in pneumococci.

1952 Hershey and Chase provide evidence that DNA carries viral genetic information.

1952 Rosalind Franklin and Raymond Gosling produce Photo 51, an X-ray diffraction image of DNA.

25 April 1953 Watson and Crick publish the double-helix model alongside papers by King's College researchers.

1958 Meselson and Stahl show that DNA replication is semi-conservative.

Chapter 18

Oral Rehydration Therapy

Severe diarrhea can drain the body of water and salts faster than a person can replace them. The resulting dehydration can lead to shock and death. For much of modern medical history, the main treatment for severe cholera was intravenous fluid: effective, but dependent on sterile supplies, trained staff, and the ability to deliver large volumes quickly.

Diarrheal disease is a group of illnesses rather than one diagnosis. Cholera, dysentery, and childhood gastroenteritis can all cause dangerous fluid loss, though they have different causes and may need different treatment. Oral rehydration does not cure the infection; it prevents dehydration from turning an otherwise survivable illness into a fatal one. Safe water, sanitation, vaccination against some causes, and prompt clinical care reduce risk at different points in the chain [51].

Oral rehydration therapy offered another way to replace losses. It uses a carefully balanced solution of water, glucose, and salts, given by mouth or through a tube into the stomach. It does not kill the organism or immediately stop diarrhea. It works by helping the intestine absorb water and electrolytes while the illness runs its course.

18.1 The intestine can still absorb fluid

Cholera can produce extraordinary watery losses, but it does not switch off all absorption in the small intestine. Researchers found that glucose stimulates sodium absorption there; water follows the absorbed salt and glucose. A solution with the right balance can therefore replace fluid even while diarrhea continues. Plain water alone does not replace lost salts, and an overly concentrated sugary drink can worsen fluid loss.

In 1968, John Fordtran, Floyd Rector, and Norman Carter described how glucose and related sugars stimulated sodium absorption in the human jejunum. That same year, David Nalin, Richard Cash, Rafiqul Islam, Majid Molla, and Robert Phillips reported a controlled

study of oral glucose-electrolyte solution in adults with cholera. Patients receiving the oral solution required far less intravenous fluid than those relying on IV replacement alone. The trials turned an intestinal transport mechanism into a practical treatment [46, 47].

18.2 A field test during the 1971 refugee crisis

The next challenge was to show that oral therapy could work outside a specialized research ward. In 1971, war in what was then East Pakistan displaced millions of people into India. Cholera spread through refugee camps, where supplies of IV fluid and trained personnel could not meet the need. In Bangaon, physician Dilip Mahalanabis and colleagues used oral fluid therapy on a large scale, with intravenous treatment available when needed. Their published account described how the approach could be delivered under field conditions and mobilize caregivers as well as clinical staff [48].

The experience helped overcome skepticism about whether patients with severe cholera could drink enough solution or whether families and less-specialized workers could administer it reliably. The answer depended on careful instructions, monitoring, and access to IV treatment for people who needed it. The field program did not make skilled medical care unnecessary; it made effective rehydration possible for many more people with the resources available.

18.3 From research wards to public-health programs

Oral rehydration therapy required more than a scientific discovery. Health workers and families needed packets that could be transported, stored, and mixed accurately, as well as clear teaching about how to use them. Bangladesh launched a national oral rehydration campaign in 1979, with UNICEF support. Community programs helped make the treatment familiar well beyond hospitals [49].

The World Health Organization and UNICEF promoted oral rehydration salts as a central response to diarrheal dehydration. The formulation changed as new evidence accumulated. In the early 2000s, WHO and UNICEF recommended a reduced-osmolarity solution, which WHO described as more effective than the earlier standard formulation [50].

18.4 What the treatment changed

Oral rehydration therapy made a powerful treatment portable. A treatment once associated with IV lines and hospital wards could reach homes, clinics, and emergency settings where

those resources were scarce. It changed the response to diarrhea from trying to stop every stool first to replacing the water and electrolytes the body was losing.

The method has limits. Severe dehydration, shock, or inability to take fluids may require urgent clinical care and intravenous rehydration. ORS addresses dehydration; it does not prevent infection, replace sanitation, or solve the social conditions that allow diarrheal disease to spread [51].

Its history is a story of collaboration between physiology, clinical trials, and community health. Researchers established why the gut could absorb solution during diarrhea; clinicians tested the treatment in patients; and public-health workers found ways to make it available at scale. The result was a simple intervention whose reach depended on a sophisticated chain of evidence and delivery.

Timeline

1968 Human intestinal studies clarify glucose-stimulated sodium absorption; a clinical cholera trial shows oral solution can reduce IV-fluid needs.

1971 Mahalanabis and colleagues use oral rehydration at scale during the cholera emergency in refugee camps in India.

1979 Bangladesh launches a national oral rehydration campaign.

2003–2005 WHO and UNICEF recommend a reduced-osmolarity ORS formulation.

Today ORS remains a key treatment for diarrheal dehydration, with IV fluids used when severe cases require them.

Chapter 19

Cholera

Cholera outbreaks once killed people within hours. The danger came from profuse watery diarrhea: the infection could empty the body's water and salts faster than a patient could replace them. In the nineteenth century, cholera repeatedly crossed borders along trade and travel routes, while physicians argued over whether it spread through air, contact, or contaminated water. Its history shows how several breakthroughs were needed: finding the route of transmission, making water safer, and replacing fluid quickly enough to keep patients alive.

19.1 Mapping an outbreak and identifying the organism

During London's 1854 outbreak, John Snow mapped deaths around the Broad Street water pump and compared households using different water supplies. The pattern pointed to contaminated water, and removing the pump handle helped interrupt exposure. Snow's investigation did not identify the bacterium, but it showed that careful comparison of cases and water sources could guide prevention before a pathogen was isolated [14].

In 1883, Robert Koch and colleagues studied cholera in Egypt and India and identified the comma-shaped bacterium later named *Vibrio cholerae*. The discovery connected the waterborne pattern to a specific organism. It strengthened the case for protecting water supplies, sewage treatment, and hygiene, while helping laboratories confirm outbreaks [278].

19.2 Rehydration turns a fatal emergency into a treatable illness

The most immediate survival breakthrough was fluid replacement. Intravenous saline could save severely dehydrated patients, but required sterile fluids, equipment, and trained staff. Research in the 1960s showed that glucose and salts given by mouth could still be absorbed by the intestine during cholera. Clinical trials and field use demonstrated that oral rehydration could treat many patients and reserve IV fluids for those who needed them. Rapid oral rehydration, with intravenous fluids for severe cases, made effective care possible on a far wider scale [47, 48, 278].

Rehydration treats the dangerous fluid loss; it does not remove the source of infection. Safe water, sanitation, hygiene, surveillance, and access to treatment work together to prevent outbreaks and deaths. Oral cholera vaccines can add protection during prevention and response programs, but do not replace these foundations [278].

Timeline

1854 John Snow uses case mapping and water-source comparisons during London's cholera outbreak.

1883 Robert Koch identifies the cholera bacterium during investigations in Egypt and India.

1968 Clinical research demonstrates that oral glucose-electrolyte solution can reduce IV-fluid needs.

1971 Oral fluid therapy is used at scale during the cholera emergency in refugee camps in India.

Today Rehydration saves lives while safe water and sanitation reduce exposure.

Chapter 20

Computed Tomography (CT)

A conventional X-ray compresses a three-dimensional body into a flat image. Bones, organs, and other structures overlap, sometimes hiding the very abnormality a physician is trying to find. Computed tomography, usually called CT, changed this view. It combines many X-ray measurements taken from different angles with computer calculations to create images of thin cross-sections through the body. The result was a new way to look inside living people without surgery.

20.1 From projections to cross-sections

The idea behind CT developed over several decades. In the 1950s, physicist Allan Cormack worked out how measurements of radiation passing through an object could be used to calculate the distribution of tissues inside it. He published experimental results in 1963 and 1964. Independently, engineer Godfrey Hounsfield at the British company EMI explored how a computer could reconstruct a picture from many X-ray projections. The two lines of work joined mathematics with practical engineering: the machine gathered the measurements, and a computer turned them into a map of internal structure [72].

Working with radiologist James Ambrose, Hounsfield produced the first CT examination of a patient on 1 October 1971 at Atkinson Morley Hospital in London. The patient had a suspected brain tumor. The first image was coarse by modern standards, and the reconstructed images were not ready for review until two days later, but they showed the lesion that surgery later confirmed. In 1972, the new scanner was introduced to the radiology community. Hounsfield and Cormack received the 1979 Nobel Prize in Physiology or Medicine for the development of computer-assisted tomography [72].

Early scanners examined the head, one slice at a time, and a full study could take a long time. Whole-body CT systems followed in the mid-1970s. As scanners gained faster

rotating parts and more detector rows, they could cover larger areas in fewer seconds. Helical scanning, introduced around 1990, collected data continuously as the patient table moved through the scanner. Multidetector systems then captured many image sections during each rotation. These changes expanded CT from a specialized brain study to a rapid tool for imaging the chest, abdomen, blood vessels, and other parts of the body [72].

20.2 How the scan makes an image

During a CT examination, an X-ray source and detectors rotate around the patient. Different tissues absorb X-rays to different degrees. A computer compares the measurements from many angles and reconstructs them as cross-sectional images, which can also be reformatted into other planes or three-dimensional views. A radiologist interprets the images alongside the person's symptoms, examination, and other test results [73].

CT is useful when clinicians need a quick, detailed view of internal injuries, bleeding, tumors, infection, or blood vessels. It can help guide biopsies and other procedures, assess disease extent, and plan treatments such as surgery or radiotherapy. A contrast material may be given to make some organs or vessels easier to distinguish. CT does not answer every medical question: for some soft-tissue problems, ultrasound or magnetic resonance imaging may provide better information, depending on the body part and the clinical need [73].

20.3 Speed, information, and dose

The scan that once required slow, separate measurements can now be completed in seconds for many examinations. Modern systems adjust image acquisition to the person and the diagnostic task, and software helps reduce radiation while retaining useful detail. However, CT uses ionizing radiation, and the dose varies with the body area and scan protocol. This led to a lasting effort to avoid unnecessary examinations and use the lowest dose that can answer the clinical question. The value of the method lies in the way it can provide important information quickly; its use is judged against the need for that information [74, 72].

CT transformed radiology by turning overlapping shadows into slices that could be examined separately. Its story is a partnership between physics, mathematics, engineering, computing, and clinical medicine. From the first slow brain images to today's rapid, detailed scans, the guiding aim has remained the same: make hidden anatomy visible enough to support diagnosis and treatment.

Timeline

1950s Allan Cormack develops the mathematical basis for reconstructing internal tissue from transmission measurements.

1963–1964 Cormack publishes experimental work on computed tomography.

1971 Hounsfield and Ambrose perform the first CT examination in a patient, imaging a suspected brain tumor.

1974 Whole-body CT scanners extend the method beyond the head.

1979 Hounsfield and Cormack receive the Nobel Prize in Physiology or Medicine.

1990s Helical and multidetector CT greatly increase scan speed and body coverage.

Today CT supports rapid diagnosis, procedural guidance, and treatment planning, with protocols designed to manage radiation dose.

Chapter 21

Magnetic Resonance Imaging (MRI)

X-rays made bones and other dense structures visible, and computed tomography turned X-ray measurements into cross-sectional images. A different imaging method grew from physics research on how atomic nuclei respond to magnetic fields. Magnetic resonance imaging (MRI) gave clinicians a way to produce detailed images of organs and soft tissues without using ionizing radiation.

21.1 From nuclear resonance to an image

In 1946, Felix Bloch and Edward Purcell independently demonstrated nuclear magnetic resonance in matter. They showed that nuclei in a magnetic field could absorb and emit radiofrequency energy at characteristic frequencies. The method became useful in physics and chemistry for studying the composition and structure of substances; its potential for medical imaging was not yet clear [124].

In 1973, chemist Paul Lauterbur published a method for locating magnetic resonance signals in space by using magnetic-field gradients. Instead of measuring one signal from an entire sample, the technique could encode where signals came from and reconstruct an image. Physicist Peter Mansfield developed mathematical and technical methods that made image acquisition faster and more practical. Lauterbur and Mansfield shared the 2003 Nobel Prize in Physiology or Medicine for discoveries concerning MRI [125, 126].

21.2 How MRI creates contrast

In an MRI scanner, a strong magnetic field influences hydrogen protons, which are abundant in water and fat. Radiofrequency pulses alter their alignment; as the protons return toward

their prior state, they produce signals that receiver coils detect. Gradient fields help encode location, and a computer reconstructs the signals into images. Different tissues return signals at different rates, creating contrast between structures that may look similar in an X-ray [127].

MRI became especially useful for examining the brain, spinal cord, muscles, joints, and many internal organs. It can image in multiple directions and often distinguishes soft tissues better than conventional radiography or CT. It also complements those methods: MRI is not always the fastest or most suitable study, and image quality can be affected by motion, metal, and the particular clinical question [127].

21.3 A different kind of safety environment

Because MRI does not use ionizing radiation, it avoids the radiation exposure associated with X-rays and CT. It has its own safety demands. The magnetic field can pull ferromagnetic objects toward the scanner, and radiofrequency energy can heat tissue or some devices. Patients are screened for implants and metal, and devices are checked for their MRI conditions before scanning. The scanner is loud, and some people find its confined space difficult. These concerns are managed through careful screening, equipment design, and trained staff [127].

MRI changed medical imaging by turning the physical behavior of atoms into anatomical information. Its development required contributions from physicists, chemists, mathematicians, engineers, radiologists, and clinical teams. The result is a powerful diagnostic method that complements X-ray, CT, and ultrasound rather than replacing them.

MRI became especially valuable when clinicians needed to distinguish soft tissues without ionizing radiation. Its use expanded from research into neurological, musculoskeletal, and cancer care as scanners became faster and image interpretation improved. It is not automatically the best test: cost, availability, scan time, implants, and the need to answer a specific clinical question all shape its usefulness [125, 127].

Timeline

1946 Felix Bloch and Edward Purcell independently demonstrate nuclear magnetic resonance.

1973 Paul Lauterbur publishes an imaging method using magnetic-field gradients.

1970s–1980s Peter Mansfield and other researchers improve image formation and scanning

speed, enabling clinical MRI.

2003 Lauterbur and Mansfield receive the Nobel Prize in Physiology or Medicine for discoveries concerning MRI.

Today MRI provides detailed imaging of soft tissues without ionizing radiation.

Chapter 22

Kidney Transplantation

Kidney failure was once a condition with few ways to sustain life. Dialysis could remove some waste from the blood, but it was demanding and not widely available. Transplant surgery offered the possibility of restoring kidney function with a donor organ. To make that possibility real, physicians had to solve two linked problems: how to connect a kidney surgically, and how to prevent the recipient's immune system from destroying it.

Kidney transplantation became practical through advances in surgery, immunology, drug treatment, tissue testing, and donor organization. The first lasting success came when the donor and recipient were identical twins. The broader treatment followed only after researchers found ways to control rejection in people who were not genetically identical.

22.1 The barrier of rejection

Surgeons had attempted kidney grafts before they could explain why a body accepted some tissues and rejected others. A transplanted kidney was not simply a replacement part: the recipient's immune system could recognize the donor organ as foreign and attack it. In the 1940s, Peter Medawar and colleagues helped establish that graft rejection was an immune response, and that this response could be altered. That work provided a scientific basis for transplant medicine [52].

Kidney surgery itself also had to become repeatable. By the early 1950s, surgeons had developed an operation in which the donated kidney could be placed in the lower abdomen and joined to the recipient's blood vessels and bladder. This accessible location allowed surgeons to connect the organ without removing the patient's own kidneys. Yet technical skill alone could not overcome rejection between unrelated people [53].

22.2 The identical-twin transplant

On 23 December 1954, a team led by surgeon Joseph Murray at Peter Bent Brigham Hospital in Boston transplanted a kidney from Ronald Herrick to his identical twin Richard, who had kidney failure. The twins' genetic similarity greatly reduced the immune barrier. The team confirmed their identity, including through skin-graft testing, before proceeding. The donated kidney functioned and the operation showed that a human kidney transplant could provide durable benefit [54].

This success was a landmark, but it did not yet make transplantation broadly available. Most people who needed a kidney did not have an identical twin. Doctors next had to find a way to suppress rejection without leaving patients too vulnerable to infection or other harms.

22.3 Medicines changed the prospects

Early efforts to suppress rejection used radiation or drugs that weakened the immune system. Results were often poor. A major advance came in the early 1960s, when clinical teams found that azathioprine combined with the corticosteroid prednisone could prevent or reverse rejection in some kidney recipients. Transplants between people who were not identical twins could then survive long enough to establish transplantation as a treatment, even though rejection, infection, and drug toxicity remained serious risks [55].

Tissue typing added another way to assess donor-recipient compatibility. Human leukocyte antigens (HLA) are markers involved in immune recognition. Matching and antibody testing could help clinicians estimate the risk of rejection, while blood-group compatibility and crossmatching added further checks. A match could improve the odds, but it could not guarantee acceptance; transplant recipients still needed medicines to suppress immune attack [56, 57].

The arrival of cyclosporine around 1980 strengthened this approach. In combination with steroids, it improved kidney-graft survival and helped make transplant programs more dependable. Later medicines further refined rejection prevention. These drugs changed transplantation from an exceptional experiment into a treatment available to increasing numbers of people, while requiring lifelong follow-up and careful management of side effects [55].

22.4 Building a donor system

Improved surgery and medicines could not help patients without donated kidneys. Living donation offered a planned route to transplantation, while deceased donation required systems for identifying donors, obtaining consent, preserving organs, and coordinating teams across hospitals. Tissue matching and blood tests helped allocate organs, and growing transplant networks connected donors with people who might otherwise wait years.

The treatment also spread beyond the first transplant centers. In India, the first successful kidney transplant was performed at Christian Medical College in Vellore in 1971, as transplant programs developed in other countries [53].

Modern kidney transplantation is therefore a coordinated process. Teams evaluate both donor and recipient, check blood type and immune compatibility, preserve and transport the organ, perform surgery, and monitor the recipient for rejection and infection. The transplant kidney is usually placed in the lower abdomen and connected to blood vessels and the bladder; the original kidneys generally remain unless there is a medical reason to remove them [57].

22.5 A treatment, not a cure

A successful transplant can restore kidney function and free a person from dialysis, but it does not cure the underlying kidney disease or end medical care. The recipient must take anti-rejection medicines for as long as the transplant functions. These medicines lower immune defenses, so clinical teams balance rejection prevention with infection risk and other side effects. Some transplants fail, and the shortage of donor kidneys means many people remain on dialysis while they wait [57].

The history of kidney transplantation shows how progress can depend on several fields advancing together. Surgery made it possible to connect the organ; immunology explained why it was rejected; medicines reduced that response; and tissue testing and donation systems expanded the pool of possible matches. No single breakthrough made transplantation routine. It became a treatment through the continuing work of laboratories, hospitals, donors, recipients, and public systems.

Timeline

1940s Medawar and colleagues show that graft rejection is an immune response.

1952 A kidney is transplanted from a mother to her son in Paris, but the graft does not provide lasting function.

1954 Joseph Murray's team performs the first successful lasting kidney transplant, between identical twins.

1962–1963
Azathioprine and prednisone help make transplants between non-identical people more practical.

1964 Prospective HLA tissue-matching trials begin for kidney transplantation.

1971 The first successful kidney transplant in India is performed at Christian Medical College, Vellore.

1980 Cyclosporine-steroid therapy improves kidney transplant outcomes.

Today Living and deceased donor transplantation combine surgery, immune management, compatibility testing, and long-term care.

Chapter 23

Heart Transplantation

Replacing a human heart had long seemed beyond the reach of surgery. Unlike a kidney, it could not be removed while the patient remained stable on another organ. Surgeons needed a way to keep blood circulating while the heart was operated on, a technique for joining the donor heart's major vessels, and a way to stop the recipient's immune system from rejecting it. They also needed a reliable system for identifying donors and deciding when death had occurred.

Heart transplantation emerged when cardiac surgery, organ preservation, immunology, intensive care, and donation systems began to work together. The first human-to-human operation in 1967 drew worldwide attention, but the years that followed showed how difficult it was to keep recipients alive. Better immunosuppressive drugs and improved post-operative care later turned an experimental operation into a treatment for selected people with end-stage heart failure.

23.1 The operation was prepared in the laboratory

In the 1950s, the heart-lung machine made it possible to keep blood moving through a patient's body while surgeons worked on the heart. At Stanford, Norman Shumway and Richard Lower used animal experiments to develop a method for removing a heart and implanting a donor heart. Their work refined the surgical connections and showed how cooling the donor heart could slow its metabolism during the operation. These years of preparation laid the technical groundwork for human transplantation [58].

Surgical technique did not solve the problem of rejection. The recipient's immune system recognized a donor heart as foreign and could damage it. Drugs available in the 1960s could reduce rejection, but they also weakened the body's defenses against infection. Early teams therefore faced a difficult balance: suppress the immune response enough to protect the graft

without leaving the patient unable to fight other illness.

23.2 The first human heart transplants

On 3 December 1967, Christiaan Barnard and his team at Groote Schuur Hospital in Cape Town transplanted a heart into Louis Washkansky, who had severe heart disease. The donor, Denise Darvall, had suffered a fatal head injury. The new heart began beating in the recipient, proving that a heart from one person could be implanted in another. Washkansky lived for 18 days before dying from pneumonia and infection while receiving strong immunosuppression. The transplant had worked surgically, but the short survival showed that surgery alone could not make the treatment reliable [59].

The operation was followed by a wave of attempts around the world. In the United States, Adrian Kantrowitz performed an infant transplant in December 1967, and Shumway's team performed the first adult heart transplant in the country at Stanford in January 1968. Many early recipients died from rejection, infection, or other complications. As results remained poor, most centers stopped, while a small number continued research and clinical care [60, 58].

India was among the early countries to attempt the operation. On 17 February 1968, P. K. Sen's team at KEM Hospital in Mumbai performed the first heart transplant in the country; the patient survived only a few hours. The first successful heart transplant in India followed at AIIMS in New Delhi in 1994, after a legal framework for deceased-organ donation had been established [64].

23.3 Death, donation, and public trust

Heart transplantation raised questions that surgery alone could not answer. To transplant a heart, doctors needed a donor whose heart could still be preserved and perfused even though recovery was no longer possible. In 1968, a Harvard Medical School committee proposed criteria for identifying irreversible coma, helping shape discussion of death determined by neurological criteria. The debate involved medical evidence, law, ethics, family consent, and public confidence in donation [63].

The meaning of death and the rules for donation differ across places and have continued to evolve. A transplant system depends on clear safeguards for donors as well as on surgical skill and medical innovation. In India, the 1994 Transplantation of Human Organs Act recognized brain-stem death and created a legal basis for deceased donation, helping make successful heart transplantation possible there [64].

23.4 Cyclosporine renewed the field

By the late 1960s and 1970s, the high rate of rejection and infection had driven heart transplantation close to abandonment. Cyclosporine, an immunosuppressive drug first used clinically in transplantation in the late 1970s, changed the balance. When Stanford teams began using it with steroids around 1980, survival improved and more centers resumed or began transplant programs. The drug did not eliminate rejection or infection, but it made longer graft survival more attainable [60].

Care improved in other ways too. Better donor-heart preservation made it possible to transport organs over longer distances. Biopsy of small samples from the transplanted heart helped clinicians detect rejection, while improved infection prevention and follow-up helped manage the effects of immunosuppression. Progress came from coordinated changes in surgery, medicines, testing, and care after the operation [58].

23.5 A treatment with limits

Today, a heart transplant can extend life and improve quality of life for selected people with advanced heart failure. Donor hearts remain scarce, so patients undergo careful evaluation and wait for a suitable organ. After transplantation, recipients need anti-rejection medicines and regular checks for rejection, infection, and damage to the graft's blood vessels. The medicines that protect the heart can also cause serious side effects [65].

Modern outcomes show both the benefit and the narrow reach of the operation. In the United States, adults transplanted in 2017–2019 had estimated one-year survival of 91.2% and five-year survival of 80.1%. Worldwide, the Global Observatory recorded 10,287 heart transplants in 2024; 59 of 92 countries reported any heart-transplant activity. These figures show that transplantation can provide years of life for many recipients, while remaining unavailable to most people with advanced heart failure because suitable donor hearts and transplant services are limited [61, 62].

Heart transplantation is therefore not a single operation but an ongoing partnership among donor families, transplant teams, recipients, laboratories, and public institutions. The early operations revealed what surgery could achieve and how quickly its promise could be limited by rejection, infection, and uncertain donation practices. The treatment became dependable only when all of those problems were addressed together.

Timeline

1959–1960

Shumway and Lower refine heart-transplant techniques in animal studies.

3 December 1967

Christiaan Barnard's team performs the first human-to-human heart transplant in Cape Town.

1968 Heart transplants are attempted in the United States and India; a Harvard committee publishes criteria for irreversible coma.

Late 1970s–1980

Cyclosporine enters clinical transplant care and improves survival.

1983 The United States approves cyclosporine for clinical use.

1994 India passes the Transplantation of Human Organs Act and records its first successful heart transplant.

Today Transplantation is an established option for selected people with end-stage heart failure, supported by lifelong monitoring and anti-rejection treatment.

Chapter 24

Immunosuppressive Therapy

The immune system protects the body from infection, but it can also attack a transplanted organ or mistakenly target the body's own tissues. Immunosuppressive therapy uses medicines to reduce that harmful activity. Its development helped turn organ transplantation from an experiment into a lasting treatment and gave doctors new ways to control autoimmune disease.

The central challenge has never been to switch immunity off completely. A patient needs enough immune defense to resist infection, while receiving enough suppression to prevent rejection or tissue damage. The history of these medicines is therefore a search for control: first broad and risky, then more effective, combined, and increasingly targeted.

24.1 From inflammation to immune rejection

Corticosteroids offered one of the earliest ways to reduce damaging inflammation. In 1948, cortisone was used to treat a patient with rheumatoid arthritis, producing striking short-term improvement. The result brought attention to the possibility of calming immune-driven disease with medicine, although doctors soon learned that sustained steroid treatment could cause serious harms [66].

At the same time, transplant researchers were learning why grafts failed. Peter Medawar's work showed that rejection was an immune response, not simply a surgical complication. This suggested that suppressing immunity might protect a transplanted organ, but early approaches such as whole-body irradiation were dangerous and difficult to control. The problem was to reduce rejection without making infection and other complications unmanageable [52, 55].

24.2 The first drug combinations

In the late 1950s, researchers found that 6-mercaptopurine, a drug developed in cancer research, could suppress antibody production in experimental animals. Its derivative azathioprine became useful in transplant medicine. In 1962, Joseph Murray's team reported an unrelated kidney graft that functioned for 17 months with azathioprine-based immunosuppression. In 1963, Thomas Starzl's group showed that combining azathioprine with prednisone could help reverse rejection and extend graft survival in people who were not identical twins. Rejection could sometimes be treated rather than accepted as inevitable. These advances gave clinical transplantation a practical, if still imperfect, foundation [55].

These medicines had broad effects. They could prevent rejection, but also weakened defenses against infections and affected other rapidly dividing cells. Physicians adjusted doses in response to rejection and toxicity. The aim was not simply to use more medicine: too little could cost the graft, while too much could endanger the recipient.

24.3 Cyclosporine and a new era

Cyclosporine, a compound isolated from a fungus, provided a more potent way to suppress parts of the immune response involved in graft rejection. Clinical trials began in the late 1970s. When cyclosporine was combined with steroids around 1980, outcomes improved for kidney and heart transplants, and the field expanded after a period when many programs had stopped [55, 60].

Later, tacrolimus became another important calcineurin inhibitor. Mycophenolate and other medicines offered additional ways to limit immune-cell activity. Antibody treatments could be used around the time of transplantation or to address rejection. By combining drugs that act at different points in the immune response, clinicians could often use lower doses of each medicine than would be needed alone [60].

24.4 Treatment beyond transplantation

Immunosuppressive therapy also became an important part of care for diseases in which the immune system attacks the body's own organs. For example, corticosteroids and other immunosuppressants can be used in severe lupus to reduce inflammation and prevent organ damage. The goal is similar to transplant care: both dampen harmful immune activity, but the target is the patient's own disease process rather than a donor organ [68].

Newer biologic medicines can target particular immune cells or signals. This has widened

treatment choices for autoimmune conditions, though targeted does not mean risk-free. The right approach depends on the disease and the organs involved; medical teams balance disease control against infection, toxicity, and other side effects.

24.5 Balancing protection and risk

Modern transplant regimens often combine medicines, such as a calcineurin inhibitor with an antiproliferative drug and, in some cases, a corticosteroid. Some recipients receive stronger induction treatment around surgery and then move to a maintenance regimen. The exact combination varies by organ, rejection risk, side effects, and the recipient's health. Regular tests and, for some drugs, blood-level monitoring help clinicians adjust treatment [67, 57].

The costs remain important. Suppressing immunity can make infections more likely, increase the risk of some cancers over time, and damage organs or affect blood pressure, blood sugar, or bone health. A transplanted organ may still be rejected, and autoimmune disease may still flare. These medicines manage immune activity; they do not erase the underlying biology [57, 68].

The history of immunosuppressive therapy is not a story of finding a single perfect drug. It is the story of learning which immune processes to restrain, how to combine treatments, and how to monitor both benefit and harm. The long-term hope is to preserve protection from infection while preventing rejection or autoimmune injury with less collateral damage.

The same medicines that protect a graft can leave a recipient more vulnerable to infection and some cancers, while long-term toxicity can affect the kidneys, blood pressure, and metabolism. Clinicians therefore balance rejection risk against drug harm and may adjust combinations over time. For autoimmune disease, the goal is similarly to control damaging inflammation while preserving enough immune defense for ordinary life [67, 68].

Timeline

1944 Medawar demonstrates that graft rejection is an immune process.

1948 Cortisone is first used clinically for rheumatoid arthritis.

1959 Experiments show that 6-mercaptopurine can suppress immune responses.

1962–1963
Azathioprine and prednisone help make non-identical kidney transplantation practical.

1976–1979 Cyclosporine's immunosuppressive effect is recognized and clinical trials begin.

1980s Cyclosporine-based regimens improve transplant outcomes; tacrolimus and other agents follow.

Today Combination and targeted therapies are adjusted to balance disease control with infection and toxicity risks.

Chapter 25

Mechanical Ventilation

Breathing moves oxygen into the blood and removes carbon dioxide. When illness or injury makes this exchange fail, mechanical ventilation can support a patient while the underlying problem is treated. It does not cure pneumonia, paralysis, or lung injury by itself; it buys time and helps keep gas exchange within a survivable range.

25.1 The iron lung and polio

The first widely used mechanical respirators were negative-pressure devices. In 1928, Philip Drinker and Louis Shaw described a tank ventilator, soon known as the iron lung. The patient's body lay inside a sealed chamber, with the head outside. Lowering pressure around the chest expanded the lungs and drew air inward; raising pressure helped air flow out. Iron lungs supported people whose respiratory muscles had been weakened by polio [128, 129].

During the 1952 Copenhagen epidemic, anesthesiologist Bjørn Ibsen proposed tracheostomy and positive-pressure ventilation for severely affected patients. Since mechanical ventilators were scarce, teams of medical students manually compressed breathing bags around the clock. The strategy helped reduce deaths and helped establish intensive care as a place where continuous respiratory support and close observation could be organized [130, 131].

25.2 From hand ventilation to modern machines

Positive-pressure ventilation became standard for many patients who need invasive support. A ventilator pushes air through a tube placed in the windpipe, or through a tracheostomy. Mask-based non-invasive ventilation offers support without an airway tube for selected conditions. Modern devices adjust pressure, airflow, oxygen concentration, and breath timing

to the patient's needs. Care teams monitor blood gases, comfort, and lung mechanics, and reduce support as recovery allows.

Ventilation can prevent death from respiratory failure, but it carries risks. An airway tube can contribute to infection or injury, and excessive pressure or volume can damage vulnerable lungs. Sedation, immobility, and critical illness also create complications. Contemporary care uses the least harmful settings that provide adequate support, alongside treatment of the cause and plans for safe removal of the breathing tube [129].

Mechanical ventilation changed what hospitals could do for patients with paralysis, anesthesia complications, and severe lung disease. Its history also connects directly to intensive care: survival depended on equipment, trained teams, and repeated adjustment as a patient's condition changed.

Timeline

1928 Drinker and Shaw describe a practical tank respirator, later called the iron lung.

1952 Ibsen's positive-pressure ventilation strategy is used during the Copenhagen polio epidemic.

1953 A dedicated intensive-care unit is established in Copenhagen after the epidemic.

Today Invasive and mask-based ventilation support patients with acute or chronic respiratory failure.

Chapter 26

Dialysis

Healthy kidneys filter waste from the blood and regulate fluid and salts. Dialysis can remove some wastes and excess fluid when the kidneys fail. It does not replace every kidney function, but it transformed kidney failure from an almost uniformly fatal condition into one that could sometimes be managed for years.

26.1 Building an artificial kidney

In the 1940s, Dutch physician Willem Kolff built a rotating-drum dialyzer using cellophane tubing as a semipermeable membrane. Blood flowed through the tubing while it rotated in a bath of cleansing solution. After earlier unsuccessful attempts, Kolff's machine helped save a patient with acute kidney failure in 1945. This established that dialysis could temporarily replace some filtering work of the kidneys [132, 133].

For years, dialysis was mainly used for acute kidney injury. Repeated long-term treatment was difficult because access to the bloodstream required repeated surgery and blood tended to clot in the apparatus. In 1960, Belding Scribner and colleagues developed an external shunt connecting an artery and a vein through Teflon tubes. The shunt made it possible to reconnect a patient to dialysis repeatedly. The Seattle Artificial Kidney Center opened in 1962 as the first outpatient center for long-term dialysis [134, 135].

26.2 Two main ways to dialyze

In hemodialysis, blood passes through a dialyzer, where wastes and extra fluid move across a membrane into dialysis solution. Treatment is often performed at a center several times each week, though home schedules are also possible. In peritoneal dialysis, the lining of the

abdomen acts as the filtering membrane; solution is placed in the abdominal cavity through a catheter and then drained. The choice depends on health, daily life, available support, and patient preference.

Dialysis is life-sustaining, but treatment can be demanding and may cause low blood pressure, infection, access problems, or fatigue. It requires ongoing clinical care and attention to fluid, medicines, and diet. For some people with kidney failure, transplantation is another option; for others, dialysis remains necessary. The technology made survival possible while also raising difficult questions about access and fair allocation when treatment was scarce [134, 135].

The technology became a long-term treatment only after engineers and clinicians solved repeated access to the bloodstream, reliable membranes, and the practical organization of treatment. Chronic dialysis extended life for people with kidney failure, but it requires recurring sessions and substantial infrastructure; transplantation can offer another option for some patients. These differences make access and quality of life central to the history, not afterthoughts [135, 133].

Timeline

1940s Willem Kolff develops a rotating-drum artificial kidney.

1945 Kolff's dialyzer is used successfully for acute kidney failure.

1960 Scribner and colleagues develop a shunt permitting repeated long-term hemodialysis.

1962 The Seattle Artificial Kidney Center opens for outpatient maintenance dialysis.

Today Hemodialysis and peritoneal dialysis provide ongoing kidney replacement therapy.

Chapter 27

Cancer Chemotherapy

Surgery and radiation can control many tumors in a specific part of the body. Chemotherapy introduced another possibility: medicines that travel through the bloodstream and affect cancer cells across the body. Early drugs were blunt tools, often harming healthy cells as well as tumors, but their successes showed that cancer could sometimes be driven into remission by chemical treatment.

27.1 The first effective drugs

During the Second World War, observations of people exposed to sulfur mustard suggested that related compounds could suppress rapidly dividing cells in bone marrow and lymphatic tissue. Researchers Alfred Gilman and Louis Goodman studied nitrogen mustard as a treatment; in 1942 it was given to a patient with lymphoma. Tumors temporarily regressed. The results, published in 1946, helped establish drug treatment as a possible approach to cancer [136, 69].

At Boston Children's Hospital, Sidney Farber pursued a different lead. In 1947, he tested aminopterin, a folic-acid antagonist developed with Yellapragada Subbarow and colleagues. Because leukemia cells depend on folate-related pathways to make DNA, the drug could interrupt their growth. Some children with acute leukemia entered temporary remission. The effect did not last, but it challenged the belief that childhood leukemia could not respond to treatment and helped launch systematic clinical research in cancer chemotherapy [69, 137].

27.2 Combination treatment and clinical progress

Many early drugs worked only briefly when used alone. Cancer cells could survive treatment, regrow, or develop resistance. Researchers therefore tested combinations that attacked cells through different mechanisms. Studies in the 1960s showed that combinations could produce remissions in acute leukemia and prolong survival. Combination therapy became a foundation of treatment for several cancers, though the right drugs and schedule differ by cancer type [69, 136].

Chemotherapy can be used before surgery to shrink a tumor, after surgery to reduce recurrence risk, or alongside radiation. Not every cancer responds, and treatment may cause infection, nausea, fatigue, infertility, nerve injury, or other serious effects because some healthy tissues also divide rapidly. Doctors weigh these risks against likely benefit and adapt treatment to the cancer and the person's overall health.

Chemotherapy's history is a story of progress without a single universal cure. Remissions, longer survival, and cures in some cancers became possible through laboratory research, clinical trials, supportive care, and combinations of treatments. The field continues to change as researchers learn which tumors are vulnerable to which medicines.

Early successes did not mean one medicine could treat every cancer. Drug combinations and carefully measured trials gradually established which cancers were curable, which could be controlled, and what toxicities patients would face. Chemotherapy transformed outcomes in selected diseases, including some childhood leukemias and lymphomas, while for other cancers its benefit may be limited or used alongside surgery and radiation [137, 69].

Timeline

1942 Nitrogen mustard is administered to a patient with lymphoma.

1947 Farber reports temporary remissions in children with acute leukemia treated with aminopterin.

1949 The United States approves mechlorethamine, a nitrogen mustard, for cancer treatment.

1960s Combination chemotherapy produces remissions in acute leukemia.

Today Chemotherapy is selected and combined with other treatments according to cancer type and patient needs.

Chapter 28

Smoking as a Cause of Lung Cancer

28.1 A familiar habit and an unfamiliar epidemic

For centuries, tobacco was used in ceremonies, social rituals, and medicine. Its dangers were not entirely unknown: some observers recognized poisoning, dependence, and irritation of the airways. Lung cancer, however, was uncommon enough in earlier medical records that many physicians considered it rare. In the twentieth century, cigarette manufacture, advertising, and mass distribution changed the scale of exposure. Cigarettes were cheap, portable, and engineered to deliver nicotine rapidly.

By the 1930s and 1940s, clinicians began to notice that lung cancer was being diagnosed more often. The observation created a puzzle. Better diagnosis and longer life could explain part of the increase, as could air pollution, occupational dust, urbanization, or changes in medical recordkeeping. Smoking was one candidate among several, but the ubiquity of the habit made it difficult to compare smokers with a genuinely unexposed population.

28.2 The first signals from clinical observation

Physicians published case series suggesting that lung cancer was associated with cigarette use. Case series were useful for recognizing a possible association, but they could not establish how large the risk was or whether smoking was responsible. Patients with lung cancer might remember their habits differently from healthy people, and doctors might be more likely to ask about smoking once they suspected a tobacco-related disease.

The question was therefore not simply whether some smokers developed cancer. That was obvious and compatible with many explanations. The stronger question was whether smokers developed lung cancer more often than comparable people who did not smoke, whether risk

increased with amount and duration, and whether the pattern persisted across places and investigators.

28.3 Case-control studies, 1950

In 1950, Richard Doll and Austin Bradford Hill in Britain and Ernst Wynder and Evarts Graham in the United States independently published case-control studies linking cigarette smoking with lung cancer [19]. They began with people who already had lung cancer and compared their smoking histories with those of control participants without the disease. The association was large enough to demand attention, particularly for heavy cigarette smokers.

Case-control research had limitations. The choice of controls could distort the comparison; recall could be imperfect; and an association could reflect a confounding factor rather than smoking itself. Yet the studies agreed across countries and used different investigators and patient groups. Agreement did not prove causation, but it made coincidence increasingly implausible.

The findings were also politically difficult. Tobacco companies questioned the methods, emphasized uncertainty, and promoted alternative explanations. Some researchers worried that the evidence was observational rather than experimental. That criticism correctly identified a limitation but sometimes treated the absence of a randomized trial as if it were evidence of safety. Randomly assigning people to smoke for decades would have been unethical; causal inference had to be built from several complementary designs.

28.4 The doctors' cohort

Doll and Hill followed British doctors and reported in 1954 that mortality was related to smoking habits [20]. Prospective cohort research reduced some problems of case-control studies: smoking was recorded before the outcome occurred, and the investigators could observe later disease and death. The design also allowed a dose-response pattern to emerge. Heavier and longer smoking was associated with greater risk, while doctors who stopped smoking began to show better outcomes than those who continued.

The cohort was not a perfect experiment. Doctors were not representative of all people, and smokers differed from nonsmokers in other ways. But the temporal sequence, dose-response relationship, consistency with case-control studies, and improvement after cessation formed a stronger causal argument than any one study could supply.

28.5 What happened inside the lung

Epidemiology identified the relationship, while pathology and chemistry explored the mechanism. Tobacco smoke contains carcinogenic compounds that can damage DNA. Repeated injury, inflammation, impaired clearance of particles, and accumulation of mutations can allow abnormal cells to survive and multiply. Laboratory studies of smoke condensates and animal tissues supplied biological plausibility, although no animal model reproduced every feature of human smoking.

The mechanism mattered because it connected exposure to disease rather than merely counting outcomes. It also explained why risk was not immediate, why long-term exposure mattered, and why reducing exposure could reduce future risk. Molecular biology later showed that tobacco-related tumors carry patterns of genetic damage consistent with chemical mutagens in smoke.

28.6 The Surgeon General and a public decision

The United States Public Health Service convened expert committees to assess the evidence. The 1964 Surgeon General's report concluded that cigarette smoking caused lung cancer in men and was a cause of chronic bronchitis, and it judged the evidence for other diseases as well [21]. The report was significant not because it discovered the association, which earlier researchers had already described, but because it evaluated the entire body of evidence for government and public use.

The review process made the reasoning visible. It considered case-control and cohort studies, pathology, animal experiments, and trends in disease. It asked whether studies agreed, whether the exposure preceded the outcome, whether greater exposure produced greater risk, and whether alternative explanations were sufficient. This was an early public demonstration of evidence synthesis in chronic disease.

28.7 Why certainty did not immediately change behavior

Scientific evidence and social change move at different speeds. Cigarettes were embedded in advertising, entertainment, military life, workplaces, and social identity. Nicotine dependence made individual choice an incomplete explanation. The tobacco industry continued to contest findings and to market filtered, light, and low-tar cigarettes in ways that could imply safety without removing the central hazard.

Public-health responses therefore developed gradually: warning labels, limits on advertising, smoke-free workplaces, taxation, cessation treatment, restrictions on youth access, and mass-media campaigns. These measures changed exposure at the population level. Their evaluation supplied another layer of evidence: when smoking prevalence fell, later disease rates also began to fall, though with the long delay expected for a disease with a long latency.

28.8 From lung cancer to a wider causal map

The same methods eventually linked smoking to chronic obstructive pulmonary disease, cardiovascular disease, stroke, adverse pregnancy outcomes, and other cancers. Passive smoke exposure extended the question from the smoker to people sharing homes, vehicles, and workplaces. The causal map became more complex, but complexity did not weaken the central finding; it showed that one exposure could damage several organ systems through several pathways.

The history also corrected a common misconception about causation. Smoking does not cause cancer in every person who smokes, and some people who never smoke develop lung cancer. Causal claims concern changes in probability across populations, not an absolute fate for each individual. Clinical intuition often looks for a single signature patient; epidemiology measures how risk shifts when exposure is present or absent.

28.9 The lesson for medical evidence

The smoking story shows why clinical intuition eventually had to be supplemented by systematic evidence. A physician could suspect tobacco after seeing many patients with lung cancer, but only comparative studies could estimate excess risk. A single study could be challenged by bias, but independent case-control and cohort studies converged. Mechanistic research explained how smoke could produce malignancy, and population interventions tested whether reducing exposure changed outcomes.

The conclusion was not based on one dramatic experiment. It was a judgment made strong by convergence: observation, comparison, prospective follow-up, dose response, biological mechanism, and the effects of cessation and policy. That combination became a model for investigating other chronic diseases whose causes cannot ethically be tested through deliberate exposure.

Timeline

1930s–1940s

Physicians report that lung cancer appears to be increasing.

1950 Doll–Hill and Wynder–Graham publish case-control studies linking smoking and lung cancer.

1954 Doll and Hill publish prospective findings from British doctors.

1964 The US Surgeon General's report identifies cigarette smoking as a cause of lung cancer.

1970s–1990s

Warning labels, smoke-free policies, taxation, and cessation programs expand.

Twenty-first century

Molecular and population studies further define tobacco-related cancers and harms from secondhand smoke.

Chapter 29

Radiotherapy

When Wilhelm Conrad Röntgen discovered X-rays in 1895, they offered a way to see structures hidden inside the body. Within a year, physicians were also testing whether these invisible rays could treat tumors. The same energy that revealed a broken bone could injure skin and destroy tissue. Radiotherapy grew from learning how to use that power deliberately: how to reach a tumor, how much radiation to give, and how to protect nearby organs.

29.1 From discovery to cancer treatment

Early X-ray machines were crude, and their output was difficult to measure. Nevertheless, experiments soon suggested that tumors could shrink after exposure. In 1899, Swedish physicians Tor Stenbeck and Tage Sjögren described skin cancers they had treated successfully with X-rays. At about the same time, Marie and Pierre Curie's discovery of radium and polonium opened a second path. Doctors could place a radioactive source in or near a tumor, an approach that later became known as brachytherapy [69, 70].

These first treatments often caused severe burns and other injuries. The problem was not simply that radiation could harm healthy tissue; clinicians also lacked dependable ways to measure and reproduce a dose. As radiation science developed, doctors began to understand that treatment needed careful planning and repeatable equipment. The field moved from brief experiments toward a specialty combining physics, biology, and clinical judgment [70].

29.2 The insight of divided doses

A major advance was to divide the total radiation dose into smaller treatments, or fractions, given over days or weeks. French radiologist Henri Coutard helped establish this approach

in the 1920s through treatment of cancers of the larynx and other head and neck sites. Fractionation gave normal tissues time to recover between sessions while radiation continued to damage tumor cells. It did not remove the risk of injury, but made it possible to deliver a substantial total dose with better control [70].

The machines also improved. More dependable X-ray tubes and techniques using several beams helped doctors treat deeper tumors while limiting the dose to any one area of skin. In the 1950s, cobalt-60 machines and medical linear accelerators produced more penetrating beams. This made tumors deep in the body more accessible and helped establish radiation as one of the principal treatments for cancer, alongside surgery and the developing drug therapies [70].

29.3 Planning treatment around the tumor

Radiation damages DNA. When the damage is too extensive to repair, cancer cells stop dividing or die, although the effect unfolds over days and weeks rather than instantly. Nearby healthy cells can also be injured, and the resulting effects depend on the part of the body treated and the dose [71].

In external beam therapy, a machine directs radiation toward a defined area from outside the body. Modern planning uses medical images to map the tumor and nearby structures. Techniques such as three-dimensional conformal radiotherapy and intensity-modulated radiotherapy shape or vary the beam so that more of the prescribed dose reaches the target. Image-guided treatment checks the patient's position and the target during treatment. For selected small tumors or lesions, stereotactic techniques deliver highly focused doses in a small number of sessions. These tools improve precision, but they cannot make radiation harmless to every normal tissue [71, 70].

Radiation can also be delivered internally. In brachytherapy, radioactive seeds or other sources are placed in or near a tumor, concentrating treatment locally. In systemic radionuclide therapy, a radioactive medicine travels through the body and can be taken up by particular tissues or tumor cells. The choice depends on the cancer's type, size, and location, as well as the person's overall health and other treatments [71].

29.4 A part of combined cancer care

Radiotherapy may be used on its own, before or after surgery, or with chemotherapy and other treatments. It can aim to cure a cancer, reduce the chance that it returns, slow its growth, or relieve symptoms such as pain from a tumor pressing on nearby structures. External beam

treatment is usually directed at a particular site, while systemic radionuclide treatments can reach cancer cells in multiple parts of the body [71].

The history of radiotherapy is a history of precision built gradually from early observation. X-rays and radium showed that radiation could destroy cancer, while fractionation, stronger machines, imaging, and dose planning made treatment more controlled. The central task remains balancing tumor control against injury to healthy tissue, with the plan tailored to the disease and the individual.

Timeline

1895 Röntgen discovers X-rays.

1898 Marie and Pierre Curie discover radium and polonium; radium is soon used in cancer treatment.

1899 Stenbeck and Sjögren report skin cancers cured by X-ray therapy.

1920s Coutard develops and popularizes fractionated X-ray treatment for cancer.

1950s Cobalt-60 units and linear accelerators bring more penetrating radiation into clinical use.

Late 20th century
Image-based planning and computer-controlled beam shaping improve the targeting of treatment.

Today External beam, brachytherapy, and systemic radionuclide treatments are selected and planned for specific cancers.

Chapter 30

Pap Smear and Cervical Screening

Cervical cancer can develop over years, often after infection with a cancer-causing type of human papillomavirus (HPV). Before screening became common, the disease was frequently found only after symptoms appeared. The Pap test changed prevention by detecting abnormal cells before they became invasive cancer, allowing follow-up and treatment at an earlier stage.

30.1 Looking at cells before symptoms

George Papanicolaou studied cells shed from the reproductive tract while investigating the menstrual cycle. In 1928, he reported that a vaginal smear could reveal cancer cells. With gynecologist Herbert Traut, he later showed how collected cells could help identify uterine cancer. The method was initially treated with skepticism, but publication and further clinical use helped establish cervical cytology as a practical screening approach [138, 69].

In a Pap test, a clinician collects cells from the cervix for examination under a microscope. Screening programs made it possible to identify precancerous changes in people who felt well. Organized programs, reliable laboratories, and follow-up of abnormal results were all needed for the test to reduce disease. Cytology screening helped lower cervical cancer incidence and mortality in populations where it was widely and regularly used [139].

30.2 From cytology to HPV testing

Research later established persistent infection with high-risk HPV as the central cause of nearly all cervical cancers. Screening therefore expanded to include tests that detect high-risk HPV itself, alone or alongside cytology. HPV vaccination can prevent many infections that lead to cancer, while screening can identify disease risk that remains. The two approaches

work at different stages and complement each other [139].

Screening is not a diagnosis of cancer. Abnormal results often require repeat testing or examination, and some abnormalities would never progress. Too frequent testing can lead to unnecessary procedures and anxiety. Recommendations depend on age, prior results, health history, and national guidance; programs also need to make screening and follow-up accessible. The achievement is a preventive pathway, not a single laboratory test.

Timeline

1928 Papanicolaou reports that cancer cells can be detected in a vaginal smear.

1941–1943
Papanicolaou and Herbert Traut publish and develop the diagnostic method.

Mid-twentieth century
Cervical cytology screening becomes established in clinical practice.

Late twentieth century
Research identifies persistent high-risk HPV as the cause of nearly all cervical cancers.

Today HPV testing, Pap testing, and vaccination contribute to cervical cancer prevention.

Chapter 31

Clinical Evidence

Medical breakthroughs require more than a plausible idea or a striking recovery. Researchers need reliable ways to compare treatments, and clinicians need ways to judge how study results apply to the people they care for. Over time, this work developed from controlled clinical trials into evidence-based practice. A newer model, the learning health system, aims to make research and routine care inform one another continuously.

31.1 Randomization tests whether treatment caused improvement

In the 1940s, streptomycin offered hope against tuberculosis, but supplies were limited and its benefits were not yet certain. In 1946, the British Medical Research Council began a trial in which patients with pulmonary tuberculosis were assigned by chance to streptomycin plus bed rest or bed rest alone. Random allocation helped balance differences between the groups; planned assessment, including chest X-rays, made the comparison more systematic [140, 141].

The shortage created a practical reason to compare treatment fairly. Published in 1948, the trial found better outcomes in the streptomycin group, while also showing the drug's limits. The study became an influential example of how randomized controlled trials (RCTs) could distinguish a treatment effect from improvement that might have occurred anyway [140].

Randomization does not make every trial reliable. Poor measurement, missing results, small samples, selective reporting, or conflicts of interest can still distort findings. Trials also study defined populations under specific conditions, and their results may not apply to everyone. Ethical review, informed consent, transparent protocols, and careful follow-up are essential.

31.2 Evidence-based medicine brings research into clinical decisions

In the early 1990s, a working group at McMaster University promoted a teaching approach that asked clinicians to find and critically assess research evidence. David Sackett and colleagues described evidence-based medicine as combining the best available research with clinical expertise and the patient's values and circumstances [142, 143].

Different questions call for different evidence. Randomized trials are useful for comparing many treatments; observational studies can help identify rare harms, long-term outcomes, and questions that cannot ethically be randomized. Systematic reviews bring studies together, but cannot remove weaknesses in the underlying research. Guidelines can summarize evidence for practice, while clinicians and patients still weigh likely benefits, harms, costs, and personal priorities [143].

31.3 Learning health systems close the feedback loop

A more recent framework extends this process from individual studies to the organization of care. In a learning health system, information from routine care and external research can generate questions, guide pragmatic studies in everyday settings, and help update clinical practice. Electronic health records and registries can make it easier to follow outcomes across larger and more diverse patient groups. The Institute of Medicine described this model as a health system that learns and improves continuously [319].

Pragmatic and adaptive trials can test care in ordinary clinical settings, and some platform trials can compare several treatments while adjusting as evidence accumulates. The NIH Health Care Systems Research Collaboratory supports pragmatic trials conducted with health-care organizations to produce evidence relevant to care decisions [320]. These approaches may connect research and care more closely, but they do not make routine data automatically reliable. Missing records, biased enrollment, privacy concerns, and unequal access to digital care can limit what a system learns. Sound study design, patient involvement, and independent review remain necessary.

The history runs from a fair comparison of treatments to a broader effort to make care responsive to evidence. Randomized trials, evidence-based practice, and learning health systems address different parts of that task; none removes the need to judge uncertainty or listen to patients.

Timeline

1946 The Medical Research Council begins its randomized trial of streptomycin for pulmonary tuberculosis.

1948 The trial reports improved outcomes and demonstrates a structured method for comparing treatments.

1992–1996 Evidence-based medicine is described as a framework for combining research, clinical expertise, and patient values.

2013 The Institute of Medicine publishes a major report on continuously learning health-care systems.

Today Pragmatic trials and routine-care data are used in efforts to connect research with clinical practice.

Chapter 32

Ultrasound Imaging

Ultrasound imaging uses high-frequency sound waves to create pictures of structures inside the body. Because the images update in real time, the method can show movement, blood flow, and the changing position of organs. It became a versatile form of medical imaging without using ionizing radiation, and it can be brought to a bedside or used to guide a procedure.

32.1 From industrial sound to medical images

Ultrasound technology grew from studies of sound and devices used to detect flaws in materials. In the 1950s, Scottish obstetrician Ian Donald worked with engineer Tom Brown and colleague John MacVicar to adapt pulsed ultrasound for diagnosis. Their 1958 report described its use in assessing abdominal masses, including distinguishing cystic from solid structures. Further work developed contact scanning and demonstrated that ultrasound could show the fetus and other soft tissues [144, 145].

32.2 How the picture is made

A transducer sends sound pulses into the body and detects echoes that return from boundaries between tissues. The time and strength of each echo help estimate its depth and properties, and a computer assembles the measurements into an image. Doppler techniques use changes in echo frequency to assess moving blood. Gel improves sound transmission between the probe and skin. Ultrasound is especially useful for organs, pregnancy, the heart, blood vessels, and image-guided procedures [146].

The method has trade-offs. Sound does not pass well through air or bone, so some

structures are difficult to image. Results can depend on probe position and operator skill, and body habitus may reduce image quality. Ultrasound does not expose the patient to ionizing radiation, but it delivers energy to tissue; prudent use avoids unnecessary exposure and keeps output appropriate to the diagnostic task [146].

Ultrasound made imaging more portable and interactive. It supports quick examinations and procedures, while CT and MRI remain better suited to many other questions. Its value comes from matching the tool to the anatomy and decision at hand.

The technology spread into obstetrics and cardiology because it could provide real-time images without ionizing radiation. It also made bedside assessment more accessible, although images depend on operator skill and may not answer every question. Its clinical value comes from choosing an appropriate scan and interpreting it alongside the person's symptoms and other findings [145, 146].

Timeline

1950s Researchers adapt industrial ultrasound equipment for medical diagnosis.

1958 Donald, MacVicar, and Brown report pulsed ultrasound for investigating abdominal masses.

Later twentieth century
Real-time, Doppler, and portable systems broaden clinical uses.

Today Ultrasound supports diagnosis, monitoring, and image-guided care across many specialties.

Chapter 33

Pacemakers

The heart depends on a timed sequence of electrical signals. When those signals slow down, pause, or fail to travel normally, the heart may not beat often enough to supply the body. Pacemakers use small electrical impulses to support the rhythm. Their development brought together cardiology, surgery, and engineering, turning a bulky machine connected to a wall outlet into a tiny implanted device that can monitor and respond to the heart's own activity.

33.1 From external stimulation to implantation

Early experiments with electrical stimulation of the heart were limited by the size of equipment and the difficulty of delivering a controlled impulse. In the 1930s, American cardiologist Albert Hyman built a hand-cranked device that he called an "artificial pacemaker." It could deliver electrical impulses to the heart through a needle, but it was awkward and did not become an established treatment. In 1952, Boston cardiologist Paul Zoll reported using an external pacemaker to treat patients with prolonged pauses in the heartbeat. The device sent current through electrodes placed on the chest; it could be lifesaving, though the shocks were painful and the equipment was cumbersome [81].

Open-heart surgery in the 1950s created an urgent need for reliable temporary pacing. Some patients developed heart block after surgery, in which signals from the upper chambers no longer reached the lower chambers normally. In 1957, engineer Earl Bakken made a battery-powered wearable pacemaker at the request of surgeon C. Walton Lillehei. It replaced the wall-powered box with a portable device that could stimulate the heart through wires. That year of engineering work laid the groundwork for longer-term pacing [81].

On 8 October 1958, surgeon Åke Senning implanted a pacemaker designed by engineer Rune Elmqvist in Arne Larsson in Stockholm. The first unit failed within hours, and a second was implanted soon afterwards. The devices needed frequent replacement, but the procedure

established that a self-contained pulse generator could be placed inside the body. Larsson lived for decades and received multiple replacement devices as technology improved [81].

33.2 Smaller devices, smarter pacing

Early implantable pacemakers had fixed rates and limited battery life. Developments in transistors, batteries, insulated leads, and sensing allowed devices to last longer and respond more appropriately to the heart's activity. Instead of pacing continuously, a demand pacemaker can detect a natural heartbeat and deliver a pulse when one is needed. Later devices could adjust their pacing rate during activity and coordinate signals between heart chambers [81].

Most traditional pacemakers have a pulse generator beneath the skin of the upper chest and one or more leads that pass through a vein to the heart. Electrodes at the end of the leads sense the heartbeat and deliver small electrical impulses. Leadless pacemakers combine the generator and electrodes in a small unit placed inside a heart chamber through a catheter. Some devices pace both ventricles to improve their timing in selected people with heart failure; this is called cardiac resynchronization therapy [82].

33.3 What pacemakers treat

Pacemakers are most often used when a person has a slow heartbeat or pauses that cause symptoms, or when an electrical block disrupts the normal sequence of heartbeats. They may also coordinate the contractions of the heart's lower chambers in certain types of heart failure. Not every irregular heartbeat requires a pacemaker. A pacemaker is also different from an implantable cardioverter-defibrillator (ICD): a pacemaker helps prevent a heart rate from becoming too slow, while an ICD can deliver a stronger shock to treat some dangerous fast rhythms [83, 82].

Implantation usually involves placing the generator under the skin and positioning the leads inside the heart using imaging. Devices are programmed for the person's needs and checked over time, including remotely in some cases. As with any procedure, complications can occur, including infection, bleeding, a displaced lead, or a problem with the device. Batteries eventually need replacement, though modern devices can often operate for years before this is required [82, 84].

The pacemaker's history shows how a treatment can advance through many small improvements rather than one final invention. Physicians identified the need to support a failing rhythm; engineers made stimulation portable, implantable, and responsive; and long-term

follow-up helped refine the technology. Today, pacemakers do not replace the heart's work. They help its electrical system maintain a rhythm that can support effective circulation.

The change in outcomes came not just from making a device small enough to implant, but from improving lead reliability, batteries, sensing, and follow-up. Modern pacemakers can be programmed to respond to a person's rhythm and activity, yet they treat selected conduction or rhythm problems rather than heart disease generally. Regular checks are part of the therapy [83, 84].

Timeline

1932 Albert Hyman describes a hand-cranked device and coins the term "artificial pacemaker."

1952 Paul Zoll reports successful external electrical pacing for prolonged cardiac standstill.

1957 Earl Bakken develops a battery-powered wearable pacemaker.

1958 Åke Senning implants a device designed by Rune Elmqvist in Arne Larsson, establishing fully implantable pacing.

1960 A self-contained implantable pacemaker provides longer-term correction of complete heart block.

Today Leadless and chamber-coordinating pacemakers expand options for selected rhythm and heart-failure conditions.

Chapter 34

Defibrillation

Some sudden cardiac arrests occur when the heart's lower chambers fall into ventricular fibrillation: electrical activity becomes chaotic and the heart cannot pump blood effectively. Defibrillation delivers a timed electrical shock that can stop this rhythm and allow an organized rhythm to resume. Its development joined electrophysiology, clinical observation, and improvements in electrical equipment.

34.1 From the operating room to the emergency room

In 1947, surgeon Claude Beck used an electrical defibrillator on an exposed human heart during an operation. The patient's heart was in ventricular fibrillation, and a shock restored an organized rhythm. This proved that defibrillation could work in people, but the device could be used only when the chest was open [148].

The next challenge was to deliver a shock through the chest wall. In the 1950s, Paul Zoll and other researchers developed external defibrillation. Early machines were heavy and used alternating current. Bernard Lown and colleagues later advanced direct-current defibrillation, which became a more controlled approach. Portable devices and automated external defibrillators (AEDs) eventually brought defibrillation beyond operating rooms [147, 148].

34.2 A treatment for a specific rhythm

Defibrillation is most effective for shockable rhythms such as ventricular fibrillation and pulseless ventricular tachycardia. It does not correct every cardiac arrest, and it is different from the lower-energy pacing used by a pacemaker. CPR and defibrillation can complement

each other: compressions help circulate blood while a defibrillator assesses and treats a shockable rhythm [147].

An AED analyzes the heart's electrical activity and guides a rescuer through its use. Public access devices, training, and emergency response systems have extended the chance of early treatment. The story shows how a hospital technology became more accessible as engineers made devices portable and rescuers learned to act within a coordinated response.

The central advance was learning to deliver a controlled electrical shock early enough to stop a lethal rhythm before the heart and brain suffered prolonged oxygen loss. Portable automated external defibrillators extended this possibility beyond hospitals, but survival still depends on rapid recognition, CPR, and emergency response; a shock cannot help every cause of cardiac arrest [147, 148].

Timeline

1947 Claude Beck reports successful defibrillation of an exposed human heart.

1950s External shocks through the closed chest are developed and used clinically.

1960s Direct-current defibrillators improve control compared with early AC machines.

1970s onward
Portable defibrillators and AEDs extend use beyond operating rooms.

Today Defibrillation is part of emergency response for selected life-threatening rhythms.

Chapter 35

Cardiopulmonary Resuscitation (CPR)

Cardiopulmonary resuscitation (CPR) combines chest compressions with breathing support in many situations to help maintain circulation during cardiac arrest. It does not correct every cause of arrest, but it can preserve blood flow to the brain and heart until normal circulation is restored or advanced treatment takes effect.

35.1 Two methods come together

For centuries, rescuers tried many methods to revive people who had drowned or stopped breathing. By the mid-twentieth century, mouth-to-mouth breathing had been shown to provide useful ventilation. Separately, engineer William Kouwenhoven and colleagues James Jude and Guy Knickerbocker found that rhythmic pressure on the chest could generate circulation without opening it. In 1960, researchers joined chest compressions with rescue breathing to form modern closed-chest CPR [147, 149].

Professional organizations began teaching the method to clinicians, then expanded training to the public. Manikins allowed people to practice compressions, and later training emphasized early recognition, prompt emergency activation, CPR, and defibrillation as links in a chain of survival. Recommendations have changed as evidence accumulated [147].

35.2 Resuscitation is a system

Cardiac arrest has many causes, and outcomes depend on how quickly it is recognized, whether CPR begins promptly, whether a shockable rhythm is defibrillated, and what care follows. CPR alone may not restore a heartbeat; it is a bridge within a broader emergency response. Training, dispatch systems, AED access, and hospital treatment all influence outcomes.

CPR also has limits. Some people do not survive despite immediate care, and resuscitation can cause injuries. Decisions about attempts at resuscitation involve the patient's wishes and clinical circumstances. The development of CPR made immediate action possible for bystanders and professionals while reinforcing the importance of organized emergency care rather than a single maneuver.

Modern CPR emerged when researchers showed that chest compressions and rescue breathing could maintain circulation until defibrillation or advanced care became available. Later guidance emphasized early compressions and simplified instructions so bystanders could act. CPR can preserve a chance of recovery, but outcomes vary with the cause of arrest, how quickly help begins, and the person's underlying health [149, 147].

Timeline

1950s Research establishes mouth-to-mouth ventilation and closed-chest cardiac massage.

1960 Researchers combine compressions and ventilation in a practical method of CPR.

1960s The American Heart Association begins clinician training in closed-chest resuscitation.

1970s onward CPR courses, emergency medical services, and AED programs expand.

Today CPR is taught as one component of a coordinated response to cardiac arrest.

Chapter 36

Heart-Lung Bypass

Some heart operations require surgeons to work inside chambers that are normally filled with moving blood. Cardiopulmonary bypass, often called heart-lung bypass, temporarily diverts blood through a machine that adds oxygen and pumps it back to the body. This gives surgeons a relatively still, bloodless field while maintaining circulation to vital organs.

36.1 A machine to take over circulation

Surgeon John Gibbon began developing an extracorporeal circulation machine after witnessing a patient die during an operation for a pulmonary embolism in 1931. With engineer and researcher Mary Gibbon and other collaborators, he developed a device to pump and oxygenate blood outside the body. Years of laboratory work and engineering refinement preceded its first successful clinical use [150, 151].

On 6 May 1953, Gibbon used his heart-lung machine during an operation to repair an atrial septal defect in an 18-year-old woman. The operation demonstrated that a machine could temporarily perform the heart's pumping and the lungs' gas exchange while a surgeon repaired the heart [150]. Early machines were complicated and not reliable enough for routine use. Teams at the Mayo Clinic and elsewhere refined pumps, oxygenators, tubing, anticoagulation, and monitoring, making bypass more reproducible and extending the range of operations possible.

36.2 The operating team and the risks

Bypass is not simply a machine switched on during surgery. A perfusionist manages the circuit with surgeons, anesthesiologists, and nurses. The team monitors blood flow, oxygenation,

temperature, and clotting. The machine can contribute to inflammation, bleeding, clotting, or organ injury, so the duration and conditions of bypass matter. Some operations use techniques that avoid the machine, depending on the procedure and patient.

Heart-lung bypass opened the way to modern open-heart surgery and helped make valve repair, congenital heart repair, and coronary surgery more widely possible. Its development depended on engineering, animal research, clinical skill, and specialized teams [151].

The machine made open-heart operations possible by temporarily taking over circulation and oxygen exchange while surgeons worked on a still heart. Gibbon's work depended on years of engineering and team development, and the first successful clinical use opened a path to repairing defects that had previously been inoperable. Bypass itself carries risks, so its use is tied to operations where the expected benefit justifies them [150, 151].

Timeline

1931 Gibbon begins work on a machine to replace heart and lung function during surgery.

1930s–1940s
Gibbon and collaborators develop and test extracorporeal circulation.

1953 Gibbon repairs an atrial septal defect using a heart-lung machine.

1950s onward
Improved pumps and oxygenators support the growth of open-heart surgery.

Today Cardiopulmonary bypass supports many complex cardiac operations.

Chapter 37

Coronary Artery Bypass Surgery

Coronary arteries supply oxygen-rich blood to the heart muscle. When atherosclerotic plaque narrows these vessels, a person may develop chest pain, breathlessness, or a heart attack. Coronary artery bypass grafting (CABG) creates a route for blood to flow around a narrowed segment. It became a major treatment through the combination of angiography, cardiac surgery, and improved bypass support.

37.1 From experimental grafts to planned surgery

Early attempts to improve coronary blood flow used several approaches, including connecting an artery or vein to the heart muscle. In 1967, Argentine surgeon René Favaloro at the Cleveland Clinic developed a systematic method using a section of the patient's saphenous vein from the leg. One end was connected to the aorta and the other to a coronary artery beyond the blockage. With the heart-lung machine, surgeons could perform the operation under controlled conditions. Favaloro and colleagues helped make coronary bypass a reproducible operation rather than an isolated experiment [152].

Surgeons also came to use the internal thoracic (mammary) artery and other vessels as grafts. Different conduits vary in durability and suitability for a particular patient. Advances in imaging, anesthesia, intensive care, and surgical technique improved outcomes and enabled treatment of more complex disease.

37.2 Choosing a treatment

CABG can relieve angina and improve outcomes for some people with severe coronary disease. It is not the only way to restore blood flow. Medicines and catheter-based procedures such

as angioplasty may be appropriate, depending on which arteries are affected, the extent of disease, heart function, other illnesses, and patient preferences. A heart team may compare the likely benefits and risks of surgery with other approaches.

The operation requires recovery and ongoing prevention of further disease. Risks include bleeding, infection, stroke, kidney injury, or complications from anesthesia. A graft can narrow over time, and bypass does not remove the underlying tendency to atherosclerosis. Medicines, rehabilitation, smoking cessation, and management of blood pressure and cholesterol remain important after surgery [152].

CABG changed care by offering a surgical route around obstructed coronary arteries. Its development depended on cardiologists who could locate blockages, surgeons who could construct grafts, and teams able to support the heart and lungs during the operation.

Bypass surgery became one option among several for coronary disease, with the choice shaped by the number and location of blockages, other illnesses, and patient goals. Its history is therefore also a history of comparing surgery with medicines and catheter procedures. The operation can relieve angina and improve outcomes for selected patients, but carries the risks of major surgery and does not remove the underlying atherosclerosis [152].

Timeline

1950s–1960s
: Surgeons explore methods to improve coronary blood flow.

1967 Favaloro establishes a systematic saphenous-vein bypass operation at the Cleveland Clinic.

Late twentieth century
: Arterial grafts, improved perioperative care, and clinical trials refine CABG.

Today CABG is considered alongside medication and catheter procedures for selected coronary disease.

Chapter 38

Coronary Angioplasty

Coronary angioplasty treats a narrowed heart artery from inside the vessel. A catheter carries a small balloon to the blockage; inflation compresses plaque against the artery wall and widens the channel for blood flow. The technique offered selected patients an alternative to open heart surgery, and it became the basis of percutaneous coronary intervention (PCI).

38.1 A balloon catheter reaches the heart

Interventional radiology had shown that catheters could widen narrowed arteries elsewhere in the body. Swiss physician Andreas Grüntzig adapted this idea, designing a balloon catheter for coronary arteries. After laboratory work and early procedures performed during heart surgery, he performed the first successful coronary balloon angioplasty in a conscious patient in Zurich on 16 September 1977. The result demonstrated that a coronary narrowing could be treated without opening the chest [153, 154].

Early angioplasty was technically demanding and suited only selected lesions. The artery could close abruptly or narrow again, and emergency bypass surgery needed to be available. Improved catheters, guidewires, imaging, and operator experience expanded the procedure to more people. Stents later helped hold treated arteries open and reduced some forms of re-narrowing [153].

38.2 A choice among treatments

PCI can restore blood flow in an acute heart attack or relieve symptoms from stable coronary disease. Its role depends on the clinical situation. For some people, medicines and risk-factor treatment are central; for others, PCI or coronary bypass surgery may offer greater benefit.

The choice considers symptoms, the location and extent of artery disease, heart function, other health conditions, and patient preferences.

Angioplasty treats a narrowed segment but does not remove the underlying atherosclerosis. Continued care may include medicines, rehabilitation, and changes that reduce cardiovascular risk. The procedure's history shows how an engineering idea became a widely used treatment through careful testing, improved tools, and follow-up of long-term outcomes.

The first procedures showed that a narrowed coronary artery could be widened without opening the chest, but early angioplasty could be followed by recoil or renarrowing. The procedure's role changed as balloon techniques, stents, imaging, and antiplatelet medicines improved. It became especially important in acute heart attacks, where opening the artery quickly can limit heart-muscle damage [154, 153].

Timeline

1960s–1970s

Catheter-based angioplasty is developed for peripheral arteries.

16 September 1977

Grüntzig performs the first successful coronary balloon angioplasty in a conscious patient.

1980s Catheter and balloon improvements expand coronary angioplasty.

Late 1980s onward

Coronary stents are added to help keep treated arteries open.

Today PCI is used for selected stable coronary disease and as a key treatment during many heart attacks.

Chapter 39

Coronary Stents

The heart depends on coronary arteries to deliver oxygen-rich blood to its own muscle. Atherosclerotic plaque can narrow these vessels and limit blood flow, causing chest discomfort or, if a vessel becomes blocked, a heart attack. Coronary stents are tiny mesh tubes that hold a treated artery open. They grew out of balloon angioplasty, a less invasive way to widen narrowed arteries, and became one of the defining technologies of interventional cardiology [89, 91].

39.1 From balloon angioplasty to a permanent scaffold

In 1977, Andreas Grüntzig performed the first successful coronary balloon angioplasty in a person. A catheter carried a deflated balloon to the narrowing; inflation compressed plaque and widened the channel. The method avoided open-chest surgery, but the artery could recoil or close suddenly, and healing tissue could gradually narrow it again. A small scaffold inside the artery promised to support the vessel after the balloon was withdrawn [89, 90].

In 1986, cardiologists Jacques Puel and Ulrich Sigwart separately implanted early coronary stents in France and Switzerland. These devices were self-expanding metal mesh designs. Other teams developed balloon-expandable stents, including the Palmaz–Schatz design, which could be delivered folded on a balloon and expanded precisely at the narrowed segment. Clinical studies in the early 1990s showed that stents reduced some of the abrupt closures and repeat narrowings seen after balloon angioplasty alone. The United States approved its first coronary stents in 1994 [90, 89].

39.2 Bare metal and drug-eluting designs

A bare-metal stent is a tiny tube made from a metal framework. During percutaneous coronary intervention (PCI), a cardiologist guides a catheter through an artery, commonly in the wrist or groin, to the coronary narrowing. A balloon expands the stent against the artery wall; the balloon and catheter are then removed, leaving the stent in place. The mesh acts as an internal support, helping maintain the opening [91].

The body's healing response created a new challenge. Tissue could grow through the mesh and narrow the vessel again, a process called in-stent restenosis. Drug-eluting stents address this by releasing a medicine from their surface to limit excessive tissue growth. The first drug-eluting coronary stent approved by the US Food and Drug Administration, the sirolimus-eluting Cypher stent, received approval in 2003. Stent designs continued to change, but restenosis and clot formation within a stent remain important complications clinicians work to reduce [89, 92].

39.3 Treatment and continuing care

Stents can restore blood flow in selected people with coronary artery disease, including some people treated during a heart attack. A stent supports a narrowed segment; it does not remove all plaque or cure the underlying tendency to atherosclerosis. Medicines and measures that address cholesterol, blood pressure, smoking, diabetes, and other risks remain part of long-term care. Whether PCI is appropriate depends on a person's symptoms, artery anatomy, and overall health [91, 89].

Because a clot can form on the implanted device, clinicians prescribe antiplatelet medicines after stenting. The combination and duration are chosen to balance clot prevention against bleeding risk, and patients should follow their clinician's instructions before changing these medicines. Other possible complications include bleeding where the catheter entered the artery, renewed narrowing, and stent thrombosis. These risks have driven improvements in stent materials, drug coatings, and the care that accompanies implantation [91, 89].

Coronary stents joined mechanical engineering with the body's own healing biology. They made it possible to hold open arteries through a catheter procedure, while each generation addressed problems revealed by the last. Their story also illustrates that a successful implant is one part of treatment: durable care includes attention to the whole cardiovascular disease that brought the patient to the catheterization laboratory.

Stents addressed a major limitation of balloon angioplasty by helping keep the artery open. Bare-metal devices reduced acute closure but could still narrow again; drug-eluting stents

lowered that risk and required careful antiplatelet treatment to prevent clotting. The history illustrates how an innovation can solve one failure mode while introducing new questions about safety, duration of treatment, and patient selection [90, 92].

Timeline

1977 Andreas Grüntzig performs the first successful coronary balloon angioplasty in a person.

1986 Jacques Puel and Ulrich Sigwart implant early coronary stents.

1994 The first coronary stents receive approval in the United States.

2003 The FDA approves the sirolimus-eluting Cypher coronary stent, an early drug-eluting design.

Today Drug-eluting stents and catheter-based PCI are used to treat selected coronary artery narrowings.

Chapter 40

Statins

Statins are medicines that lower low-density lipoprotein (LDL) cholesterol and reduce the risk of cardiovascular events for many people. They work by blocking HMG-CoA reductase, an enzyme involved in cholesterol production in the liver. Their discovery linked microbiology to a new approach in preventive cardiology: reducing a risk factor before it causes a heart attack or stroke.

40.1 A mold leads to a new drug class

In the 1970s, Japanese biochemist Akira Endo screened fungi for substances that might inhibit cholesterol synthesis. His team isolated compactin (also called mevastatin) from a mold. It inhibited HMG-CoA reductase, the rate-limiting enzyme in cholesterol production. The result opened a new research path, although development of compactin for clinical use was halted after concerns arose in animal studies. Researchers elsewhere developed lovastatin, which became the first statin approved for use in the United States in 1987 [155, 156].

40.2 From cholesterol numbers to clinical outcomes

Early studies established that statins lower LDL cholesterol. Later large trials tested whether that change translated into fewer heart attacks, strokes, and cardiovascular deaths. Evidence from primary- and secondary-prevention studies helped establish statins as a foundation of care for people at elevated risk and for those with known atherosclerotic cardiovascular disease [155].

Statins are not interchangeable in every circumstance, and the expected benefit depends on a person's cardiovascular risk and treatment goals. Some people experience muscle

symptoms or other adverse effects, and clinicians consider interactions and medical history when selecting a drug. Lifestyle measures and other medicines may also be part of care. The drug class gave clinicians a well-studied tool to lower cardiovascular risk over time.

Large clinical trials established that statins reduce cardiovascular events, not merely cholesterol measurements. The size of benefit depends on baseline risk and the degree of LDL reduction, while muscle symptoms and rare serious adverse effects influence treatment choices. Their broader public-health effect depends on identifying people likely to benefit and maintaining affordable access to long-term prevention [155, 156].

Timeline

1976 Endo reports discovery of compactin, the first identified statin.

1979 Researchers isolate lovastatin from a different fungus.

1987 Lovastatin becomes the first statin approved in the United States.

1990s onward Large clinical trials show that statin treatment can reduce cardiovascular events in appropriate patients.

Today Statins are selected according to a person's cardiovascular risk, health history, and treatment goals.

Chapter 41

Antihypertensive Drugs

High blood pressure often causes no immediate symptoms, yet over time it can damage the heart, brain, kidneys, and blood vessels. For much of the early twentieth century, treatment options were limited and severe hypertension was often considered difficult to change. New medicines made it possible to lower blood pressure more reliably and prevent complications across large populations.

41.1 From severe illness to long-term treatment

Before effective drugs, doctors tried restrictive diets, surgery on sympathetic nerves, and agents that were difficult to tolerate. In the 1940s and 1950s, ganglion-blocking drugs and hydralazine offered some patients treatment, but side effects and complicated dosing limited their use. The introduction of chlorothiazide in 1958 marked a major change: oral thiazide diuretics were easier to use and could lower blood pressure with fewer burdens than earlier approaches [157].

Additional classes followed, including beta blockers, calcium-channel blockers, and medicines that inhibit the renin-angiotensin system. These drugs act through different pathways and can be combined when one alone does not adequately control pressure. Large trials later showed that treating hypertension could reduce risks of stroke, heart failure, and other cardiovascular complications [158, 157].

41.2 A chronic condition needs follow-up

Antihypertensive medicines turned treatment into a long-term partnership between patients and clinicians. Blood pressure readings guide care, and the choice of medicine depends on

a person's health, other conditions, side effects, and response. Diet, activity, and other risk factors also matter. Because hypertension can be silent, measurement and continued follow-up are central to recognizing whether treatment is working.

No single medicine suits everyone. Some people need more than one drug, and treatment plans may change over time. The historical advance came through a growing range of effective options and evidence that controlling blood pressure can prevent serious disease.

The decisive shift was from treating very high blood pressure as a symptom to testing whether sustained reduction prevented stroke, heart failure, and premature death. Long-running trials established that benefit, while subsequent studies compared drug classes and combinations. Effective medicines only improve population health when blood pressure is detected, treatment is tolerated, and people can continue care over time [157, 158].

Timeline

1940s–1950s
: Ganglion blockers and other drugs are used for severe hypertension, with substantial limitations.

1958
: Chlorothiazide introduces a more practical oral diuretic treatment.

1960s onward
: Beta blockers, calcium-channel blockers, and renin-angiotensin system drugs broaden therapy.

Later twentieth century
: Large trials confirm that treatment reduces cardiovascular complications.

Today
: Drug choice and follow-up are tailored to blood pressure, health history, and individual response.

Chapter 42

Electrocardiography

The heartbeat is coordinated by electrical signals that trigger the heart muscle to contract. An electrocardiogram (ECG or EKG) records changes in these signals from electrodes placed on the skin. It gives clinicians a quick, non-invasive view of heart rhythm and electrical patterns, and has become one of the most familiar diagnostic tests in medicine.

42.1 Measuring tiny currents

In the late nineteenth century, British physiologist Augustus Waller showed that the heart's electrical activity could be recorded using a capillary electrometer. The instrument was cumbersome and produced difficult traces. Dutch physiologist Willem Einthoven developed a more sensitive string galvanometer, reported in 1903, that could record clearer electrical waveforms from the body. He named their main deflections P, Q, R, S, and T, labels still used in ECG interpretation. Einthoven received the 1924 Nobel Prize in Physiology or Medicine for his work on the electrocardiogram [159, 160].

The early galvanometer was large and required patients to immerse their hands and feet in saline solution connected to the instrument. Later designs became smaller and easier to use. The standard 12-lead ECG records electrical activity from different directions across the heart, helping clinicians assess rate, rhythm, conduction, and signs that may accompany ischemia or prior injury.

42.2 A tracing in clinical care

An ECG is a snapshot. Some heart problems come and go, so a normal tracing does not rule out every condition. Longer monitoring can capture intermittent rhythm changes. ECG

findings also need to be interpreted alongside symptoms, examination, and other tests. Its speed and low cost make it valuable in emergency care, routine evaluation, and monitoring of some treatments.

Electrocardiography translated an invisible process into a visible trace. The instrument evolved through improvements in physics, engineering, and clinical interpretation, and it remains a basic tool for understanding how the heart's electrical system is functioning.

Einthoven's string galvanometer made electrical changes in the heart visible and recordable, turning a brief bedside observation into a tool that could be compared over time. The ECG later became central to rhythm assessment and urgent evaluation for heart attack. It is powerful but not definitive: a normal tracing does not exclude every cardiac problem, and interpretation depends on symptoms and other tests [160].

Timeline

1887 Waller records a human electrocardiogram using a capillary electrometer.

1903 Einthoven develops the string galvanometer and publishes a detailed electrocardiographic record.

1924 Einthoven receives the Nobel Prize for his work on the electrocardiogram.

Twentieth century
Portable machines and multiple leads make ECG a routine clinical test.

Today ECGs support rapid evaluation of rhythm and possible cardiac injury.

Chapter 43

Modern Heart-Attack Treatment

A heart attack occurs when blood flow to part of the heart muscle is reduced or blocked, usually because a clot forms at a narrowed coronary artery. Treatment has changed from prolonged bed rest and observation to rapid diagnosis, continuous monitoring, and urgent efforts to restore blood flow. Modern care combines emergency systems, medicines, and procedures chosen for the type of heart attack and the patient's condition.

43.1 From observation to coronary care

In 1912, James Herrick described the clinical pattern of coronary thrombosis and myocardial infarction. For decades, treatment focused on rest and managing complications. The establishment of coronary care units in the 1960s allowed continuous ECG monitoring and prompt treatment of dangerous rhythms such as ventricular fibrillation. This reduced early deaths from arrhythmias and heart block [161].

In the 1970s, attention shifted toward reperfusion: reopening the blocked artery to limit heart muscle injury. Thrombolytic medicines and later catheter-based PCI made this possible. Treatment also expanded to include antiplatelet medicines, anticoagulants in selected settings, cholesterol- lowering drugs, and rehabilitation. Regional emergency networks helped shorten the time from symptoms to treatment [161, 162].

43.2 A coordinated response

Current care begins with rapid assessment, including an ECG and blood tests for heart muscle injury. In a major heart attack caused by a fully blocked artery, clinicians seek prompt reperfusion through PCI or, when timely PCI is unavailable and the patient is eligible,

thrombolytic therapy. Other heart attacks may be treated with medicines and an early invasive evaluation. The precise approach depends on the ECG, symptoms, bleeding risk, other illnesses, and local resources [162].

Treatment does not end when the artery is reopened. Monitoring, medicines, cardiac rehabilitation, and changes to reduce future risk support recovery. The history of heart-attack care shows how survival improved through faster recognition, better monitoring, and therapies that address both the immediate blockage and the underlying coronary disease.

The change was systemic: coronary-care units, continuous monitoring, rapid ECG diagnosis, emergency transport, and reperfusion treatments all contributed. Each step shortened the time between artery blockage and restored flow. The improved outlook did not come from one procedure alone, and outcomes still depend on rapid access, prevention after discharge, and rehabilitation [161, 162].

Timeline

1912 Herrick describes the clinical features of myocardial infarction caused by coronary obstruction.

1960s Coronary care units improve monitoring and treatment of fatal arrhythmias.

1970s–1980s
Reperfusion with thrombolytic drugs and angioplasty changes acute care.

1990s onward
Primary PCI and regional emergency systems expand rapid artery-opening treatment.

Today Heart-attack care combines rapid diagnosis, reperfusion when indicated, medicines, and rehabilitation.

Chapter 44

Thrombolytic Therapy

Thrombolytic therapy uses medicines that help dissolve blood clots. In acute heart attacks caused by a blocked coronary artery, a thrombolytic drug can reopen the vessel and restore blood flow to threatened heart muscle. Its benefit depends strongly on time: injury grows while the artery remains blocked, so treatment must be considered quickly and balanced against bleeding risk.

44.1 From local infusion to large trials

Researchers began testing drugs such as streptokinase to dissolve coronary clots during heart attacks in the 1950s and 1960s. Early studies were small and gave conflicting results. Some delivered drug directly into a coronary artery through a catheter, while later approaches used intravenous treatment. The large Italian GISSI trial, published in 1986, randomly assigned more than 11,000 patients to streptokinase plus usual care or usual care alone. It showed a reduction in early mortality, especially when treatment began soon after symptoms started [163, 164].

Other large studies confirmed that thrombolysis could save lives when given early to eligible patients. Aspirin and other treatments were studied alongside clot-dissolving drugs. The trial evidence changed practice by establishing reperfusion as a priority in many ST-elevation heart attacks [161].

44.2 Benefit and bleeding risk

Thrombolytic drugs act throughout the circulation and can cause serious bleeding, including bleeding in the brain. Clinicians therefore assess whether a heart attack is likely to benefit

and whether a person's bleeding risk or medical history makes treatment unsafe. When timely catheter-based PCI is available, it is generally preferred for many patients with an acute blocked coronary artery; thrombolysis remains an important option in settings where transfer for PCI would cause a substantial delay [165].

Thrombolysis marked a turning point: medicines could treat a coronary blockage without an operation. The treatment also established a central lesson in acute cardiology: the time between symptom onset and restored blood flow can shape how much heart muscle survives.

Trials in the 1980s and 1990s established that clot-dissolving drugs could reduce early deaths in acute myocardial infarction when given promptly. The benefit was balanced against serious bleeding, especially brain hemorrhage, and treatment could fail to open the artery. This evidence helped define when rapid drug treatment was useful and when transfer for catheter-based intervention was preferable [163, 161, 165].

Timeline

1950s–1970s
: Researchers test intracoronary and intravenous drugs to dissolve coronary clots.

1986
: The GISSI trial demonstrates a survival benefit from early intravenous streptokinase.

1988
: ISIS-2 strengthens evidence for early thrombolysis and aspirin in suspected heart attack.

1990s onward
: Primary PCI increasingly replaces thrombolysis where rapid catheter treatment is available.

Today
: Thrombolysis remains an option when timely PCI cannot be delivered to an eligible patient.

Chapter 45

Primary PCI for Heart Attacks

Primary percutaneous coronary intervention (primary PCI) is an emergency catheter procedure used to reopen a coronary artery during a heart attack. A cardiologist guides a catheter to the blocked vessel, crosses the obstruction with a wire, and restores flow with a balloon and usually a stent. The procedure can directly identify the blockage and treat it in the same session.

45.1 From elective angioplasty to emergency treatment

After Andreas Grüntzig demonstrated coronary balloon angioplasty in 1977, researchers began testing whether the procedure could treat acute heart attacks. Early work in the 1980s showed that opening the infarct-related artery could restore blood flow. Larger randomized comparisons later found that primary angioplasty could produce better outcomes than thrombolysis in many settings, although access to an experienced catheter team was essential [167, 166].

Primary PCI required hospitals to coordinate emergency departments, ambulance services, cardiologists, nurses, and catheter laboratories. Regional systems developed to route patients directly to PCI-capable centers or transfer them quickly. The procedure's value depends in part on how soon it can be delivered. Current guidelines emphasize rapid reperfusion and systems that reduce delays for eligible patients [165].

45.2 A precise tool within broader care

Primary PCI is especially important for many heart attacks in which an artery is acutely occluded. It does not eliminate the need for medicines, monitoring, and rehabilitation after

the procedure. Some patients have anatomy or medical risks that call for a different approach, and in places without timely PCI, thrombolysis may be used before transfer. The care plan depends on the type of heart attack, symptoms, ECG findings, and available services [165].

Primary PCI changed emergency cardiology by turning a catheter procedure into a time-critical treatment. Its success relies on the combination of trained teams, reliable transport, round-the-clock services, and continued care after the artery has been opened.

In the DANAMI-2 randomized trial, the 30-day composite of death, reinfarction, or disabling stroke occurred in 8.013.7reinfarctions; mortality alone was not significantly different. The findings supported organized transfer to PCI centers when treatment could be delivered promptly, while keeping timely fibrinolysis important where transfer would cause excessive delay [166, 165].

Timeline

Late 1970s

Balloon angioplasty demonstrates that coronary narrowing can be treated by catheter.

1980s Clinicians begin using emergency angioplasty to reopen arteries during heart attacks.

1990s Comparative trials establish primary PCI as a preferred reperfusion strategy where timely treatment is available.

2000s onward

Regional networks organize rapid transfer to PCI-capable hospitals.

Today Primary PCI is delivered through coordinated emergency systems for eligible heart-attack patients.

Chapter 46

Anticoagulant Therapy

Blood must clot to stop bleeding, but clots inside blood vessels can block circulation or travel to the lungs, brain, or other organs. Anticoagulants reduce the blood's ability to form or extend clots. They are used in many conditions, including some heart attacks, irregular heart rhythms, and venous clots. Their history is a story of identifying the chemistry of coagulation and finding ways to intervene while managing bleeding risk.

46.1 From heparin to oral medicines

In 1916, medical student Jay McLean identified an anticoagulant substance in canine liver while working in William Howell's laboratory. Howell later named it heparin. Purification made it possible to use heparin in patients and in surgery. In the 1920s and 1930s, a bleeding disorder in cattle that had eaten spoiled sweet clover led researchers to dicoumarol and then warfarin, which was approved for medical use in 1954 [168].

For decades, heparin and vitamin K antagonists such as warfarin were the main anticoagulants. Heparin acts rapidly and is given by injection; warfarin is taken orally but requires monitoring because its effect can vary with diet, medicines, and illness. Later, low-molecular-weight heparins and direct oral anticoagulants targeting thrombin or factor Xa added options for some patients [168].

46.2 Different pathways, different purposes

Anticoagulants interfere with the clotting cascade. Antiplatelet drugs, such as aspirin and P2Y12 inhibitors, reduce platelet activity and are often used in coronary artery disease. Thrombolytic drugs dissolve an existing clot. These treatments affect related parts of clot

formation, but they are not interchangeable. In acute heart attacks, doctors select antiplatelet and anticoagulant medicines according to the treatment strategy and the person's risks [162].

Because anticoagulants can cause serious bleeding, clinicians balance their benefits against bleeding risk and adjust treatment to the indication. For some medicines, laboratory tests help guide dosing; others use fixed doses for defined groups. The development of safer and more targeted medicines expanded treatment while preserving the need for careful selection and follow-up.

Anticoagulation has different purposes in conditions such as venous thrombosis, atrial fibrillation, and acute coronary syndromes, so the history is not a single progression toward one universal drug. Newer oral agents simplified treatment for many patients, while kidney function, interactions, bleeding risk, and access to reversal or monitoring still matter. The central clinical tradeoff is prevention of dangerous clots against the risk of bleeding [168, 162].

Timeline

1916 Jay McLean identifies heparin in laboratory research.

1920s–1930s
Spoiled sweet clover leads to discovery of dicoumarol and the coumarin anticoagulants.

1954 Warfarin is approved for medical use.

Late twentieth century
Low-molecular-weight heparins broaden anticoagulant options.

Twenty-first century
Direct oral anticoagulants become available for selected indications.

Chapter 47

HIV as the Cause of AIDS

47.1 A note on names

HIV is the virus; AIDS is the advanced clinical syndrome that can result from untreated HIV infection. The terms are related but not interchangeable. Modern antiretroviral treatment can suppress HIV, preserve immune function, and greatly reduce the likelihood that a person will develop AIDS. This distinction also matters historically: early clinicians encountered the syndrome before they could identify its cause.

47.2 Before AIDS had a name

The ancestry of HIV-1 is traced to simian immunodeficiency viruses circulating in primates in west-central Africa. Genetic and historical studies indicate that cross-species transmission occurred in the early twentieth century, after which the virus adapted to human populations and spread through social and transport networks. The exact story is reconstructed from molecular evidence; there was no single contemporaneous event recognized as the beginning of the epidemic.

For decades, infection could remain clinically inconspicuous. HIV gradually damages immune cells, especially CD4 T lymphocytes, while an infected person may appear well. This long interval helped the virus move through communities before the syndrome was recognized as a distinct public-health problem.

47.3 The first recognized cases, 1981–1982

On 5 June 1981, the US Centers for Disease Control and Prevention (CDC) reported five previously healthy young men in Los Angeles with Pneumocystis pneumonia, an opportunistic infection rarely seen in people with intact immunity [1]. Around the same period, unusual cases of Kaposi sarcoma were reported among gay men in New York and California. The clustering of rare illnesses suggested an acquired disorder of immunity rather than an isolated clinical curiosity.

Cases soon appeared among people who injected drugs, people with hemophilia, infants, and recipients of blood products. These observations were crucial: they showed that the illness was not limited to one community and pointed toward transmission through sex, blood, and from mother to child. In 1982, the CDC adopted the name *acquired immune deficiency syndrome*, or AIDS [2].

The early language of the epidemic was often stigmatizing. Labels such as “GRID” (gay-related immune deficiency) wrongly implied that the illness belonged to a particular group. Community organizations and clinicians pushed public health authorities toward language based on routes of transmission and clinical evidence rather than identity. That shift was medically important as well as ethically necessary.

47.4 Finding the cause, 1983–1986

In 1983, a team at the Institut Pasteur reported a new retrovirus isolated from a patient with lymphadenopathy. Independent work in the United States identified the related virus in people with AIDS and in people at risk of developing the syndrome. The scientific dispute over priority was eventually followed by agreement that the virus caused AIDS; Françoise Barré-Sinoussi and Luc Montagnier were later awarded the 2008 Nobel Prize for their role in its discovery.

The virus was initially called by several names, including LAV and HTLV-III. In 1986, an international naming decision established *human immunodeficiency virus* (HIV) as the standard term [5]. Naming the virus made it possible to build reliable surveillance around infection rather than waiting for AIDS to appear.

47.5 Testing, blood safety, and prevention

Once HIV was identified, researchers developed antibody tests. In 1985, the US Food and Drug Administration licensed the first commercial blood test, and blood banks began

screening donations [3]. This transformed transfusion safety, although the earlier absence of screening had already caused profound harm, particularly among people with hemophilia.

Public-health practice also changed. Education about condoms, sterile injection equipment, testing, and safer sex gradually replaced speculation and fear as the central tools of prevention. In clinical settings, universal precautions made blood-safety procedures routine for every patient rather than attaching a particular identity to presumed risk.

47.6 The first treatments and the crisis of the 1980s

The first antiretroviral drug, zidovudine (AZT), received approval in 1987, after an unusually rapid development and review process. It offered an important proof of principle but, when used alone, HIV commonly developed resistance. Treatment was also difficult to tolerate, expensive, and not available equally across countries.

The epidemic exposed a mismatch between the speed of illness and the speed of medical institutions. Patients and activists demanded access to experimental drugs, meaningful participation in research, protection from discrimination, and honest public communication. Groups such as ACT UP changed the politics of clinical trials and helped establish patient advocacy as a force in modern medicine.

47.7 Combination therapy and a turning point, 1995–1997

The development of protease inhibitors and the use of several antiretroviral drugs together changed the prognosis of HIV infection. By 1996–1997, highly active antiretroviral therapy (HAART) had become the new standard of care in many well-resourced settings. AIDS diagnoses and deaths fell sharply where effective treatment was accessible [4].

This was not a simple end to the epidemic. Treatment required sustained access, adherence, monitoring, and systems capable of delivering medicines over a lifetime. The benefits arrived unevenly, revealing how strongly the history of HIV was shaped by poverty, racism, gender inequality, criminalization, and the geography of health-care infrastructure.

47.8 The global response

By the 1990s, HIV was recognized as a global pandemic, with the greatest burden in sub-Saharan Africa. International coordination expanded through the creation of UNAIDS in

1996. In the early twenty-first century, large-scale initiatives including the Global Fund and the US President's Emergency Plan for AIDS Relief (PEPFAR) helped expand access to testing and antiretroviral treatment in regions that had previously been priced out of it.

Prevention and treatment became increasingly connected. Antiretroviral therapy was shown to reduce the risk of transmission, and treatment during pregnancy substantially reduced transmission to infants. In 2012, the US FDA approved Truvada for pre-exposure prophylaxis (PrEP), adding a biomedical prevention tool for people who did not have HIV but faced a substantial risk of acquiring it [6].

47.9 HIV history in the present

The history of HIV has moved from mysterious clusters of fatal illness to a chronic, manageable infection for people who can obtain and continue effective treatment. Viral suppression also has a public-health meaning: a person with a sustained undetectable viral load does not sexually transmit HIV, summarized by the phrase "undetectable equals untransmittable," or U=U.

Yet scientific success has not erased the disease's social history. Stigma, unequal access to diagnosis and treatment, criminalization, and discrimination continue to influence who is diagnosed early and who benefits from advances. The central lesson of the HIV era is therefore twofold: biomedical innovation can transform a disease, but the transformation becomes real only when the innovation is made accessible and trusted by the people it is meant to serve.

Timeline

1981 CDC publishes the first report of the illness later called AIDS.

1982 The syndrome is named acquired immunodeficiency syndrome (AIDS).

1983 A retrovirus associated with AIDS is reported by Institut Pasteur researchers.

1985 The first commercial HIV antibody test is licensed; blood screening begins.

1986 The standard name HIV is adopted internationally.

1987 Zidovudine (AZT) becomes the first approved HIV treatment.

1996–1997 Combination antiretroviral therapy produces a major decline in AIDS deaths where available.

2003 PEPFAR is created to support the global HIV response.

2012 The first drug for HIV pre-exposure prophylaxis is approved in the United States.

Chapter 48

Combination Antiretroviral Therapy

The first HIV medicines showed that treatment could slow the virus, but a single drug could not suppress it for long. HIV reproduces rapidly and can acquire mutations that make a medicine less effective. Combining antiretroviral drugs that act at different stages of viral replication made suppression far more durable and transformed the outlook for people with HIV.

48.1 Three drugs change the trajectory

Zidovudine (AZT), approved in 1987, was initially used alone. Viral levels and illness could improve, but resistance often developed and the benefit was temporary. In the mid-1990s, protease inhibitors introduced a new class of medicines. When combined with drugs that blocked reverse transcriptase, they could attack HIV at multiple stages of its life cycle. By 1996–1997, this approach, called highly active antiretroviral therapy (HAART), had become standard care in well-resourced settings [169, 6].

The ACTG 320 trial offered clear clinical evidence for the combination strategy. It compared indinavir plus two older antiretroviral drugs with the two-drug regimen alone in people with advanced HIV. The study was stopped early after the three-drug group showed fewer AIDS-related events and deaths. Combination therapy could suppress the virus and improve immune function more effectively than earlier regimens [170].

48.2 A chronic treatment that keeps evolving

Early HAART regimens could involve many pills, strict schedules, and substantial side effects. Drug development produced simpler combinations with better tolerability, and many current

regimens can be taken as a single daily tablet. Modern antiretroviral therapy generally uses more than one active medicine, selected for effectiveness, resistance history, other health conditions, interactions, and patient preference [171].

Treatment suppresses HIV but does not remove every copy of the virus from the body, so therapy must continue. Sustained viral suppression protects immune health and prevents sexual transmission of HIV. Yet the benefit depends on timely diagnosis, access, continuity, and trust. The arrival of combination therapy changed the natural history of HIV; equitable access determines who can share in that change [171].

The treatment shift was measured in both survival and daily life: people who once faced rapid progression could maintain health for decades with ongoing therapy. That gain depends on dependable access and continued use, and treatment does not eradicate HIV. Combination therapy also reduces transmission when viral suppression is sustained, linking individual care with prevention [170, 171].

Timeline

1987 Zidovudine becomes the first approved antiretroviral treatment for HIV.

1995–1996
Protease inhibitors introduce a new drug class for combination regimens.

1996–1997
HAART becomes standard care in many well-resourced settings.

1997 ACTG 320 reports improved clinical outcomes with a three-drug regimen.

Today Combination regimens suppress HIV, preserve health, and prevent sexual transmission when viral suppression is maintained.

Chapter 49

HIV Pre-Exposure Prophylaxis

Antiretroviral medicines first transformed care for people living with HIV. Researchers then tested whether these medicines could also prevent infection in people who did not have HIV but faced a substantial chance of exposure. This approach, called pre-exposure prophylaxis (PrEP), extended HIV medicine from treatment into prevention.

49.1 From treatment drugs to prevention

Clinical trials showed that antiretroviral medicines taken before exposure could reduce the chance that HIV would establish infection. In 2012, the US Food and Drug Administration approved the combination of tenofovir and emtricitabine as the first medication specifically indicated for HIV PrEP. Its effectiveness depended on taking the medication as studied, and PrEP was framed as one part of a broader prevention strategy that included regular testing and other prevention tools [272, 273].

49.2 More than a daily tablet

Daily oral PrEP made prevention possible without requiring a partner to take treatment, but it also made adherence and access central to its impact. Long-acting cabotegravir injections became another option after FDA approval in 2021. In 2025, the FDA approved lenacapavir for PrEP, adding an injectable option administered twice a year in the United States. These formulations show how delivery technology can respond to varied needs and reduce the burden of frequent dosing [274, 275].

PrEP does not treat established HIV infection, so testing before and during use is essential to distinguish prevention from treatment and to reduce the risk of selecting drug-resistant

virus if infection is missed. PrEP also does not prevent other sexually transmitted infections. Its population impact depends on voluntary access, informed choice, regular clinical follow-up, and addressing barriers such as stigma, cost, and unequal access to health services [274].

PrEP marked a major change in HIV prevention: people at risk gained a biomedical tool they could use before exposure. Combined with condoms, testing, treatment that suppresses viral load, and other prevention approaches, it helped shift the response toward a broader prevention toolkit.

PrEP extended prevention beyond condoms and behavioral counseling by allowing HIV-negative people to use antiretroviral medicines before exposure. Clinical trials and subsequent implementation established strong protection when medication is taken as directed; long-acting options may reduce daily-pill burden. Testing, follow-up, affordability, and equitable availability remain essential to its impact [274, 275].

Timeline

2010 A major clinical trial reports evidence that oral antiretroviral PrEP can prevent HIV acquisition.

2012 The FDA approves tenofovir/emtricitabine for HIV PrEP.

2021 Long-acting injectable cabotegravir is approved for PrEP in the United States.

2025 The FDA approves twice-yearly injectable lenacapavir for PrEP.

Chapter 50

Hepatitis B Vaccine

Hepatitis B can cause acute hepatitis and lifelong infection that may lead to cirrhosis and liver cancer. Its vaccine became a milestone because it was the first vaccine designed to prevent a major human cancer. The path to it joined epidemiology, protein chemistry, biotechnology, and public-health programs aimed at protecting infants and people exposed to blood or body fluids.

50.1 Finding a marker in the blood

In the 1960s, Baruch Blumberg and colleagues studying inherited differences in blood proteins identified an unusual antigen in the blood of an Australian Aboriginal person. The antigen, first called Australia antigen, was later recognized as hepatitis B surface antigen (HBsAg). Its discovery helped researchers distinguish hepatitis B from other forms of viral hepatitis and develop tests to identify infected donors [172].

50.2 A vaccine made from surface antigen

In the 1970s, Maurice Hilleman and colleagues developed a vaccine using HBsAg purified from the plasma of people with chronic hepatitis B. The antigen stimulated protective antibodies without containing infectious virus. The United States licensed this plasma-derived vaccine in 1981. It was effective, but concerns about using a human blood product limited acceptance [172].

Recombinant DNA technology offered another way to make the antigen. Researchers inserted the gene for HBsAg into yeast cells, which produced the protein for purification and formulation. The first recombinant hepatitis B vaccine was licensed in the United States

in 1986. This approach removed dependence on human plasma and showed how genetic engineering could supply vaccine antigen at scale [172, 173].

50.3 Prevention across a lifetime

Vaccination programs expanded from groups considered at higher risk to routine infant vaccination. Protecting newborns is especially important because infection acquired around birth is much more likely to become chronic than infection acquired in adulthood. Screening during pregnancy and timely prevention for newborns complement vaccination. In places where hepatitis B is common, broad infant immunization has reduced chronic infection and its later complications.

The hepatitis B vaccine linked cancer prevention to infection control. It demonstrated that a vaccine could prevent a cancer by preventing the virus that causes it, an idea later extended by HPV vaccines. Its history also showed the value of moving from a biological marker to a reliable test, a manufactured vaccine, and a delivery program that reaches children before exposure.

The vaccine's public-health value is greatest when protection begins at birth and continues through the recommended series. Preventing chronic hepatitis B also prevents downstream cirrhosis and liver cancer, though coverage and timely birth doses vary widely. Vaccination prevents new infection; people with established chronic hepatitis B need separate evaluation and care [173].

Timeline

1965 The Australia antigen, later identified as hepatitis B surface antigen, is described.

1970s Plasma-derived hepatitis B vaccine is developed.

1981 The first hepatitis B vaccine is licensed in the United States.

1986 The first recombinant hepatitis B vaccine is licensed.

Chapter 51

HPV Vaccine

Human papillomavirus (HPV) is a group of viruses, some types of which cause cervical and other cancers. Establishing the role of persistent high-risk HPV infection made prevention through vaccination possible. The HPV vaccine brought together cancer biology, structural virology, and decades of research on virus-like particles.

51.1 From an association to a cause

Epidemiologic and laboratory research in the late twentieth century linked specific HPV types to cervical cancer. Harald zur Hausen's work helped establish HPV as a causal agent, while studies of viral proteins E6 and E7 explained how high-risk strains can disrupt normal cell growth. This shifted cervical cancer prevention toward addressing infection before precancerous changes developed [174].

51.2 Virus-like particles

In the early 1990s, Douglas Lowy, John Schiller, and other researchers showed that the HPV L1 capsid protein could self-assemble into virus-like particles. These particles resembled the virus shell and prompted strong antibody responses, but contained no viral genetic material and could not cause infection. The finding supplied the central technology for prophylactic HPV vaccines [174].

The first HPV vaccine, quadrivalent Gardasil, was approved in the United States in 2006. It targeted HPV types 16 and 18, which cause most cervical cancers, and types 6 and 11, which cause most genital warts. A bivalent vaccine followed in 2009, and a nine-valent vaccine later broadened protection against cancer-associated types. Initial programs focused on girls,

then expanded to boys as evidence and policy developed [175, 174].

51.3 A new kind of cancer prevention

HPV vaccines prevent new infections; they do not treat an infection or cancer already present. Their introduction did not make cervical screening unnecessary, because vaccines do not cover every cancer-causing HPV type and not everyone is vaccinated. In combination with screening, vaccination represents a major shift: a cancer can be prevented by immunizing against its infectious cause.

The scientific achievement required identifying the cause, building a safe antigen, testing protection, and organizing vaccination at population scale. Its public-health impact depends on access and delivery before exposure.

The key public-health test is whether vaccination reduces precancer and cancer, not simply whether it prevents detectable infection. Follow-up studies have shown substantial reductions in vaccine-type HPV disease among vaccinated populations, while the largest benefit comes from immunizing before exposure. Coverage, gender-inclusive programs, and continued screening remain important [174, 175].

Timeline

1970s–1980s
: Research establishes the association between high-risk HPV and cervical cancer.

Early 1990s
: HPV L1 proteins are shown to form noninfectious virus-like particles.

2006 The first HPV vaccine is approved in the United States.

2009 A bivalent vaccine is approved; quadrivalent vaccine use expands to males.

2014 A nine-valent HPV vaccine is approved in the United States.

Chapter 52

Hepatitis C Direct-Acting Antivirals

For years, hepatitis C was known as non-A, non-B hepatitis: a blood-borne liver disease that could persist for decades, cause cirrhosis and liver cancer, and evade identification. The discovery of the virus and the development of medicines that directly block its replication turned a chronic infection into a curable disease for most people treated.

52.1 Finding the virus

In 1989, researchers reported the identification of hepatitis C virus (HCV) using molecular cloning methods. The discovery enabled more specific blood tests and screening of donated blood, reducing transfusion-associated transmission. It also gave scientists a target for treatments designed to interfere with viral replication [270].

Early treatments used interferon, later combined with ribavirin. They could clear infection in some patients, but treatment lasted many months, caused difficult side effects, and did not work reliably for everyone. The first direct-acting antiviral drugs, approved in 2011, targeted viral proteins but still had to be combined with interferon and ribavirin. Their success showed that specific viral enzymes could be blocked with medicines [270].

52.2 Short-course, all-oral treatment

The approval of sofosbuvir in 2013 marked another turning point. Combinations of direct-acting antivirals could target different parts of the virus's replication machinery. As all-oral regimens developed, treatment became shorter and far easier to tolerate than interferon-based therapy. Current direct-acting antiviral courses cure more than 95 percent of treated cases, often in 8 to 12 weeks [270, 271].

An effective cure does not create immunity: a person can be reinfected after successful treatment. There is also no licensed HCV vaccine. Finding people with infection, linking them to care, and making treatment accessible remain necessary to reduce liver disease and transmission. The HCV story illustrates how molecular virology can lead from identifying a pathogen to drugs that target its life cycle with high precision.

Direct-acting antivirals changed hepatitis C from a usually chronic infection managed with difficult, prolonged regimens into an infection that can often be cured with a short oral course. Cure prevents ongoing liver injury, but reinfection remains possible and established liver damage may still need monitoring. Diagnosis is now a major bottleneck: people must be tested and linked to treatment to benefit [270, 271].

Timeline

1989 Researchers identify HCV, the cause of most non-A, non-B hepatitis.

1990s Diagnostic testing and screening of donated blood expand.

2011 The first direct-acting antivirals for HCV receive approval.

2013 Sofosbuvir is approved, beginning a rapid shift toward more effective combination regimens.

2010s onward
All-oral combinations make short-course treatment and cure possible for most patients.

Chapter 53

Typhoid Fever

Typhoid fever spread through food and water contaminated with *Salmonella* Typhi. Before reliable water treatment and antibiotics, the infection could cause prolonged fever, intestinal bleeding or perforation, and death. It was among the diseases that made urban water systems and food safety matters of life and death. Improvements in sanitation lowered exposure; vaccines and antibiotics gave clinicians additional ways to prevent and treat infection [282].

53.1 Water systems interrupt transmission

Typhoid spreads only among humans, including through people who continue to carry and shed the bacteria after recovering. Filtration, chlorination, sewage disposal, handwashing, and safer food handling reduced the chance that human waste would contaminate drinking water. These infrastructure changes helped make typhoid uncommon in places that built and maintained reliable systems. They also prevent other intestinal infections, so their benefits extend far beyond one disease [282].

53.2 Vaccination and antimicrobial treatment

Almroth Wright developed an early heat-killed typhoid vaccine in the 1890s, which was used among British troops. Later vaccines improved protection and ease of delivery. Modern typhoid conjugate vaccines can be given to young children as a single dose and are recommended for use in endemic settings. Vaccination is particularly valuable where clean water and sanitation are not yet reliably available [282, 293].

Antibiotics made typhoid treatable and sharply improved the outlook for patients. As resistant strains spread, however, familiar drugs became less reliable. Diagnosis, susceptibility

testing, appropriate antibiotics, and completion of treatment remain essential; vaccination also reduces the number of infections that need antibiotic treatment. The survival story is therefore shared between infrastructure, immunization, and clinical care [282].

Timeline

1896 Wright develops an early typhoid vaccine.

Late nineteenth and twentieth centuries
Water filtration, chlorination, and sewage systems reduce transmission in many cities.

Mid-twentieth century
Antibiotics make typhoid fever treatable.

2017 onward
WHO prequalifies typhoid conjugate vaccines for wider use in high-burden countries.

Chapter 54

Plague

Plague continued to cause deadly outbreaks in parts of the world during the twentieth century. Caused by the bacterium *Yersinia pestis*, it can spread through infected fleas and animals; pneumonic plague can also pass between people through respiratory droplets. Before effective antibiotics, case-fatality was high, especially for pneumonic and bloodstream infection. Recognizing the organism and its routes of spread supported both prevention and the rapid treatment that transformed the disease's outlook [283, 284].

54.1 Laboratory diagnosis and outbreak control

The identification of the plague bacillus in 1894 by Alexandre Yersin and Shibasaburo Kitasato connected outbreaks to a specific pathogen. Subsequent work clarified the role of rodents and fleas in transmission. Public-health responses combined surveillance, isolation of pneumonic cases, safer handling of infected animals, and control of rodent fleas. These measures helped prevent local outbreaks from spreading, though plague persisted in animal reservoirs and could not be eliminated through human vaccination campaigns alone [283].

54.2 Antibiotics change survival

Sulfonamide drugs in the 1930s and later antibiotics transformed plague treatment. US surveillance records show a marked fall in case fatality after effective antibiotics became available. In pneumonic plague, early treatment is especially important because illness can worsen rapidly. Modern treatment uses antibiotics alongside supportive care and urgent public-health measures for close contacts [283, 284].

The history is a warning against waiting for certainty during a fast-moving infection.

Clinicians must recognize the possibility of plague, obtain appropriate tests, begin effective treatment promptly, and notify public health teams. Antibiotics can save individual patients; surveillance and infection control protect others.

Timeline

1894 Yersin and Kitasato independently report the plague bacillus.

1890s onward
Research identifies the rodent-flea cycle in plague transmission.

1930s Sulfonamide treatment begins to improve survival.

1942 onward
Effective antibiotics become a major part of plague care in the United States.

Chapter 55

Yellow Fever

Yellow fever repeatedly devastated tropical cities and posed a serious threat to travelers, soldiers, and local communities. Severe infection can cause jaundice, bleeding, shock, and death. Physicians long debated how it spread. Two breakthroughs changed prevention: proving that mosquitoes transmit the virus and developing a vaccine that could protect people before exposure. There is still no specific antiviral cure, so early supportive care remains important for people who become severely ill [285, 286].

55.1 Finding the vector

Cuban physician Carlos Finlay proposed in 1881 that mosquitoes transmitted yellow fever. In 1900, the US Army Yellow Fever Commission led by Walter Reed tested the idea and demonstrated that infected mosquitoes could transmit the disease. The evidence helped public-health teams target *Aedes aegypti* through mosquito control, removal of breeding sites, and protection from bites. These measures helped control outbreaks, especially in urban settings [285, 286].

55.2 The 17D vaccine

Max Theiler and colleagues developed the live-attenuated 17D vaccine in the 1930s. It provided durable protection and could be produced at a scale useful for public-health programs. Theiler received the 1951 Nobel Prize for this work. Vaccination became the central defense in areas at risk, alongside mosquito surveillance and control. It also made it possible to protect workers and travelers who might otherwise carry the disease into new areas [285, 286].

Yellow fever's history shows that prevention can be more powerful than treatment when

a dangerous infection has no specific cure. Vector control reduced exposure, and vaccination offered strong protection to individuals and populations. Outbreaks still occur where vaccination coverage is low or the virus circulates in forested areas, so both approaches remain necessary.

Timeline

1881 Carlos Finlay proposes mosquito transmission of yellow fever.

1900 Walter Reed's commission demonstrates mosquito transmission.

1937 Theiler and colleagues develop the 17D vaccine.

1938 17D vaccination begins at large scale.

Chapter 56

Syphilis

For centuries, syphilis could cause chronic pain, disability, pregnancy loss, and damage to the brain, heart, and other organs. Before effective treatment, patients faced remedies that were toxic and uncertain. The discovery of the syphilis bacterium, development of blood tests, and introduction of effective antibiotics transformed the disease from a feared lifelong infection into one that can usually be cured [287, 288].

56.1 A cause, a test, and the first targeted drug

In 1905, Fritz Schaudinn and Erich Hoffmann identified the spiral-shaped bacterium *Treponema pallidum* as the cause of syphilis. The Wassermann test, introduced soon afterward, made it possible to detect infection through a blood test and expanded screening. In 1910, Paul Ehrlich and Sahachiro Hata introduced arsphenamine, also known as Salvarsan. It was an early chemotherapy designed to attack a specific pathogen, but treatment was complicated and the drug could be toxic [287].

56.2 Penicillin brings a reliable cure

Penicillin proved effective against syphilis in the 1940s and became the preferred treatment. It shortened treatment from prolonged courses of toxic compounds to a much more effective antibiotic regimen. Screening during pregnancy and treatment of infected patients also help prevent congenital syphilis and interrupt transmission [287, 288].

The benefit of an effective medicine depends on people receiving it. The Tuskegee study, in which Black men with syphilis were deceived and denied available penicillin, is a grave example of how racism and research abuse blocked a proven treatment. A history of medical

breakthroughs must include who was denied care, and why trust and informed consent are essential to public health [288].

Timeline

1905 Researchers identify *Treponema pallidum* as the cause of syphilis.

1906 The Wassermann blood test helps make diagnosis and screening possible.

1910 Salvarsan becomes an early targeted treatment.

1940s Penicillin becomes an effective, widely adopted cure.

Chapter 57

Rabies

Rabies is almost always fatal once symptoms begin. For most of history, a bite from a rabid animal left families and physicians with little to offer beyond wound care and desperate observation. Louis Pasteur's development of a post-exposure vaccine in the nineteenth century changed that prospect: a person bitten by a rabid animal could be treated before the virus reached the brain. Modern wound washing, vaccines, and rabies immunoglobulin make prompt care even more effective [289, 290].

57.1 Pasteur's post-exposure experiment

In 1885, Pasteur and his collaborators treated Joseph Meister, a nine-year-old boy severely bitten by a rabid dog, with a series of injections made from attenuated rabies virus. Meister survived. The case helped establish that vaccination could prevent a disease even after exposure had occurred, a major conceptual advance in immunology and emergency medicine [289].

The early vaccine was difficult to produce and could cause serious reactions. Later cell-culture vaccines became safer and more effective. Modern post-exposure prophylaxis includes immediate thorough wound washing, rabies vaccine, and rabies immunoglobulin for people who have not previously been vaccinated. These steps must begin quickly, before symptoms develop [290].

57.2 Protecting communities and animals

Human treatment after bites saves lives, but vaccinating dogs and other reservoir animals prevents exposures in the first place. Surveillance, community education, and access to

post-exposure care are especially important where rabies remains common. The central breakthrough is extraordinary: a disease that is nearly always fatal after symptoms start can usually be prevented if exposed people receive care in time.

Timeline

1885 Pasteur's team administers post-exposure rabies vaccination to Joseph Meister.

Twentieth century
Safer vaccines and immunoglobulin improve post-exposure treatment.

Today Wound washing, prompt vaccination, immunoglobulin when indicated, and animal vaccination prevent deaths.

Chapter 58

Leprosy

Leprosy, also called Hansen disease, has caused chronic nerve injury, blindness, disability, and social exclusion. It was often treated through lifelong isolation because effective medicines were unavailable and fear shaped policy. The disease is curable. Early diagnosis and multidrug therapy can stop infection, prevent many disabilities, and make it possible to integrate care into ordinary health services [291, 292].

58.1 From isolation to treatment

Gerhard Armauer Hansen identified the leprosy bacillus in 1873, providing evidence that the disease had a bacterial cause. For decades, patients were confined to colonies, while the infection's slow course and limited diagnostics complicated care. Dapsone became an important treatment in the mid-twentieth century, but using one drug for years led to resistance in some strains [292, 294].

58.2 Combination therapy prevents resistance and disability

In 1981, WHO recommended multidrug therapy combining dapsone, rifampicin, and clofazimine. Using several drugs reduces the chance that resistant bacteria will survive, and the regimen cures both major forms of leprosy. WHO and partners later made treatment available free of charge. Early detection and prompt therapy can prevent progressive nerve damage; contact screening and preventive treatment can reduce new cases [291, 292].

Leprosy is not usually a mass-killing disease, but it belongs in a history of medical breakthroughs because untreated infection can permanently alter a person's mobility, sensation,

vision, work, and place in society. Effective medicines changed what could be done for patients, while respectful care challenged the assumption that isolation was the only response.

Timeline

1873 Hansen identifies the bacterium associated with leprosy.

1940s Dapsone becomes an important treatment.

1981 WHO recommends multidrug therapy to cure leprosy and limit drug resistance.

2000 onward
WHO and partners make multidrug therapy available free of charge worldwide.

Chapter 59

Tuberculosis Chemotherapy

For centuries, tuberculosis (TB) was managed with rest, fresh air, nutrition, and isolation in sanatoriums. These measures could support patients and limit spread, but they could not reliably eliminate infection. Antibiotics changed the outlook, yet TB's slow-growing bacteria and ability to develop resistance meant that treatment required combinations of drugs taken for months.

59.1 The first trials reveal resistance

Streptomycin became available in the 1940s, and the British Medical Research Council conducted a landmark randomized trial in pulmonary TB. Patients given streptomycin initially improved, but resistance commonly emerged when the drug was used alone. A subsequent trial found that adding para-aminosalicylic acid reduced resistance. The studies established both the potential of chemotherapy and the need to use more than one active drug [176].

Isoniazid, introduced in the early 1950s, made combination treatment more effective. Early regimens still required prolonged treatment, sometimes lasting 18 months or longer. Rifampicin, introduced in the 1960s, and pyrazinamide helped researchers shorten therapy by attacking bacterial populations that survived initial treatment. Trials in East Africa, Hong Kong, Singapore, and elsewhere tested combinations and durations under real-world conditions [176, 177].

59.2 Short-course therapy and public-health delivery

By the 1970s and 1980s, regimens combining isoniazid, rifampicin, pyrazinamide, and ethambutol made treatment over roughly six months possible for many cases of drug-susceptible TB. Combination therapy reduces the chance that naturally resistant bacteria will survive and multiply. Completing a long course became a public-health challenge as well as a pharmacological one: programs developed systems for reliable drug supply, follow-up, and support for people facing treatment barriers [177].

Drug resistance remains a central concern. When treatment is interrupted, incorrectly prescribed, or affected by resistant organisms, standard regimens may fail. Multidrug-resistant TB prompted additional drugs, diagnostic testing, and newer regimens for some patients. The history of TB chemotherapy continues to evolve alongside laboratory diagnostics and public-health infrastructure.

59.3 What changed

TB chemotherapy transformed a disease once associated with years in a sanatorium into an infection that can often be cured with a defined course of medicines. This depended on trials that measured resistance and relapse, combinations that addressed bacterial persistence, and programs that helped patients complete treatment.

Combination treatment made cure possible for many people with active tuberculosis and reduced the need for prolonged sanatorium isolation. The regimen must be completed over months because surviving bacteria can lead to relapse and resistance. The burden of drug-resistant TB and unequal access show why the breakthrough still depends on reliable diagnosis, drug supply, and patient support [176, 177].

Timeline

1940s Streptomycin becomes the first effective antibiotic treatment for TB.

1948 A randomized trial demonstrates benefit and the problem of resistance to streptomycin alone.

1950s Isoniazid and combination regimens improve treatment.

1960s Rifampicin becomes available and enables shorter regimens.

1970s–1980s

Trials establish short-course combination chemotherapy.

Chapter 60

Malaria Transmission

60.1 A disease with a misleading name

Malaria was one of the clearest examples of medicine mistaking a correlation for a cause. The word came from the Italian *mal'aria*, or bad air, because the disease was associated with marshes, low-lying ground, and warm evenings. The association was real: wetlands supported the mosquitoes that transmitted malaria. The explanation was wrong. Physicians could see where the disease flourished without knowing what moved the parasite from one person to another.

Malaria also resisted simple clinical reasoning because its symptoms appeared intermittently. Fever, chills, sweating, anemia, and enlarged spleen could follow a pattern, but the patient might look well between attacks. The disease was blamed variously on climate, soil vapors, stagnant water, and constitutional weakness. These explanations sometimes identified risky environments while leaving the actual chain of transmission invisible.

60.2 The parasite appears in blood

The first decisive change came from microscopy. In 1880, the French army physician Alphonse Laveran observed pigmented bodies in the blood of patients with malaria in Algeria. He proposed that a protozoan parasite, rather than a poisonous atmosphere, caused the disease. This was an important advance but not the entire answer. A parasite in blood explained illness inside the patient; it did not explain how the parasite reached the next patient.

Quinine supplied another piece of evidence. It could reduce fever and parasites in the blood, yet treatment did not establish the route of infection. A person could recover and become ill again, and communities could continue to suffer even when quinine was known.

The unresolved question was a transmission question: what connected the infected person's blood to the next human host?

60.3 Manson proposes an insect

Patrick Manson, working from his experience with filarial disease, suggested that mosquitoes might carry parasites between people. His hypothesis was partly analogical: if an insect could acquire a parasite from one host and transmit it to another, perhaps the same could happen with malaria. Manson encouraged Ronald Ross, a surgeon in the Indian Medical Service, to investigate the idea.

The proposal faced practical obstacles. Human malaria was difficult to study under controlled conditions, mosquito species were easily confused, and the parasite changed form during its life cycle. A negative dissection could mean that the wrong insect had been used, that it had fed at the wrong time, or that the investigator had missed a structure too small to recognize. The experiment needed not only a microscope but also careful insect breeding and timing.

60.4 Ross's experiments in India

Ross began by examining mosquitoes that had fed on patients with malaria. At first, his results were frustrating and his institutional support was limited. He shifted to bird malaria, using a parasite that could be studied in birds. This model allowed him to control the host, the insect, and the timing of feeding without abandoning the central biological question.

In 1897, Ross found pigmented forms of the parasite in the stomach wall of a mosquito that had fed on an infected bird. He then traced changes in the parasite inside the insect. The observation suggested a developmental stage outside the human body, exactly what the mosquito hypothesis required. In 1898 and 1899, further work connected the mosquito's infection with the ability to transmit malaria to birds.

The logic was cumulative. The parasite was present in infected blood; a mosquito acquired it by feeding; the organism changed inside the mosquito; and the insect could pass infection onward. Ross's success was not a flash of intuition but the product of a model, repeated trials, and attention to the insect's life cycle.

60.5 The human parasite and disputed credit

Ross's bird-malaria work established the principle, but the human story required further evidence. Giovanni Battista Grassi and colleagues in Italy showed that specific mosquito species could transmit human malaria and helped clarify the parasite's development. Disputes over priority became intense, especially because scientific credit, colonial institutions, and national prestige were entangled.

Those disputes should not obscure the structure of the finding. Laveran showed the parasite in human blood; Manson articulated an insect hypothesis; Ross provided an experimentally tractable model and followed the parasite through a mosquito; and Grassi's group helped establish the human transmission cycle. The result was stronger because different forms of evidence converged, not because one investigator's account defeated all others.

In 1902, Ross received the Nobel Prize for showing how malaria enters the human organism and for laying the foundation for methods of combating it [18]. Laveran later received the 1907 Nobel Prize for his discovery of the role of protozoa in disease. The sequence of awards reflects how the causal chain was assembled in stages.

60.6 From discovery to prevention

Once mosquitoes became recognized as vectors, public health could target the link rather than merely treating fever. Drainage of standing water, screening of buildings, bed nets, larval control, and insecticides all became possible interventions. Military campaigns and colonial administrations used malaria control to protect troops and labor forces, but their programs sometimes imposed burdens on local populations and treated disease control as an instrument of imperial administration.

The history also showed the danger of translating a true mechanism into an overconfident intervention. Not every mosquito transmits malaria, and vector species differ by region. Insecticide resistance, parasite resistance to drugs, seasonal ecology, housing quality, and human movement all alter the outcome. The discovery of the vector therefore did not yield one universal solution; it created a framework for locally tested combinations of interventions.

60.7 What clinical intuition could and could not do

Experienced clinicians had correctly noticed the relationship between malaria and marshy places. Their intuition contained a useful signal, but it bundled the mosquito, the parasite, the climate, and the social environment together. Only a chain of experiments could separate

them. An experienced observer may identify a reliable association long before science can specify its mechanism, but the association becomes portable only after the causal steps are made visible.

The mosquito finding changed the definition of prevention. Before it, prevention meant avoiding bad air or strengthening the body. After it, prevention could mean blocking a vector, interrupting parasite development, protecting the sleeping environment, or treating infected people promptly. A new causal model reorganized daily behavior, engineering, government budgets, and the design of field studies.

60.8 The continuing evidence

Malaria research combines laboratory biology, clinical trials, entomology, and population surveillance. New vaccines and improved vector control are assessed against the same old question: which link in the chain is being interrupted, and how durable is that interruption? The answer must be measured in communities, not inferred from a striking individual case.

The history of malaria demonstrates why medical evidence became more rigorous than clinical intuition alone. Intuition located the problem in the marsh; microscopy located the parasite in blood; experimental entomology located the parasite in the mosquito; and public-health trials tested whether controlling the vector reduced disease. Each layer corrected and extended the one before it.

Timeline

1880 Laveran observes malaria parasites in human blood.

1894–1896
Manson develops the hypothesis that mosquitoes transmit malaria.

1897 Ross finds parasite forms in mosquitoes fed on infected birds.

1898–1899
Ross traces mosquito infection and transmission in bird malaria.

1902 Ross receives the Nobel Prize for his malaria research.

1907 Laveran receives the Nobel Prize for identifying the role of protozoa in disease.

Twentieth century

Vector control, bed nets, insecticides, and antimalarial drugs become central prevention tools.

Chapter 61

Antimalarial Therapy and Malaria Control

Understanding how malaria spreads made prevention possible, but treatment also depended on medicines that could kill the parasite. The history of malaria control has repeatedly joined drug discovery with mosquito control, diagnosis, and public-health services. It has also shown how quickly resistance can weaken a tool used too widely or without alternatives.

61.1 From quinine to synthetic drugs

For centuries, cinchona bark was used to treat malaria; its active medicine, quinine, became a key treatment in the nineteenth century. During and after the Second World War, chloroquine and other synthetic antimalarial medicines expanded treatment and prevention. In 1955, the World Health Organization launched a global eradication campaign based largely on indoor insecticide spraying, drug treatment, and surveillance. The campaign succeeded in some places but could not be sustained everywhere, and the 1969 World Health Assembly shifted emphasis toward control rather than a single global eradication effort [266, 267].

Chloroquine resistance spread in important malaria parasites, undermining a widely used medicine. In the late 1960s and 1970s, Chinese scientist Tu Youyou and her colleagues identified artemisinin from the plant *Artemisia annua*. Artemisinin derivatives act rapidly against malaria parasites. Combining them with a second antimalarial medicine helps improve treatment and limit the risk that resistance will emerge to one drug alone. Tu received the 2015 Nobel Prize in Physiology or Medicine for discoveries concerning a novel therapy against malaria [268, 269].

61.2 Control combines several tools

Insecticide-treated bed nets and indoor residual spraying reduce contact between people and mosquitoes. Prompt diagnosis and effective treatment reduce severe illness and the time people carry parasites that mosquitoes can acquire. Surveillance helps public-health teams identify outbreaks, measure whether interventions work, and respond when drug or insecticide resistance develops. No single method has been sufficient in every setting; control depends on combining tools suited to local transmission and health services [267, 269].

The global eradication campaign demonstrated the reach of organized prevention, while uneven coverage and resistance exposed its limits. Modern malaria control builds on that experience through sustained local programs, effective treatment, vector control, and continued research.

The control story is a combination of interventions: prompt diagnosis and effective treatment reduce deaths, while insecticide-treated nets and other vector measures reduce exposure. Artemisinin-based combinations became central as resistance undermined older medicines, but resistance and drug quality remain continuing threats. Gains depend on delivering tools to communities with the highest disease burden [268, 269, 267].

Timeline

1820 Quinine is isolated from cinchona bark.

1955 WHO launches the Global Malaria Eradication Programme.

1969 WHO shifts from a global eradication campaign to sustained control.

1970s Tu Youyou and colleagues identify artemisinin as an antimalarial medicine.

2000s Artemisinin-based combination therapies become central to malaria treatment.

Chapter 62

Malaria Vaccines

For decades, malaria prevention relied on mosquito control, bed nets, diagnosis, and anti-malarial medicines. Developing a vaccine proved difficult because the malaria parasite passes through several forms in the human body and can evade immune responses. The arrival of effective vaccines added a new prevention tool for children in regions where malaria remains a major cause of illness and death.

62.1 Two vaccines reach public-health programs

The RTS,S vaccine, developed over many years of research, targets the *Plasmodium falciparum* parasite before it establishes infection in the liver. A pilot program in Ghana, Kenya, and Malawi beginning in 2019 assessed how the vaccine worked in routine childhood immunization services. In 2021, the World Health Organization recommended RTS,S for children in areas with moderate-to-high transmission. A second vaccine, R21/Matrix-M, received a WHO recommendation in 2023 [276, 277].

Both vaccines reduce malaria cases and severe outcomes, but neither provides complete or lifelong protection. WHO recommends them as part of a broader malaria-control strategy that includes insecticide-treated nets, diagnosis, effective treatment, and other locally appropriate measures. Pilot implementation also showed that vaccine delivery through routine services could reach children and contribute to fewer severe cases and deaths [277].

62.2 A milestone with practical limits

Malaria vaccines represented a scientific and public-health milestone: they provided protection against a complex parasite for which decades of research had struggled to produce a useful

vaccine. Their impact depends on reliable supply, funding, delivery through health systems, and continued use of other prevention tools. The experience also shows that a vaccine can become most valuable as one layer in a coordinated public-health program rather than as a standalone solution.

These vaccines add partial protection to existing malaria control rather than replacing nets, diagnosis, or treatment. Their public-health value depends on completing multiple doses and reaching children in high-transmission regions; effectiveness can vary by setting and over time. The historical story is therefore both a scientific advance and a test of delivery through routine immunization systems [276, 277].

Timeline

2019 RTS,S pilot vaccination begins in Ghana, Kenya, and Malawi.

2021 WHO recommends RTS,S for children in malaria-endemic areas with moderate-to-high transmission.

2023 WHO recommends the R21/Matrix-M malaria vaccine.

2024 onward
National programs expand malaria vaccination alongside established control measures.

Chapter 63

Measles Vaccine

Before vaccination, measles infected nearly every child in many countries and caused pneumonia, encephalitis, blindness, and death. The virus is so contagious that even brief contact can transmit it. Vaccination interrupts the chain of infection by building immunity before exposure.

63.1 Growing the virus and weakening it

John Enders and Thomas Peebles isolated measles virus in tissue culture in 1954, using samples collected during an outbreak in Boston. Enders and colleagues developed a vaccine from the Edmonston strain. In 1963, the United States licensed both an inactivated vaccine and a live attenuated vaccine. The inactivated product did not provide durable protection and was withdrawn. Further attenuation produced live vaccines with fewer reactions [178, 179].

In 1971, measles vaccine was combined with mumps and rubella vaccines as MMR. After outbreaks among school-aged children who had received one dose, public-health authorities recommended a second dose in 1989. The second dose protects many people who did not develop immunity after the first [179].

63.2 Control and elimination

Vaccination sharply reduced measles incidence in countries with broad coverage. The United States declared measles eliminated in 2000, meaning continuous transmission had stopped, although cases could still be imported and spark outbreaks in under-immunized communities. Elimination depended on high and sustained coverage, surveillance, and rapid response; it did not mean the virus had disappeared worldwide.

Measles remains a measure of how connected communities are. Because it spreads so efficiently, small immunity gaps can permit outbreaks. Its vaccine story combines a laboratory achievement with continuing work on access, public trust, and reliable immunization services.

The vaccine changed measles from a near-universal childhood infection into a preventable cause of illness and death where coverage is high. Because measles spreads so readily, even small gaps can permit outbreaks, and two-dose coverage is needed to protect those who do not respond to the first dose. Outbreaks continue wherever coverage falls, showing why sustained immunization matters [179, 178].

Timeline

1954 Measles virus is isolated in tissue culture.

1963 The first measles vaccines are licensed in the United States.

1971 Measles, mumps, and rubella vaccines are combined as MMR.

1989 A second dose is recommended after outbreaks among one-dose recipients.

2000 Measles is declared eliminated in the United States.

Chapter 64

Diphtheria, Tetanus and Pertussis Vaccines

Diphtheria, tetanus, and pertussis are very different diseases, but childhood vaccination brought their prevention together. Diphtheria and tetanus vaccines use inactivated toxins, called toxoids, to teach the immune system to neutralize the poisons. Pertussis vaccines use bacterial components or inactivated whole cells to prevent whooping cough. Combining them made routine immunization more practical.

Before vaccination, these infections killed or permanently harmed many children. Diphtheria toxin could block breathing and damage the heart; pertussis could cause fatal coughing illness in infants; and tetanus toxin could produce painful muscle spasms after a wound or during newborn infection. Vaccination made prevention routine, while clean delivery and proper wound care remain important for tetanus because it does not spread between people [180, 181].

64.1 Toxoids and the combined vaccine

Diphtheria and tetanus toxoids were developed in the early twentieth century. Whole-cell pertussis vaccines became available in the United States in 1914, and the combined diphtheria, tetanus, and pertussis (DTP) vaccine became available in 1948. Together, these vaccines prevented diseases that had caused severe childhood illness and death [180].

Whole-cell pertussis vaccines were effective but often caused fever and local reactions. Researchers developed acellular vaccines containing selected pertussis antigens. The first pediatric DTaP products were licensed in the United States in 1991 for later childhood doses, then extended to the infant series in 1996. Acellular vaccines changed tolerability while preserving combined protection against diphtheria and tetanus [180].

64.2 Boosters and changing protection

Protection against pertussis wanes over time, and older children and adults can transmit infection to infants who are too young to have completed their early series. Tdap formulations, with reduced quantities of diphtheria and pertussis antigens, extended booster vaccination to adolescents and adults. Recommendations have changed as evidence, vaccine products, and disease patterns have changed [181].

Combined vaccines show how one visit can protect against multiple infections. They also show why vaccines need continuing evaluation: protection can wane, pathogens can circulate despite high coverage, and formulations may need revision. Sustained impact requires uptake, booster programs, surveillance, and access to care.

These vaccines illustrate distinct prevention problems: diphtheria and pertussis spread between people, while tetanus follows environmental exposure and is not transmitted person to person. Protection can wane, particularly for pertussis, so booster schedules and vaccination during pregnancy help protect infants before their own series begins. Their different routes of transmission require different prevention strategies [180, 181].

Timeline

1920s Diphtheria and tetanus toxoids are developed for vaccination.

1948 Combined DTP vaccine becomes available in the United States.

1991 Acellular pertussis vaccine is licensed for later childhood doses in the United States.

1996 Pediatric DTaP is licensed for the infant series.

2000s Tdap booster vaccination is introduced for adolescents and adults.

Chapter 65

1918 Influenza Pandemic

The influenza pandemic of 1918–1920 spread around a world already strained by war, troop movement, crowded transport, and limited medical resources. An estimated one third of the world's population became infected. Death estimates vary, but the pandemic killed at least tens of millions worldwide. It was especially severe among young adults, as well as the very young and old [281].

65.1 A deadly virus with few effective tools

The virus was not identified at the time. Doctors could recognize influenza as a contagious illness, but they could not test for the responsible virus, and no influenza vaccine or antiviral medicine was available. Care relied on rest, nursing, hydration, isolation, and attempts to limit contact. Masks, school closures, quarantine, and restrictions on public gatherings were used in some places, with uneven timing and effect. Many patients died with secondary bacterial pneumonia, which could not yet be treated with antibiotics [281].

The pandemic made respiratory infection a central public-health concern and showed the limits of medicine when it could not identify the pathogen or prevent bacterial complications. It also left a lasting lesson: surveillance and timely communication matter because epidemic control depends on action before hospitals are overwhelmed.

65.2 Later breakthroughs changed the response

Researchers isolated influenza viruses in the 1930s, then developed vaccines that became available to civilians during the 1940s. Antibiotics later made many secondary bacterial infections treatable, although they do not act on influenza virus itself. International surveillance,

updated vaccines, diagnostic testing, oxygen, and intensive care added further protection. These advances cannot make a future pandemic harmless, but they give health systems tools absent in 1918 [281, 182].

The story therefore links the catastrophe to later survival gains without crediting the 1918 response with tools that did not yet exist. Vaccination prevents some severe influenza; antibiotics treat bacterial complications; and surveillance helps guide vaccine composition and public-health response.

Timeline

1918–1920 Influenza spreads worldwide and kills tens of millions.

1933 Influenza A virus is isolated from humans.

1940s Inactivated influenza vaccines are developed and enter use.

1952 WHO establishes a global influenza surveillance network.

Today Vaccination, surveillance, antibiotics for bacterial complications, and supportive care reduce risk.

Chapter 66

Influenza Vaccine

Influenza vaccination presents a recurring challenge: the virus changes, and the strains that circulate in one season may differ from those in the next. Vaccine development therefore requires continuing surveillance and regular updates. The history began with experiments during the 1930s and accelerated when the military sought to protect troops during the Second World War.

66.1 Identifying the virus

In 1933, Wilson Smith, Christopher Andrewes, and Patrick Laidlaw isolated influenza A virus, showing that influenza was caused by a virus rather than a bacterium. Thomas Francis and colleagues isolated influenza B in 1936. Researchers learned to grow influenza virus in fertilized chicken eggs, which provided a practical method for producing vaccine material [182].

With support from the US Army, Thomas Francis Jr. and Jonas Salk developed an inactivated influenza vaccine during the 1940s. It was first used in military settings and licensed for civilian use in 1945. The first formulation protected against influenza A; after influenza B was identified, a bivalent vaccine was developed to include both types [182].

66.2 A vaccine that must be revised

The 1947 influenza season showed that a vaccine made from earlier strains could fail when circulating viruses had changed. This helped establish the need to collect viruses around the world, characterize them, and update vaccine composition. The World Health Organization's global influenza surveillance network, established in 1952, made international monitoring a

routine part of vaccine development.

Later technologies expanded production beyond traditional egg-based methods, including cell-culture and recombinant vaccines. Live attenuated vaccines also became available for some populations. The core task remains the same: predict which strains are likely to circulate, manufacture a matching vaccine in time, and evaluate its performance after each season.

Influenza vaccination is an iterative public-health program built around a changing virus. Its history connects virology, manufacturing, international cooperation, and epidemiologic surveillance.

Influenza vaccination requires recurring global surveillance because circulating strains change and vaccine composition must be selected ahead of each season. Protection is therefore variable, but vaccination can reduce severe illness and is especially important for people at higher risk. This recurring process makes the history a story of international coordination as much as vaccine discovery [182].

Timeline

1933 Influenza A virus is isolated from humans.

1936 Influenza B virus is isolated.

1940s Francis and Salk develop an inactivated influenza vaccine.

1945 Influenza vaccine is licensed for civilian use in the United States.

1952 WHO establishes a global influenza surveillance network.

1947 onward
Seasonal strain changes make regular vaccine review essential.

Chapter 67

Monoclonal Antibodies

Antibodies are proteins made by the immune system that bind particular targets. For much of the twentieth century, scientists could obtain antibodies from blood serum, but these mixtures varied and could be difficult to reproduce. Monoclonal antibody technology made it possible to produce a uniform antibody that recognizes one target, opening new paths in laboratory diagnosis and treatment.

67.1 A reproducible antibody

In 1975, Georges Köhler and César Milstein described the hybridoma method. They fused an antibody-producing B cell with an immortal myeloma cell. The resulting hybrid cell could grow continuously while making one antibody, creating a reproducible supply directed at a selected target. Niels Jerne, Köhler, and Milstein received the 1984 Nobel Prize for discoveries concerning the specificity of the immune system and monoclonal antibody production [183].

The first therapeutic monoclonal antibody approved in the United States, muromonab-CD3 (OKT3), was licensed in 1986 to help prevent rejection after organ transplantation. Early antibodies were made in mice, which could provoke immune reactions in people. Genetic engineering later produced chimeric, humanized, and fully human antibodies, improving the ability to use them repeatedly [184].

67.2 From laboratory tool to medicine

Monoclonal antibodies became valuable in cancer treatment, autoimmune disease, infectious disease, and diagnostics. Rituximab, approved in 1997, targeted CD20 on B cells and established a new approach to some lymphomas. Other antibodies block signaling molecules,

neutralize inflammatory proteins, or deliver a drug or radioactive payload to selected cells [185].

The technology's strength is also a constraint: an antibody is useful only when its target is sufficiently important and accessible. Antibodies are large molecules, often given by injection or infusion, and can cause immune reactions or other target-related effects. Their development reflects the growing ability to translate molecular knowledge into treatments aimed at defined biological structures.

Hybridoma methods made it possible to produce antibodies with a defined target, first transforming laboratory diagnosis and later enabling targeted therapies. Therapeutic antibodies now include products for cancer, autoimmune disease, and infection, but each depends on the target, disease, and evidence for benefit. Production complexity and cost have also shaped who can receive these medicines [184, 185].

Timeline

1975 Köhler and Milstein describe hybridoma production of monoclonal antibodies.

1984 The hybridoma discovery is recognized with the Nobel Prize.

1986 OKT3 becomes the first therapeutic monoclonal antibody approved in the United States.

1997 Rituximab is approved for treatment of B-cell lymphoma.

Chapter 68

Cancer Checkpoint Immunotherapy

The immune system can recognize and attack abnormal cells, but tumors may exploit signals that restrain immune responses. Checkpoint proteins such as CTLA-4 and PD-1 normally help prevent excessive immune activity. Cancer checkpoint therapy uses antibodies to block these inhibitory signals, allowing some immune cells to attack tumors more effectively.

68.1 Understanding immune brakes

Research on CTLA-4 and PD-1 began decades before checkpoint medicines reached clinics. These proteins help regulate T cells, but cancer cells and surrounding tissues can use checkpoint pathways to dampen antitumor responses. James Allison's work on CTLA-4 blockade and Tasuku Honjo's work on PD-1 helped establish that interrupting these signals could produce durable tumor responses in some patients.

Ipilimumab, an antibody that blocks CTLA-4, became the first checkpoint inhibitor approved in the United States in 2011 for advanced melanoma. Antibodies directed against PD-1 and its ligand PD-L1 followed. Pembrolizumab was first given to patients in 2011 and received approval for advanced melanoma in 2014 after clinical studies demonstrated meaningful responses [186, 187].

68.2 A different pattern of benefit and risk

Checkpoint inhibitors can produce long-lasting responses in some cancers, including melanoma and several lung and kidney cancers. They do not work for everyone, and benefit varies with tumor biology and immune context. By interrupting normal immune brakes, these drugs can cause inflammation in healthy organs, requiring careful recognition and management by

clinical teams.

Checkpoint therapy changed cancer treatment by demonstrating that durable control could come from mobilizing a patient's own immune system. It also reinforced the value of basic research: the clinical strategy emerged from studies of immune regulation rather than from a direct search for a conventional tumor poison.

Checkpoint drugs produced durable remissions for some people with advanced cancers that had previously been difficult to treat, a major change in expectations. They do not work for everyone, and releasing immune restraints can cause inflammation that damages healthy organs. Biomarkers and clinical trials help identify likely benefit, but prediction remains imperfect [186, 187].

Timeline

1980s–1990s CTLA-4 and PD-1 are identified as regulators of T-cell activity.

2011 Ipilimumab becomes the first approved checkpoint inhibitor.

2014 Pembrolizumab is approved for advanced melanoma.

2018 Allison and Honjo receive the Nobel Prize for checkpoint research.

Chapter 69

Targeted Cancer Therapy

Traditional chemotherapy attacks rapidly dividing cells, which can damage healthy tissues as well as cancer. Targeted cancer therapies emerged from a different idea: identify a molecular change that helps a tumor grow, then design a medicine to block it. The approach depends on understanding the biology of an individual cancer, not only its location in the body.

69.1 A drug designed for a defining abnormality

Chronic myeloid leukemia (CML) is often driven by the BCR-ABL fusion protein, produced by a chromosomal rearrangement called the Philadelphia chromosome. Researchers identified the abnormal signal as a potential treatment target. In the 1990s, Brian Druker and colleagues helped develop imatinib, a small molecule that blocks the BCR-ABL kinase. Clinical trials showed striking responses, and the US Food and Drug Administration approved imatinib for CML in 2001 [188].

Imatinib became a landmark because it linked a cancer's defining molecular lesion to a treatment designed to inhibit it. The same drug later proved useful in gastrointestinal stromal tumors with a different susceptible molecular target. Subsequent targeted therapies were developed against altered receptors, signaling proteins, DNA-repair pathways, and other tumor dependencies.

69.2 Testing the target

Targeted treatment requires tests that determine whether a tumor carries the relevant molecular feature. A drug can be highly effective for a biologically selected group yet offer little benefit when its target is absent or resistant. Tumors can also evolve under treatment,

making repeat testing and new strategies important.

Targeted therapy did not replace chemotherapy, surgery, or radiotherapy. It expanded cancer treatment by making molecular diagnosis part of therapeutic planning. Its history shows how advances in genetics, laboratory measurement, and drug design can combine to change the prognosis of particular cancers.

The approach depends on validated tests and access to matching treatment. Many cancers lack a clear target, and tumors can develop resistance even when the initial response is strong. Molecular testing and the cost of long-term therapy therefore influence who benefits from precision treatment [188].

Timeline

1960 The Philadelphia chromosome is described in chronic myeloid leukemia.

1990s BCR-ABL is established as a drug target and imatinib enters clinical testing.

2001 Imatinib is approved for Philadelphia chromosome-positive CML.

2017 Tumor molecular features begin to guide some tissue-agnostic cancer drug approvals.

Chapter 70

CAR-T Cell Therapy

CAR-T therapy turns a patient's T cells into engineered cancer-fighting cells. The treatment combines cell biology, gene transfer, and immunotherapy: clinicians collect T cells, introduce genetic instructions for a chimeric antigen receptor (CAR), expand the modified cells, and return them to the patient. The cells can then recognize a selected marker on cancer cells.

70.1 Engineering a new receptor

Early work on genetically redirecting T cells developed over several decades. Researchers refined CAR designs so the engineered receptor could recognize a surface antigen and activate the T cell. Clinical studies showed that modified T cells targeting CD19 could produce deep remissions in some patients with relapsed B-cell leukemia and lymphoma.

In 2017, the US Food and Drug Administration approved tisagenlecleucel for certain children and young adults with acute lymphoblastic leukemia. It was the first approved CAR-T therapy. A second product for adult lymphoma followed that year. Later approvals extended CAR-T to selected adults with other lymphomas and multiple myeloma. Its benefit is substantial for eligible people with certain advanced blood cancers, but it remains limited to particular cancers and treatment stages [189].

70.2 A living medicine

CAR-T cells can multiply in the body and persist after infusion. For some people with blood cancers that returned after other treatments, they have produced long remissions and, in some cases, apparent cures. That is a major change for patients who once had few effective options, even though CAR-T remains a treatment for a limited group rather than a population-wide

cancer therapy [189].

Treatment can also trigger severe cytokine release syndrome or neurologic effects, so patients need close monitoring and experienced clinical teams. In the usual approach, cells are collected from each patient, sent for modification and expansion, then returned; this individualized manufacturing usually takes about three to five weeks. Patients also receive chemotherapy to prepare for the infusion and need follow-up for infections and other immune effects. The time, specialist-center requirements, eligibility rules, and treatment risks limit how many patients can receive it [189].

70.3 What could widen its reach

CAR-T has been most successful in blood cancers whose cells display accessible targets such as CD19. Extending the approach to solid tumors remains difficult: targets may also appear on healthy tissue, and tumors can suppress immune activity. Researchers are testing safer designs, new targets, and donor-derived cells that could be prepared in advance. These approaches may simplify or broaden treatment, but they remain under study; they are not yet proof that CAR-T can treat most cancers or reach most patients [189].

Timeline

1980s–1990s Early approaches to redirecting T cells by genetic engineering are developed.

2010s Clinical trials show major responses in some B-cell cancers.

2017 The first CAR-T therapy is approved in the United States.

Chapter 71

Bone Marrow and Stem Cell Transplantation

Bone-marrow transplantation replaces blood-forming stem cells after disease or intensive treatment has damaged the marrow. Donor stem cells can also provide a new immune system capable of attacking certain cancers. This made transplantation a treatment for leukemia, marrow failure, inherited blood disorders, and immune deficiencies, but success depended on matching donors and controlling immune complications.

71.1 From marrow rescue to transplantation

Early experiments showed that marrow cells could restore blood formation after radiation injury. In the 1950s and 1960s, E. Donnall Thomas and colleagues developed methods for transplanting marrow between people. The first successful allogeneic transplants for severe immune deficiencies and blood diseases were reported in 1968. These procedures required conditioning treatment, compatible donors, and close monitoring for infection and graft-versus-host disease [190].

Human leukocyte antigen (HLA) typing improved donor selection. Transplants from matched siblings became more successful, while registries later made unrelated-donor transplantation possible for many patients. The source of stem cells also expanded: cells could be collected from bone marrow, mobilized peripheral blood, or umbilical-cord blood.

71.2 A changing role in medicine

Transplantation can restore blood production after high-dose chemotherapy and can replace diseased marrow with healthy donor cells. In some cancers, donor immune cells contribute a graft-versus-leukemia effect. These benefits must be balanced against graft-versus-host disease, relapse, infections, and the risks of conditioning therapy.

The term stem-cell transplant now reflects a broader technology than the original marrow harvest. The central principle remains the same: transfer blood-forming cells capable of rebuilding the marrow. Advances in tissue matching, supportive care, infection prevention, and donor recruitment have made the procedure available to more people, though its demands remain substantial.

Timeline

1950s Experimental work establishes marrow as a source of blood-forming cells.

1968 Successful allogeneic transplants are reported for severe immune deficiency and blood disorders.

1970s HLA typing and transplant registries improve donor selection.

1980s onward Peripheral blood and cord blood expand stem-cell sources.

Chapter 72

Mammography

Mammography uses low-dose X-rays to produce images of the breast. It can reveal changes too small to feel, including masses and clusters of tiny calcium deposits. The development of screening mammography brought radiology into preventive care and created a new challenge: evaluating whether earlier detection could reduce deaths, not merely identify more abnormalities.

72.1 From diagnostic X-ray to population screening

Physicians experimented with breast radiographs in the early twentieth century, but image quality and technique improved gradually. In 1963, the Health Insurance Plan of Greater New York began a randomized trial comparing periodic mammography and clinical examination with usual care. The trial helped establish screening as a subject for rigorous evaluation and stimulated improvements in dedicated mammography equipment [191, 192].

Screening programs expanded during the 1970s and 1980s as trials and observational studies accumulated. Film mammography later gave way to digital imaging, which stores images electronically and supports computer-assisted review. More recent technologies include digital breast tomosynthesis, which reconstructs a series of images into thin slices.

72.2 Benefits, limits, and judgment

Mammography can detect cancers earlier, but screening can also produce false alarms, lead to biopsies of benign findings, or identify abnormalities that would never have caused harm during a person's lifetime. These trade-offs made communication and evidence review important parts of screening policy. Mammography illustrates that a diagnostic technology

has value through the outcomes it improves, not through image detail alone.

The history of mammography therefore joins engineering with population research. Screening recommendations have been shaped by trials, changing technology, age-specific risk, and differing judgments about the balance of benefits and harms [192].

Screening only achieves benefit when an abnormal result leads to timely imaging, biopsy, and treatment. Age and baseline risk shape the balance between earlier detection, false alarms, and overdiagnosis, so trial findings must be considered alongside current screening guidance. Access to diagnostic follow-up is part of an effective program [191, 192].

Timeline

1910s Early X-ray studies describe breast tumors.

1963 The Health Insurance Plan breast screening trial begins in New York.

1970s Trial results help establish mammography as a screening method.

2000 Digital mammography is approved in the United States.

Chapter 73

Colonoscopy

Colonoscopy allows clinicians to inspect the lining of the colon and rectum with a flexible instrument carrying a light and camera. Its contribution to medicine is both diagnostic and preventive: it can find cancer and can also locate and remove many precancerous polyps during the same procedure.

73.1 A flexible instrument reaches the whole colon

Early examinations of the colon relied on rigid instruments, X-rays, or limited views of the lower bowel. Improvements in flexible fiber-optic instruments made it possible to navigate the turns of the colon and examine its length. In 1969, William Wolff and Hiromi Shinya began modern retrograde colonoscopy; Shinya also developed a wire-loop snare to remove polyps through the instrument's channel [193, 194].

Polypectomy changed the role of endoscopy. A suspicious growth no longer had to be left in place until open surgery: it could often be removed during the examination and sent for pathology. This linked visualization, intervention, and prevention in one procedure.

73.2 A platform for screening

Over time, colonoscopy became one of several approaches to colorectal cancer screening. It requires preparation and trained staff, and it carries procedural risks, but it can inspect the entire colon and remove certain lesions. Research into screening tests and intervals has continued to compare the benefits and burdens of available methods [195].

The development of colonoscopy depended on advances in optics, steering controls, illumination, and working channels. Its history shows how a diagnostic tool can change

disease prevention when it also allows removal of early lesions.

Its preventive value comes from finding and removing some precancerous polyps before they become cancer, making the procedure both a diagnostic test and an intervention. The benefit depends on preparation quality, complete examination, removal of lesions, and follow-up at appropriate intervals. Bleeding and perforation are uncommon but important risks, and screening access remains uneven [194, 195].

Timeline

1950s Flexible fiber-optic endoscopy is developed.

1969 Wolff and Shinya perform early modern full-colon examinations and polypectomies.

1973 Their report describes polyp removal through the fiber-optic colonoscope.

Late twentieth century
Colonoscopy becomes part of colorectal cancer screening programs.

Chapter 74

Endoscopy

Endoscopy uses a lighted instrument to examine the inside of the body. Early endoscopes were rigid tubes that could inspect only limited areas and were uncomfortable to use. Flexible fiber-optic instruments transformed the field by carrying light around bends and allowing clinicians to see deeper within the digestive tract.

74.1 The fiber-optic breakthrough

In the 1950s, gastroenterologist Basil Hirschowitz and physicists Larry Curtiss and Wilbur Peters developed a flexible fiberscope at the University of Michigan. Glass fibers clad to keep light within the bundle made it possible to transmit an image through a bendable instrument. Hirschowitz tested a prototype gastroscope in 1957, helping establish flexible upper gastrointestinal endoscopy as a practical clinical method [196].

Later endoscopes added channels for biopsy forceps, suction, and instruments. Physicians could take tissue samples, stop bleeding, remove polyps, and treat some obstructions without open surgery. Video chips eventually replaced direct viewing through fiber bundles in many instruments, improving image display and documentation.

74.2 A family of procedures

Endoscopy now includes procedures for the esophagus, stomach, bowel, airways, urinary tract, and joints. Each depends on a combination of optics, flexible materials, steering mechanisms, and specialized instruments. The technology expanded medicine's ability to diagnose disease directly and to intervene through natural openings or small incisions.

Endoscopy is not one procedure with one risk profile. Its evolution has created a family

of diagnostic and therapeutic methods that require training, careful instrument processing, and appropriate patient selection. The central change was the ability to see and work inside the body with far less disruption than open surgery.

Flexible scopes turned visualization into a tool for both diagnosis and treatment: clinicians could biopsy suspicious tissue, control bleeding, and remove some lesions without open surgery. Camera, lighting, and instrument improvements broadened the procedures, but endoscopy still requires training, infection control, and careful selection because sedation and procedural complications carry risk [196].

Timeline

1800s Rigid endoscopes are developed for limited examinations.

1957 Hirschowitz and colleagues introduce a practical flexible fiber-optic gastroscope.

1960s–1970s
Biopsy and therapeutic channels expand endoscopic procedures.

1980s onward
Video endoscopy improves image display and recording.

Chapter 75

Laparoscopic Surgery

Laparoscopic surgery allows surgeons to operate inside the abdomen using a camera and long instruments passed through small incisions. The approach developed from diagnostic laparoscopy, used for decades in gynecology, into a way to remove organs and perform complex operations. It changed surgery by reducing the size of the access incision while demanding new skills and equipment.

75.1 From inspection to operation

In the early twentieth century, physicians used laparoscopes to inspect the abdominal cavity. Fiber-optic illumination and improved optics made the view brighter and clearer. The arrival of video cameras that projected the operative field onto a monitor allowed the surgeon and operating team to share the image and use instruments through separate ports.

Erich Mühe performed an early laparoscopic gallbladder removal in Germany in 1985; Philippe Mouret performed a widely recognized procedure in France in 1987. The operation spread rapidly after surgeons showed it could be taught and performed beyond a few pioneering centers. By the early 1990s, laparoscopic cholecystectomy had become a major example of minimally invasive surgery [197].

75.2 A new operating technique

Laparoscopy requires coordination while viewing a two-dimensional image, working with instruments that pivot at the abdominal wall, and operating in a space created by insufflated gas. Surgeons adapted their movements and teams redesigned training, equipment, and operating-room workflows. Patients often benefited from smaller incisions and shorter recovery,

while safe use depended on experience and appropriate case selection.

The approach expanded to gynecologic, urologic, and many general surgical operations. Laparoscopy did not eliminate open surgery; it created another route to reach internal organs, with different advantages and limitations.

After early skepticism, improved cameras, instruments, and surgeon training helped laparoscopic cholecystectomy become a widely adopted operation. Smaller incisions often mean less pain and faster recovery, although outcomes depend on the operation and experience of the team. The transition also changed surgical training because operators had to learn to work from a screen rather than directly view the organs [197].

Timeline

Early 1900s
: Laparoscopy is used primarily to inspect the abdomen.

1950s–1970s
: Fiber-optic light and video technology improve visualization.

1985 Mühe performs an early laparoscopic cholecystectomy.

1987 Mouret performs a laparoscopic cholecystectomy in France.

1990s Laparoscopic surgery spreads across general surgery.

Chapter 76

Robotic-Assisted Surgery

Robotic-assisted surgery places a surgeon at a control console linked to instruments at the operating table. The system translates the surgeon's hand movements into precise instrument motion and provides a magnified view. The robot does not make independent decisions or perform an operation on its own; the surgeon remains in control throughout the procedure.

76.1 Teleoperation enters the operating room

Research into surgical robots grew from efforts to improve precision, enable remote operation, and make minimally invasive procedures easier. Early systems helped position cameras or guide instruments. The da Vinci system received US Food and Drug Administration clearance for surgical use in 2000, adding a three-dimensional view and wristed instruments that could move within the body with more degrees of freedom than straight laparoscopic tools [198].

Robotic systems were adopted first in specialties where precise dissection and suturing through small openings were valuable, including urology, gynecology, and cardiac surgery. Over time, manufacturers improved cameras, instrument articulation, and surgeon controls.

76.2 Capability and evidence

Robotic assistance can help surgeons perform intricate movements through small incisions, but it adds cost, equipment, and training demands. Whether it improves outcomes depends on the operation, the team, and comparison with established laparoscopic or open approaches. The technology's history is therefore a story of engineering and clinical evaluation together.

The ROLARR randomized trial illustrates why results must be assessed by procedure. Among 471 patients having rectal-cancer surgery, conversion to open surgery occurred in 8.1%

of the robotic group and 12.2% of the laparoscopic group; the difference was not statistically significant. The trial also found no significant difference in quality of life at six months. This does not settle the value of robotics for every operation, but it shows that greater instrument articulation does not automatically produce better patient outcomes [199].

Robotic systems have expanded the available surgical toolkit. Their clinical value depends on meaningful benefits for particular procedures, measured against the extra cost, operating time, training, and maintenance required.

Timeline

1980s Research prototypes explore computer-assisted and remote surgery.

1990s Robotic camera and instrument systems enter clinical testing.

2000 The da Vinci system receives US FDA clearance for surgical procedures.

2000s onward
Robotic-assisted techniques expand into multiple specialties.

Chapter 77

Hip and Knee Replacement

When cartilage at a joint wears away, movement can become painful and difficult. Medicines, exercise, and other treatments help many people, but advanced arthritis or injury can damage a joint so severely that replacing its worn surfaces becomes an option. Hip and knee replacements are among the most visible achievements of modern orthopaedic surgery. Their history is a story of trial and error in materials, mechanics, and surgical design, aimed at relieving pain while preserving useful movement.

77.1 The modern hip replacement

Early surgeons tried to restore damaged joints with tissue or artificial materials, but many implants wore out or loosened. In the 1930s, London surgeon Philip Wiles described a metal hip replacement. The modern era gathered pace in the work of British surgeon John Charnley. Beginning in 1959, he tested a metal femoral component against a plastic socket secured with acrylic bone cement. Early versions using Teflon wore rapidly. In 1962, Charnley adopted high-density polyethylene for the socket and developed what he called low-friction arthroplasty. A small metal ball moving against the plastic liner reduced friction and helped create a durable replacement [85].

The Charnley design was not the work of one person alone: surgeons, engineers, technicians, and manufacturers contributed to its development. Its basic arrangement remains familiar today. In a total hip replacement, the damaged ball at the top of the thigh bone is replaced with a stem and ball, while a cup and liner replace the damaged socket. Components may be fixed with cement or held by a close fit that allows bone to grow onto their surface. Metals, ceramics, and durable plastics are selected in different combinations [85, 87].

77.2 Replacing the knee's surfaces

The knee presented a different mechanical problem. It bends and straightens while also rotating slightly, and it depends on ligaments for stability. Early implants included hinged designs that constrained the joint, but they could place substantial stress on the bone and fixation. By the late 1960s and early 1970s, surgeons and engineers developed designs that resurfaced the ends of the thigh and shin bones while allowing the ligaments to guide motion. Frank Gunston's polycentric design and later condylar designs helped shape this transition. John Insall and colleagues introduced the total condylar knee prosthesis in the 1970s, refining the way the implant reproduced the joint surface and managed stability [86].

A total knee replacement is better understood as resurfacing than as removing the whole knee. The surgeon removes damaged cartilage and a small amount of underlying bone from the thigh and shin bones, then places metal components with a durable plastic spacer between them. The kneecap may also be resurfaced, depending on the case. When arthritis is limited to one compartment, a partial knee replacement may preserve more of the natural joint [88].

77.3 From pain relief to rehabilitation

Osteoarthritis is a common reason for replacement, though inflammatory arthritis, previous injury, and other conditions can also damage the hip or knee. Surgery is generally considered when pain and loss of function remain substantial despite other treatments. The decision takes account of the affected joint, overall health, daily activities, and the person's goals [87, 88].

Replacement can reduce pain and improve everyday function, but it does not make a joint identical to a natural one. Recovery includes movement and strength exercises, and implant materials can wear or loosen over time. Infection and blood clots are uncommon but serious complications; hip implants can also dislocate. Improved materials, fixation methods, and surgical techniques have made many replacements last for years or decades, though some eventually need revision surgery [87, 88, 85].

The evolution of joint replacement shows how closely medicine can connect to engineering. Early devices proved that an artificial joint was possible, but lasting success depended on reducing friction, preventing wear, anchoring components, and respecting the body's biomechanics. Hip and knee replacements do not restore youth or unlimited activity. They offer many people a way to move with less pain and greater independence.

Long-term studies show why these operations are judged by pain relief, mobility, implant survival, and the need for revision, not by the operation alone. Hip and knee replacements

have different mechanics and recovery patterns: hip replacement often restores walking with less joint pain, while knee replacement outcomes depend more on regaining motion and strength. Both can substantially improve function for appropriately selected people, but neither guarantees pain-free movement [87, 88].

Timeline

1938 Philip Wiles reports an early metal total hip replacement.

1959 John Charnley begins clinical trials of a cemented low-friction hip replacement.

1962 Charnley adopts a polyethylene socket, establishing the durable low-friction hip design.

1968 Total knee replacement is first performed.

Early 1970s Newer knee surface and condylar designs improve the fit and function of knee prostheses.

Today Modern hip and knee replacements use refined materials and fixation methods, followed by rehabilitation and long-term monitoring.

Chapter 78

Cataract Surgery with Intraocular Lenses

A cataract clouds the eye's natural lens and can gradually impair vision. For much of history, surgery removed the opaque lens but left the eye without its natural focusing power. Patients needed thick spectacles or contact lenses to compensate. The intraocular lens (IOL) changed cataract surgery by replacing the removed lens inside the eye.

78.1 Replacing the natural lens

During the Second World War, ophthalmologist Harold Ridley observed that fragments of acrylic aircraft canopies could remain in injured eyes without the severe reaction caused by many other materials. He reasoned that a carefully designed lens made from the same material might be tolerated inside the eye. Ridley implanted an early IOL in 1949, despite skepticism from colleagues and complications with the first designs [200].

The idea matured as surgeons refined lens shape, material, and placement. Charles Kelman's introduction of phacoemulsification in 1967 used ultrasonic energy to break up the cataract through a smaller incision. Improvements in foldable lenses and surgical technique later allowed IOLs to be inserted through small openings, making cataract surgery one of the most common operations worldwide [201].

78.2 A major change in visual rehabilitation

Modern cataract surgery removes the cloudy lens and usually implants an IOL to restore focusing power. Lens options have expanded, but every design involves trade-offs in optical

correction and visual effects. The historical transformation came from treating lens removal and visual rehabilitation as parts of the same operation.

Cataract surgery demonstrates how an implanted device can turn an operation that once left a major optical disability into a routine means of restoring useful sight. Its success required biocompatible materials, reliable instruments, surgical training, and access to care.

Timeline

1949 Harold Ridley implants an early intraocular lens after cataract removal.

1967 Charles Kelman introduces phacoemulsification.

1970s–1980s

Improved lens materials and designs make IOL implantation increasingly reliable.

Late twentieth century

Smaller-incision techniques become widely adopted.

Chapter 79

Laser Refractive Eye Surgery

Eyeglasses and contact lenses correct refractive errors by changing how light is focused. Laser refractive surgery alters the curvature of the cornea itself. Excimer lasers can remove microscopic amounts of tissue with precision, creating a new way to correct nearsightedness and other refractive errors.

79.1 Excimer lasers and corneal reshaping

Photorefractive keratectomy (PRK) was among the first excimer-laser procedures approved in the United States. The FDA approved an excimer laser for PRK in 1995. LASIK developed in the 1990s by combining a corneal flap with laser reshaping of the tissue beneath it; FDA approvals for LASIK systems followed in the late 1990s [202, 203].

The procedures offered an alternative to glasses and contact lenses, but they required careful measurement and stable corneal tissue. Equipment evolved to include computerized treatment profiles, eye tracking, and newer methods for creating corneal flaps. Laser refractive surgery became an elective procedure whose results depend on patient selection, technology, and surgical skill.

79.2 Precision with limits

Refractive surgery can reduce dependence on corrective lenses, but it does not prevent age-related changes in vision or remove all need for glasses. Dry eye, glare, undercorrection, overcorrection, and other complications remain possible. The FDA's published information emphasizes that suitability and outcomes vary, making informed evaluation an important part of the technology's use.

The history of laser eye surgery illustrates the translation of precise energy delivery into a new form of surgery. It also shows why technical capability must be paired with evidence, clear communication, and long-term follow-up.

Timeline

1980s Excimer laser research establishes controlled corneal reshaping.

1995 The first excimer laser is approved in the United States for PRK.

Late 1990s
FDA approvals expand to LASIK procedures.

2000s onward
Tracking, imaging, and laser platforms continue to advance.

Chapter 80

Anti-VEGF Retinal Therapy

Some retinal diseases cause abnormal blood vessels to grow and leak beneath or within the retina. In neovascular age-related macular degeneration, this process can rapidly damage central vision. Anti-VEGF medicines block vascular endothelial growth factor, a signal that promotes abnormal vessel growth and leakage.

80.1 A shift in retinal treatment

Before anti-VEGF drugs, treatments such as laser photocoagulation and photodynamic therapy could slow damage in some patients but often could not restore or preserve vision as effectively. Ranibizumab, an antibody fragment that blocks VEGF, was approved for neovascular age-related macular degeneration in 2006. Clinical trials showed that regular injections could prevent vision loss and improve visual acuity for many patients [204, 205].

Bevacizumab, developed initially for cancer, was also used off-label in the eye after clinicians observed its potential. The National Eye Institute's Comparison of AMD Treatments Trial found similar outcomes between bevacizumab and ranibizumab under the studied schedules, prompting a major discussion about evidence, cost, and access. Aflibercept later provided another anti-VEGF option [204].

80.2 Treatment over time

Anti-VEGF therapy changed neovascular retinal disease from a condition that often led to severe central vision loss into one that could frequently be managed over years. The treatment commonly requires repeated injections and ongoing monitoring. The need for continued care makes adherence and access central to preserving benefit.

This therapy also illustrates how a biological mechanism can direct treatment: identifying VEGF as a driver of abnormal vessel growth led to medicines that interrupt that signal. Clinical trials then established whether the mechanism translated into useful vision outcomes.

Before anti-VEGF treatment, abnormal vessel growth in wet macular degeneration often led to progressive central vision loss. Trials showed that repeated injections could preserve or improve vision for many patients, changing the outlook for a major cause of blindness. Treatment requires ongoing monitoring and injections, so adherence and access shape the long-term result [205, 204].

Timeline

2004 Bevacizumab is approved for cancer treatment.

2006 Ranibizumab is approved for neovascular age-related macular degeneration.

2008 The NEI Comparison of AMD Treatments Trial begins.

2011 Aflibercept becomes another treatment option for retinal disease.

Chapter 81

Modern Glaucoma Treatment

Glaucoma is a group of diseases that damage the optic nerve and can cause permanent vision loss. Treatment cannot restore nerve tissue already lost, so modern care aims to slow further damage. Lowering pressure inside the eye became the central strategy, using medicines, lasers, and surgery.

81.1 From pressure measurement to pressure lowering

For centuries, physicians recognized that some blind eyes felt unusually hard, but the relationship between pressure and optic-nerve damage took time to establish. In the nineteenth century, investigators increasingly linked elevated intraocular pressure with glaucoma. Treatments evolved from early medicines to safer eye drops that reduce fluid production or increase its drainage.

Laser treatment expanded the options. The Glaucoma Laser Trial, reported in 1990, suggested that argon laser trabeculoplasty could serve as an effective initial treatment for some people with newly diagnosed open-angle glaucoma [206]. Trabeculectomy and drainage implants created alternative routes for fluid to leave the eye when medicines or laser were insufficient. More recently, minimally invasive glaucoma surgeries have added procedures designed to lower pressure with smaller interventions [207].

81.2 Long-term monitoring

Because glaucoma can progress without obvious early symptoms, treatment depends on repeated measurement of eye pressure, optic nerve structure, and visual function. Drops, laser, and surgery can reduce risk but do not cure the disease or reverse existing vision

loss. Modern treatment therefore combines intervention with long-term monitoring and adjustment.

The history of glaucoma care reflects a broader shift toward preventing irreversible damage. A measurable risk factor became a treatment target, and several therapies developed around the goal of preserving remaining vision.

Glaucoma treatment cannot restore vision already lost, so the advance lies in detecting risk and lowering eye pressure before further optic-nerve damage occurs. Evidence for laser and surgical approaches expanded the options beyond drops, particularly when medicines were hard to use or insufficient. Long-term follow-up is essential because the disease may progress without noticeable symptoms [206, 207].

Timeline

1860s Pressure-lowering medicines enter early glaucoma treatment.

Twentieth century
Eye drops and filtration surgery improve pressure control.

1990 The Glaucoma Laser Trial supports laser trabeculoplasty as a treatment option.

Twenty-first century
Minimally invasive procedures expand the surgical toolkit.

Chapter 82

In Vitro Fertilization (IVF)

For much of history, infertility offered few treatments beyond surgery or the hope that conception might occur later. In vitro fertilization (IVF) changed that outlook by moving one stage of conception into a laboratory. An egg is fertilized with sperm outside the body, and an embryo may then be transferred to the uterus. The method joined reproductive biology with careful clinical timing and laboratory work, creating a new path to pregnancy for some people who could not conceive without assistance.

82.1 A long scientific effort

IVF did not begin with one sudden discovery. Scientists had to learn how eggs mature, when they can be fertilized, how to retrieve them safely, and how to support early embryos outside the body. In the 1960s, British researcher Robert Edwards advanced the laboratory study of human eggs and fertilization. He later collaborated with Patrick Steptoe, a surgeon who used laparoscopy to view the ovaries and retrieve eggs, and Jean Purdy, a nurse and embryologist who helped develop and maintain the laboratory methods. Their work brought together expertise that had previously been separate [75].

The early attempts were uncertain. A pregnancy achieved by the team in 1976 ended when the embryo implanted in a fallopian tube rather than the uterus. They refined their approach, including the timing of egg collection and the use of a natural menstrual cycle. On 25 July 1978, Lesley Brown gave birth to Louise Brown at Oldham, England. She was the first child born after successful human IVF. The birth demonstrated that an egg could be fertilized outside the body, develop as an embryo, and lead to a live birth after transfer to the uterus. Edwards received the 2010 Nobel Prize in Physiology or Medicine for the development of human IVF; Steptoe and Purdy were central members of the clinical and laboratory team [75, 76].

India's early IVF history includes a parallel and contested account. On 3 October 1978, a team led by Subhas Mukherjee in Calcutta announced the birth of Kanupriya, known as Durga, after an IVF procedure. Contemporary authorities rejected the claim amid disputes over its evidence and documentation. Later researchers revisited Mukherjee's papers and records, and recent historical work has treated the case as an important, contested milestone in Indian reproductive medicine. The dispute shows how scientific recognition depends not only on a result, but also on which institutions record, assess, and credit it [79, 80].

82.2 From a rare breakthrough to a treatment

The first IVF cycle used a single egg collected by laparoscopy. Later advances made treatment more practical. Hormone medicines could stimulate the ovaries to mature several eggs in one cycle, while ultrasound-guided retrieval avoided the need for abdominal surgery. Improved culture conditions allowed embryos to develop in the laboratory for several days. Freezing made it possible to preserve embryos for later transfer, and intracytoplasmic sperm injection (ICSI), introduced in the early 1990s, allowed an embryologist to inject a single sperm into an egg in some cases of male-factor infertility or prior fertilization difficulty [76, 77].

These developments changed IVF from a treatment primarily associated with blocked fallopian tubes into a range of options used for different fertility problems. IVF may be considered when other approaches have not worked, when the fallopian tubes are blocked or damaged, or when sperm-related factors make fertilization difficult. It can also be used with donor eggs or sperm, and in fertility preservation. The appropriate approach depends on the circumstances of the people seeking care and local clinical and legal frameworks [78].

82.3 How IVF works today

A typical treatment cycle begins with assessment and medicines that stimulate the ovaries to develop several eggs. Clinicians monitor the response with blood tests and ultrasound. The eggs are collected, usually with a needle guided by ultrasound, and brought to the laboratory. They are either mixed with sperm or fertilized using ICSI. Embryologists observe development over the next several days. An embryo may be transferred to the uterus through a thin catheter; suitable additional embryos can be frozen for later use. A cycle may not produce an embryo for transfer, and transfer does not guarantee pregnancy or birth [78].

Modern care often aims to transfer one embryo at a time. Transferring more than one can increase the chance of twins or higher-order pregnancy, which can carry added risks for the pregnant person and babies. Fertility medicines can also cause ovarian hyperstimulation

syndrome, and IVF is associated with risks such as ectopic pregnancy. Success varies with factors including age and the cause of infertility, and treatment can be physically, emotionally, and financially demanding [78].

82.4 New possibilities and difficult questions

IVF made possible forms of family building that had not existed before, including the use of donor eggs, sperm, or embryos and the freezing of reproductive cells and embryos. It also raised questions about consent, embryo storage and disposition, donor identity, access, and the limits of genetic testing. Laws and cultural expectations differ across countries, so these questions are addressed through regulation as well as individual and family decisions.

The story of IVF is not only a laboratory achievement. It reflects the experiences of people seeking to have children, the work of clinicians and embryologists, and changing ideas about reproduction and family. Its success expanded medical possibilities, while its limits remind us that a technical advance can offer hope without guaranteeing an outcome.

Timeline

1960s Robert Edwards develops laboratory methods for human egg maturation and fertilization, and begins collaborating with Patrick Steptoe.

1976 A pregnancy following IVF is reported, but it is ectopic.

1978 Louise Brown is born in England after IVF by Edwards, Steptoe, and Jean Purdy's team.

1980s Ovarian stimulation, ultrasound-guided egg retrieval, and embryo freezing broaden treatment options.

1992 Successful pregnancies using intracytoplasmic sperm injection are reported.

2010 Robert Edwards receives the Nobel Prize in Physiology or Medicine for the development of human IVF.

Today IVF includes laboratory fertilization, embryo transfer, freezing, and related techniques tailored to different circumstances.

Chapter 83

Hormonal Contraception

Hormonal contraception made it possible to prevent pregnancy through medicines that alter the hormonal signals governing ovulation and reproduction. The development of the oral contraceptive pill joined reproductive physiology, clinical research, pharmaceutical manufacturing, and public debate about women's autonomy and family planning.

83.1 From hormone research to the pill

By the mid-twentieth century, researchers understood that ovarian hormones could suppress ovulation. Gregory Pincus, John Rock, Margaret Sanger, and other researchers helped advance clinical studies of hormonal contraception, while activists and funders shaped the project and the public context in which it developed. In 1960, the US Food and Drug Administration approved Enovid for use as an oral contraceptive, after it had earlier been marketed for menstrual disorders [208].

The pill offered a highly effective method that did not require an action at the time of intercourse. Its introduction changed clinical practice and social life, while also raising questions about consent, side effects, medical oversight, and access. Early formulations contained higher hormone doses than later products; evidence of cardiovascular risks led to dose reductions and changes in prescribing information.

83.2 A broader set of methods

Hormonal contraception expanded beyond the pill to include injections, implants, patches, vaginal rings, and hormonal intrauterine devices. These methods use different delivery schedules and formulations, but all rely on reproductive hormones to prevent pregnancy.

Their history reflects continuing efforts to improve convenience, effectiveness, safety, and reproductive choice.

The contraceptive pill became a landmark not only because of its pharmacology. It made reproductive planning more controllable for many people and altered the relationship between medicine, sexuality, and public policy. Access and decision-making remain influenced by law, cost, gender, culture, and health systems.

The pill's historical significance extended beyond pharmacology: it gave many people a reliable way to separate sexual activity from the immediate risk of pregnancy and helped reshape family planning. Its development was shaped by researchers, advocates, clinicians, and trial participants, while questions about consent, side effects, and unequal access accompanied adoption. FDA approval was one milestone in a broader history shaped by advocates, researchers, clinicians, and trial participants [208].

Timeline

1957 Enovid is marketed in the United States for menstrual disorders.

1960 FDA approves Enovid as an oral contraceptive.

1960s–1970s
Lower-dose pills and additional hormonal methods are introduced.

Late twentieth century
Implants, patches, rings, and hormonal intrauterine devices expand options.

Chapter 84

Condoms

Condoms occupy a distinct place in the history of contraception. They can reduce the chance of pregnancy and also reduce transmission of many sexually transmitted infections, including HIV. Unlike a pill or an intrauterine device, a condom is used at the time of sex; its reach depends on access, consistent use, and whether partners can agree to use it.

84.1 From handmade sheaths to mass production

Claims about condom use in the ancient world are difficult to verify. A sixteenth-century European medical text by Gabriele Falloppio described a linen sheath intended to help prevent syphilis. Animal-membrane sheaths were also used in later centuries. These early products varied in construction and were far removed from the standardized disposable products used today [209].

Vulcanized rubber made it possible to manufacture more durable condoms in the nineteenth century, although early rubber products were thick and often reusable. Latex manufacturing in the 1930s enabled thinner, disposable condoms that could be produced at scale. Later material and design changes, including internal condoms, broadened the available options [209, 210].

84.2 Two public-health roles

Condoms became more than a method for spacing or preventing pregnancies. Correct and consistent use reduces the risk of HIV and many other sexually transmitted infections as well as unintended pregnancy. During the HIV/AIDS epidemic, condom promotion and distribution became important parts of public health prevention programs, alongside testing,

treatment, and other measures.

A modeling study estimated that increased condom use from 1990 through 2019 averted about 117 million new HIV infections worldwide. This is a modeled comparison with a scenario in which condom use did not increase, not a direct count of individually identified infections. It illustrates how a simple barrier method, when adopted across populations, can change the course of an epidemic [212, 210].

84.3 Reach, agency, and limits

Condoms are relatively inexpensive, do not require a medical procedure, and can be distributed through community programs as well as health services. Internal condoms offer a barrier method that can be inserted into the vagina. Access still varies with reliable supply, affordability, education, stigma, and the power to negotiate use. These social conditions help determine whether an effective method becomes an effective public-health program [210, 211].

Condoms reduce risk but do not eliminate it. Their protection is strongest when they are used correctly for every act of sex; they provide less protection against infections that can spread from skin not covered by the condom. Their history therefore includes both a major population benefit and the limits of relying on a method that requires consistent use and access [210, 211].

Timeline

Sixteenth century
: Falloppio describes a linen sheath intended to help prevent syphilis.

1840s–1850s
: Vulcanized rubber enables more durable manufactured condoms.

1930s Latex manufacturing makes thinner, disposable condoms practical at scale.

1980s onward
: Condoms become a central tool in many HIV-prevention programs.

Chapter 85

Vaginal, Assisted, and Caesarean Delivery

Birth can occur through vaginal delivery, assisted vaginal delivery with forceps or a vacuum device, or caesarean section. These methods developed to meet different needs during labor. Safer childbirth also depended on anesthesia, infection control, blood transfusion, monitoring, and newborn care. The appropriate approach depends on the circumstances and the pregnant patient's informed preferences.

85.1 Assisted vaginal delivery

Forceps were developed in the early seventeenth century and changed over time to help guide the baby's head through the birth canal in selected difficult births. Vacuum extraction was described centuries ago, but a practical modern vacuum extractor emerged in the 1950s through the work of Swedish obstetrician Tage Malmström. Forceps and vacuum devices can help complete a vaginal birth when labor needs to be expedited, but each has specific indications and risks. Safe use requires an appropriately positioned baby, skilled clinicians, and the ability to respond if the attempt does not succeed [321, 322].

85.2 Making caesarean delivery safer

Caesarean delivery is an operation to deliver a baby through incisions in the abdomen and uterus. For centuries, it was associated with catastrophic maternal outcomes. Early operations often followed prolonged labor, when infection and blood loss had already become severe. Anesthesia in the nineteenth century reduced suffering, while antiseptic technique

lowered the risk of surgical infection. Surgeons refined uterine incisions and methods for controlling bleeding. By the mid-twentieth century, antibiotics and reliable access to banked blood further reduced mortality and morbidity [213, 214].

Improvements in obstetric monitoring, surgical instruments, and newborn resuscitation added to safety. Caesarean delivery can be essential when vaginal birth poses serious risks to the mother or baby. Safe use requires appropriate surgical, anesthesia, blood-bank, and newborn support. Its rise also makes thoughtful use and equitable access to emergency surgery important public-health concerns.

85.3 Choosing a route of birth

Vaginal, assisted vaginal, and caesarean births are not successive steps in a simple march of progress. They are different approaches with different benefits and risks. A caesarean can save the life of a pregnant patient or baby, while an unnecessary or poorly supported operation can cause harm. Assisted vaginal birth may avoid surgery in selected situations, but it also requires expertise and careful follow-up. Safer childbirth grew from a system of skilled attendants, surgical and anesthesia teams, infection control, blood services, and newborn care rather than from one procedure alone [213, 214, 321].

Timeline

Early seventeenth century
: The Chamberlen family develops obstetric forceps.

1847 Anesthesia is first used in obstetric surgery.

1880s Antisepsis enters surgical practice.

Early twentieth century
: Improved uterine incisions reduce operative risk.

1950s Tage Malmström's vacuum extractor helps establish modern vacuum-assisted birth.

After 1945
: Antibiotics and blood transfusion contribute to safer caesarean delivery.

Chapter 86

Prenatal Ultrasound

Prenatal ultrasound uses sound waves to create images of the developing fetus and the pregnant uterus. It gave clinicians a way to observe pregnancy without ionizing radiation or surgery. Ultrasound made fetal growth, position, and some structural abnormalities visible before birth, changing obstetric diagnosis and the experience of pregnancy.

86.1 From sonar to obstetric imaging

In the 1950s, Scottish obstetrician Ian Donald worked with engineer Tom Brown and colleagues to adapt pulsed ultrasound for medical imaging. Their 1958 Lancet paper described using ultrasound to examine abdominal masses, including pregnancy-related structures. As equipment improved, clinicians developed ways to measure fetal growth and assess development [215].

A key step came in 1972, when researchers reported prenatal ultrasound diagnosis of anencephaly. Real-time imaging later allowed observation of fetal movement and heart activity. Doppler techniques added information about blood flow, while improvements in resolution expanded the range of visible structures.

86.2 Information and uncertainty

Prenatal ultrasound supports dating, assessment of fetal growth, and identification of some abnormalities, but it cannot detect every condition. Its images can reassure, raise uncertain findings, or prompt further testing. The development of prenatal imaging therefore required both technical improvements and counseling about what a scan can and cannot show.

The technology changed pregnancy care by making the fetus visible as a patient for obser-

vation and diagnosis. Its use continues to raise questions about access, privacy, interpretation, and the emotional meaning of seeing a pregnancy on a screen.

Timeline

1958 Ian Donald and colleagues publish an early clinical account of pulsed ultrasound for abdominal masses.

1960s Ultrasound begins to be used to assess fetal measurements and development.

1972 Prenatal diagnosis of a fetal abnormality by ultrasound is reported.

Late twentieth century
Real-time and Doppler imaging become routine parts of obstetric practice.

Chapter 87

Neonatal Intensive Care

Neonatal intensive care brings specialized staff, monitoring, and technology together to support newborns who are premature or seriously ill. Before the development of neonatal units, care for fragile infants was often scattered across general nurseries. Specialized units made it possible to coordinate temperature control, breathing support, nutrition, infection prevention, and family care.

87.1 A new specialty takes shape

Incubators and organized premature-infant care developed in the late nineteenth and early twentieth centuries. By the mid-twentieth century, advances in respiratory support and monitoring made intensive care for newborns increasingly possible. Yale neonatologist Louis Gluck established an early US neonatal intensive care unit in 1960, helping create a model for concentrated, specialized newborn care [216, 217].

The NICU brought together pediatricians, nurses, respiratory therapists, laboratory staff, and social workers. As survival improved for smaller and sicker infants, neonatology developed as a specialty and hospitals built systems for neonatal transport, ventilation, nutrition, and infection control.

87.2 Technology and family care

Modern neonatal care uses incubators, monitors, ventilators, imaging, and specialized nutrition. The same intensive environment that supports fragile infants can separate families and create stressful decisions. Family-centered care and skin-to-skin contact became important alongside technological support.

The NICU is not a single machine but a coordinated system. Its history shows how concentrated expertise and technology changed survival, while also making communication with families and attention to long-term development essential parts of care.

Organized newborn intensive care brought temperature control, respiratory support, infection management, and continuous observation together for infants who previously had few options. Survival improved as these measures and obstetric care advanced, especially for premature infants, though survival does not guarantee absence of disability. Family support and long-term follow-up are part of the history of care [216, 217].

Timeline

Late 1800s
: Incubators and specialized premature-infant care are introduced.

1960
: Louis Gluck establishes an early neonatal intensive care unit at Yale.

1960s–1970s
: Neonatal units spread as respiratory support and monitoring improve.

1975
: Neonatal-perinatal medicine becomes a recognized pediatric subspecialty in the United States.

Chapter 88

Surfactant Therapy for Premature Infants

Many premature infants are born before their lungs have made enough surfactant, a substance that helps keep the tiny air sacs open. Without it, the lungs are stiff and breathing can become severely difficult. Respiratory distress syndrome was once a major cause of death among premature newborns.

88.1 Replacing a missing lung substance

In 1959, Mary Ellen Avery and Jere Mead linked the syndrome in premature infants to a lack of pulmonary surfactant. After years of laboratory work, Tetsuro Fujiwara and colleagues reported in 1980 that surfactant extracted from bovine lungs and given to premature infants could improve respiratory distress. This first successful clinical treatment helped launch a new era in neonatal medicine [218, 219].

Clinical trials during the 1980s and 1990s established the value of surfactant replacement. Commercial products became available, and treatment was incorporated into neonatal intensive care alongside respiratory support. Different formulations and dosing approaches were developed as researchers studied when and how best to give the substance.

88.2 Changing survival

Surfactant therapy reduced the severity of respiratory distress and improved survival for many premature infants. It did not remove all risks of prematurity, and outcomes still depended on gestational age, other illnesses, ventilation strategies, and the resources of the neonatal unit.

The treatment illustrates a direct path from physiology to therapy: identifying a missing substance led researchers to replace it. Its success depended on basic science, careful clinical trials, pharmaceutical preparation, and the intensive-care teams able to deliver it.

Because surfactant is most useful when given early in respiratory distress, its benefit depends on rapid recognition, trained neonatal staff, and a reliable supply of the treatment. These requirements make access to a capable newborn unit part of the therapy itself; the drug cannot substitute for respiratory support and broader care of premature infants [218, 219].

Timeline

1959 Surfactant deficiency is linked to respiratory distress syndrome in premature infants.

1980 Fujiwara and colleagues report successful surfactant treatment in newborns.

1980s–1990s Controlled trials establish surfactant replacement in neonatal care.

1990 A synthetic surfactant product becomes available in the United States.

Chapter 89

Rh(D) Immune Globulin

Rh(D) immune globulin is an antibody preparation that prevents an Rh-negative person from becoming sensitized to Rh-positive red blood cells. In pregnancy, fetal red cells can enter the maternal circulation. If the mother forms anti-D antibodies, a later Rh-positive fetus may develop hemolytic disease, in which maternal antibodies destroy fetal red cells.

89.1 Preventing an immune response

Before preventive treatment, Rh incompatibility could cause severe newborn anemia, jaundice, brain injury, or death. Researchers recognized that giving anti-D antibodies could clear fetal red cells from the maternal circulation before the mother's immune system formed its own response. Rh immune globulin was licensed in the United States in 1968, transforming prevention of Rh hemolytic disease [220, 221].

The intervention became part of prenatal and postpartum care for eligible Rh-negative mothers. Its success depended on identifying blood type, preventing sensitization after relevant pregnancies or exposures, and maintaining access to a reliably manufactured plasma-derived product.

89.2 A preventive antibody

Rh immune globulin is an example of passive immunization: ready-made antibodies provide temporary protection by preventing the recipient's immune system from becoming primed against the target. Unlike a vaccine, it does not train the body to make long-term immunity. The distinction between active and passive immunity is central to understanding why timing and follow-up matter in prevention programs.

The use of Rh immune globulin dramatically reduced a once-common cause of fetal and newborn illness. Its history connects blood-group science, immunology, obstetric care, and preventive medicine.

Routine prevention during pregnancy and after delivery sharply reduced sensitization and the severe fetal and newborn disease that could follow in later pregnancies. This is a preventive treatment whose success often goes unnoticed because the harm it averts does not occur. Correct identification of Rh status and timely administration are essential; the medicine is given when a pregnant patient is at risk of Rh sensitization, not as a universal treatment [220, 221].

Timeline

1940 The Rh blood-group system is described.

1950s–1960s
Researchers develop anti-D preparations to prevent maternal sensitization.

1968 Rh(D) immune globulin is licensed in the United States.

Late twentieth century
Prevention becomes routine in obstetric care.

Chapter 90

Folic Acid and Neural-Tube Defect Prevention

Neural-tube defects such as spina bifida and anencephaly develop very early in pregnancy, often before a person knows they are pregnant. In the late twentieth century, research showed that folic acid, a B vitamin, could prevent many of these defects when adequate amounts were present around conception.

90.1 A prevention trial changes practice

Observational studies suggested a relationship between folate intake and neural-tube defects. The British Medical Research Council's randomized trial, published in 1991, showed that folic acid supplementation before conception and during early pregnancy substantially reduced recurrence among women with a previous affected pregnancy. A Hungarian trial also found a protective effect against first occurrences [222, 223].

In 1992, the US Public Health Service recommended that women capable of becoming pregnant consume folic acid to reduce neural-tube defect risk. Because neural-tube closure occurs early, public-health measures extended beyond prenatal care. Food fortification later made folic acid available through enriched grain products and helped reduce the population burden [222].

90.2 From clinical evidence to food policy

Folic acid prevention illustrates how a trial finding can become a population intervention. Supplements and food fortification reach people who may not be planning pregnancy, while

education helps explain the timing of the benefit. Fortification policy has required coordination among researchers, food producers, and public-health agencies.

The history also shows the limits of prevention: folic acid does not prevent every neural-tube defect, and risk has multiple causes. Still, the discovery created one of the clearest examples of a vitamin intervention reducing serious birth defects at a population scale.

Timeline

1960s Studies begin to examine folate and neural-tube defect risk.

1991 The MRC randomized trial demonstrates prevention of recurrent neural-tube defects.

1992 US Public Health Service recommends folic acid for women capable of pregnancy.

1998 Folic acid fortification of enriched grain products begins in the United States.

Chapter 91

Vitamin C Deficiency and Scurvy

91.1 A new kind of medical history

Vitamins are not a single disease but a group of essential organic nutrients. Their history belongs in a history of disease because the concept emerged from attempts to explain and prevent illnesses such as scurvy, beriberi, rickets, pellagra, and night blindness. These conditions were once described separately; the discovery of vitamins connected them through a common cause: the absence of small but necessary substances in food.

91.2 Practical remedies before nutritional science

People had long observed that particular foods could prevent illness even when they did not know why. Sailors learned that fresh fruits and vegetables could protect against scurvy. In the eighteenth century, James Lind's controlled comparison of treatments aboard HMS *Salisbury* showed that oranges and lemons were remarkably effective against the disease [7]. Adoption was slow, however. Naval policy, food preservation, cost, and competing theories all affected whether an observation became routine practice.

Rickets was associated with poor diet and lack of sunlight, while beriberi was common in populations whose staple diet relied heavily on polished rice. Pellagra occurred especially where poverty and restricted diets limited access to varied foods. Physicians often interpreted these illnesses through the dominant theories of their time, including infection, toxins, and an imbalance of the known macronutrients.

91.3 The accessory food factors, 1880s–1910s

Nineteenth-century nutrition science focused on proteins, fats, carbohydrates, and mineral salts. Experiments by Nikolai Lunin and others showed that animals could not thrive on purified mixtures of these apparently complete ingredients; something present in natural foods was missing. In 1906, Frederick Gowland Hopkins described these substances as “accessory food factors,” emphasizing that they were required in very small amounts but were not sources of energy [8].

Research on beriberi provided especially strong evidence. Christiaan Eijkman observed that chickens developed a beriberi-like illness when fed polished rice but improved when given rice bran. He initially believed the bran neutralized a toxin or corrected a dietary error, but later work showed that the missing factor was essential to normal nerve function. Eijkman and Hopkins shared the 1929 Nobel Prize for work that established the importance of vitamins [9].

91.4 Funk and the word “vitamin”

In 1912, the Polish chemist Casimir Funk proposed that several deficiency diseases were caused by substances he called “vital amines,” shortened to “vitamines” and later to “vitamins.” The name became influential even though not all of the substances were chemically amines. Funk’s terminology gave a wide-ranging research program a memorable identity, but the scientific achievement was collective: physicians, epidemiologists, nutrition scientists, laboratory workers, and affected communities all contributed evidence [10].

The alphabetic system developed as researchers separated different factors. Vitamin A was linked to growth and protection against xerophthalmia; vitamin B initially referred to a group later divided into several vitamins; vitamin C was associated with protection from scurvy; and vitamin D became central to the prevention of rickets. The letters record the order and circumstances of discovery more than a complete theory of human nutrition.

91.5 Isolation, chemistry, and the defeat of deficiency diseases

During the 1920s and 1930s, researchers isolated and chemically characterized several vitamins. Albert Szent-Györgyi and colleagues clarified the chemistry of vitamin C, while work on vitamin D connected dietary substances with the effects of ultraviolet light and cod-liver oil. Goldberger’s investigations of pellagra, followed by the identification of niacin as the

protective factor, showed that a disease affecting poor communities could be a consequence of food systems and social conditions rather than contagion.

Chemistry changed prevention. Once vitamins could be concentrated or synthesized, governments and manufacturers could add them to staple foods. Iodized salt, vitamin D-enriched milk, and enrichment of flour with B vitamins helped reduce deficiency disease on a population scale. Fortification was not merely a laboratory success: it depended on manufacturing, regulation, public trust, and decisions about which communities and foods would be prioritized.

91.6 Supplements, public health, and changing meanings

By the mid-twentieth century, the major vitamins had been identified and many could be produced industrially. Vitamins moved from hospital wards and nutrition laboratories into household advice, advertising, and the expanding market for dietary supplements. This widened access to useful preventive products but also created opportunities for exaggerated claims.

The medical history of vitamins therefore includes both deficiency and excess. A nutrient can be essential at one dose and harmful at another, particularly with concentrated fat-soluble vitamins. Modern nutrition guidance distinguishes between treating a diagnosed deficiency, meeting ordinary dietary requirements, and taking supplements without a clear indication. In the United States, the regulatory history of supplements developed separately from the history of prescription medicines; the FDA does not establish a supplement's effectiveness before it is marketed in the same way it evaluates a new drug [11].

91.7 The continuing history

Vitamin research transformed the understanding of disease by demonstrating that health could depend on substances present in quantities too small to see or measure with early methods. It also revealed that biology and social structure cannot be separated: famine, poverty, restricted diets, food processing, fortification policy, and access to sunlight all shaped who became ill.

Today, severe deficiency diseases are less common in many regions, yet they have not disappeared. They persist where food insecurity, displacement, restrictive diets, malabsorption, chronic illness, or inadequate public-health programs limit access to essential nutrients. The

history of vitamins remains a reminder that a medical discovery is most successful when it is translated into safe, affordable, and culturally workable food practices.

91.8 How the evidence changed clinical reasoning

The vitamin story also explains why a familiar clinical pattern was not enough to establish a cause. Scurvy had recognizable signs and responded to citrus, but sailors could improve for several different reasons: a change in water, a change in work, a seasonal improvement, or the specific food itself. Lind's comparison mattered because it placed several treatments beside one another under similar conditions. The experiment did not identify vitamin C, but it made a nutritional explanation harder to dismiss.

The later history followed the same logic. In beriberi, rice polishing created a natural comparison between diets, while animal experiments made it possible to control food ingredients. In pellagra, Joseph Goldberger challenged the idea of an infectious agent by studying household and institutional patterns and by showing that dietary change could prevent illness. Critics demanded a cultured organism; Goldberger's evidence instead showed why the infection hypothesis made poor predictions. The eventual identification of niacin strengthened the case, but the public-health lesson was already visible: a disease could be caused by what a population lacked rather than by what invaded it.

The laboratory then added a new level of precision. Researchers separated extracts, tested their activity in animals, measured their chemical properties, and eventually synthesized the active compounds. Each procedure corrected a different weakness in earlier reasoning. Clinical observation identified the syndrome; dietary comparison suggested the missing factor; animal models demonstrated a reproducible effect; and chemistry made the factor measurable. No single method could have carried the whole argument.

This sequence is why the word "discovery" can be misleading. The history was not a race in which one person suddenly found a capsule hidden in food. It was a long process of converting practical knowledge into a causal account. It also depended on people whose suffering made the pattern visible: sailors, prisoners, children with rickets, patients with anemia, and families living with restricted diets were not merely background to the laboratory story. Their experiences supplied the observations that researchers learned, sometimes very slowly, to interpret.

Finally, the history of vitamins warns against treating a successful theory as a universal prescription. A nutrient may prevent a deficiency disease without improving every symptom in a well-nourished person. Fortification may prevent population-level illness while requiring careful monitoring of dose and equity. Supplements may be useful for a diagnosed need

but misleading when advertising turns a deficiency-prevention discovery into a promise of general health. The same demand for comparison, mechanism, and outcome that established vitamins must continue to govern their use.

Timeline

1747 James Lind compares scurvy treatments and finds citrus fruit effective.

1897 Eijkman reports that polished rice is associated with a beriberi-like illness in chickens.

1906 Hopkins describes essential “accessory food factors.”

1912 Casimir Funk introduces the term “vitamine.”

1920 The spelling is shortened to “vitamin” as non-amine factors are recognized.

1929 Eijkman and Hopkins receive the Nobel Prize for foundational work on vitamins.

1930s–1940s Vitamins are isolated, synthesized, and increasingly used in food fortification.

1937 Niacin is identified as the factor that prevents pellagra.

Chapter 92

Corticosteroid Therapy

Corticosteroids are medicines modeled on hormones produced by the adrenal cortex. They can reduce inflammation and suppress immune activity, while other members of the family help regulate salt balance and stress responses. Their discovery created treatments for conditions that previously caused severe disability, but their broad effects also made careful use essential.

92.1 Cortisone reaches the clinic

In the 1930s and 1940s, Edward Kendall, Tadeus Reichstein, Philip Hench, and their collaborators isolated and studied adrenal hormones. Hench had observed that rheumatoid arthritis symptoms could improve during pregnancy and jaundice, and he hypothesized that a body substance might suppress inflammation. After cortisone became available for clinical testing, Mayo Clinic clinicians treated a patient with rheumatoid arthritis in September 1948 and saw dramatic improvement [224].

The discovery led to intense research and publicity. Cortisone and related glucocorticoids helped control inflammatory and autoimmune disease, but prolonged treatment could cause serious effects, including infection risk, bone loss, and metabolic changes. Clinicians learned to distinguish the dramatic short-term benefit from the hazards of sustained exposure.

92.2 A family of treatments

Synthetic corticosteroids were developed with different anti-inflammatory strength and duration. They became important in rheumatology, allergy, asthma, dermatology, transplantation, and many other fields. Local formulations such as inhaled, topical, and joint-injected steroids allowed treatment to be directed more narrowly in some settings.

The history of corticosteroid therapy is a story of therapeutic power and restraint. These medicines can rapidly change inflammation, but their effects reach many organs. Better formulations and an expanding evidence base helped clinicians balance benefit against harm.

Timeline

1930s–1940s
: Adrenal cortical hormones are isolated and characterized.

1948 Cortisone is first given to a patient with rheumatoid arthritis.

1950 Hench, Kendall, and Reichstein receive the Nobel Prize.

1950s onward
: Synthetic corticosteroids expand treatment across specialties.

Chapter 93

Modern Psychiatric Medications

Before the mid-twentieth century, psychiatric care relied heavily on institutional support, psychotherapy, sedation, and physical treatments. The introduction of medicines that changed psychosis, mood, and anxiety opened a new era in mental-health care. These drugs did not resolve the many causes of mental illness, but they offered new ways to reduce symptoms and support life outside institutions.

93.1 Lithium and chlorpromazine

In 1949, Australian psychiatrist John Cade reported that lithium salts could reduce manic excitement. A few years later, chlorpromazine, first developed in France in the context of anesthesia, was used in psychiatric patients. French psychiatrists Jean Delay and Pierre Deniker described its effects in 1952. Chlorpromazine could lessen agitation and psychotic symptoms, and its introduction stimulated the development of other psychotropic medicines [225, 226].

The 1950s also brought antidepressants, including imipramine and monoamine oxidase inhibitors. Over subsequent decades, research produced benzodiazepines, antipsychotics with different receptor profiles, and selective serotonin reuptake inhibitors. The medicines differed in their effects, benefits, and risks, so psychiatric treatment became more varied rather than centered on a single drug.

93.2 Treatment and social change

Psychiatric medicines changed hospital care and contributed to the movement toward community treatment. Their effects were neither uniformly beneficial nor sufficient by themselves:

side effects, unequal access, coercive use, and inadequate social support remained concerns. Medication could reduce symptoms, but recovery also depended on relationships, housing, rehabilitation, and access to continuing care.

Modern psychopharmacology has evolved through clinical observation, controlled trials, and research on brain signaling. Its history reminds us that a medicine's role depends on how it is prescribed, monitored, and combined with psychological and social support.

Medicines contributed to a broader shift toward community care, but deinstitutionalization also depended on policy, housing, and social services. Where those supports were absent, discharge could replace hospital confinement with isolation or unstable living conditions. Drug treatment changed options for many patients without making psychiatric care self-sufficient [225, 226].

Timeline

1949 John Cade publishes his report on lithium for manic excitement.

1952 Chlorpromazine is introduced into psychiatric treatment.

1950s Early antidepressants and additional antipsychotics become available.

1980s Selective serotonin reuptake inhibitors begin to reshape depression treatment.

Chapter 94

Levodopa for Parkinson Disease

Parkinson disease involves the loss of dopamine-producing neurons in the brain. Dopamine itself does not readily cross from the bloodstream into the brain, but its precursor, levodopa, can. The discovery that levodopa could improve movement transformed treatment for people with Parkinson disease.

94.1 From dopamine biology to clinical treatment

In the 1950s, Arvid Carlsson and colleagues showed that dopamine was a neurotransmitter and that dopamine depletion was linked to parkinsonian symptoms. In 1961, neurologist Oleh Hornykiewicz and collaborators reported that levodopa could improve symptoms in patients. George Cotzias and colleagues later demonstrated that gradually increasing oral doses could produce sustained improvements in a larger group of patients [227, 228].

Adding a peripheral dopa-decarboxylase inhibitor such as carbidopa reduced the amount of levodopa broken down outside the brain and helped limit some peripheral effects. Levodopa became a central treatment because it could improve slowness and rigidity, though responses could fluctuate and involuntary movements could emerge over time.

94.2 A major advance with limits

Levodopa does not stop the underlying loss of neurons, and it does not relieve every feature of Parkinson disease. Its effect can vary over the course of the illness, prompting development of other medicines, delivery methods, and surgical options. Yet levodopa remains one of the clearest examples of a treatment designed from a biochemical understanding of a brain disorder.

The history also shows how laboratory neuroscience can lead to a therapy that meaningfully changes daily function. The key insight was not simply finding a chemical difference, but identifying a precursor that could cross into the brain and restore signaling.

Timeline

1910–1913
: Levodopa is isolated from plant material.

1950s
: Dopamine is established as a neurotransmitter and linked to parkinsonism.

1961
: Early clinical reports describe benefit from levodopa.

1967–1969
: High-dose oral levodopa regimens demonstrate sustained benefit.

Chapter 95

Antiseizure Medications

For much of history, epilepsy was surrounded by fear and stigma, and effective long-term treatment was limited. Antiseizure medicines changed care by reducing the frequency and severity of seizures for many people. The first drugs were discovered by clinical observation; later advances increasingly came from laboratory testing and knowledge of how brain cells generate electrical signals.

95.1 From bromides to modern medicines

In 1857, Charles Locock reported using potassium bromide to reduce seizures. Phenobarbital's antiseizure effect was recognized in 1912, and phenytoin followed in 1938 after researchers deliberately tested related compounds in animal seizure models. These drugs were more useful than earlier sedatives for some seizure types, although they had significant side effects [229].

New medicines became available through the second half of the twentieth century, including carbamazepine, valproate, and ethosuximide. From the 1990s, additional drugs such as lamotrigine, gabapentin, and levetiracetam broadened options. Some treatments target particular seizure types or work through distinct mechanisms, and individual response can vary.

95.2 A more treatable condition

Antiseizure medicines do not cure every form of epilepsy, but many people achieve good seizure control. Treatment also expanded beyond medication to include surgery, dietary therapy, and implanted neurostimulation for selected cases. The development of drug therapy

helped change epilepsy from a condition commonly hidden or feared into a neurological disorder that can often be managed over time.

The story reflects a shift from chance discovery to systematic pharmacology. It also shows that a drug's usefulness depends on matching its effects and risks to the person's seizure pattern and life circumstances.

Timeline

1857 Potassium bromide is reported as a treatment for seizures.

1912 Phenobarbital's antiseizure effect is recognized.

1938 Phenytoin is introduced after systematic testing.

1960s onward
New drug classes expand treatment options.

Chapter 96

Stroke Thrombolysis and Mechanical Thrombectomy

Most strokes occur when a clot blocks blood flow to part of the brain. For decades, treatment was largely supportive because clinicians lacked a proven way to reopen an artery quickly without causing unacceptable bleeding. Brain imaging, organized stroke systems, and clinical trials eventually made time-sensitive reperfusion treatment possible.

96.1 Thrombolysis changes acute care

The 1995 National Institute of Neurological Disorders and Stroke trial showed that intravenous tissue plasminogen activator (tPA, alteplase) given within three hours of symptom onset improved the chance of a better outcome in selected patients with ischemic stroke. The result established that stroke could be a medical emergency for which rapid diagnosis and treatment might reduce disability [230].

Thrombolysis required urgent brain imaging to distinguish ischemic stroke from bleeding and to assess whether treatment was appropriate. Later studies extended the treatment window for selected patients using imaging and clinical criteria. Stroke teams, emergency transport, and rapid scanning became central parts of the treatment system.

96.2 Mechanical clot removal

For some severe ischemic strokes, a large artery remains blocked despite intravenous treatment or is unlikely to reopen with medicine alone. In 2015, several randomized trials demonstrated that endovascular thrombectomy, in which specialists retrieve a clot through a catheter,

improved outcomes for selected patients with large-vessel occlusion. The ESCAPE trial was among the studies that helped establish this approach [231].

Thrombectomy depends on coordinated imaging, expert teams, and rapid transfer to a capable center. Thrombolysis and thrombectomy are not suitable for every stroke, and time, anatomy, and imaging findings shape clinical decisions. Together, these advances changed stroke from a condition with few acute options into one where restoring circulation can be attempted in selected cases.

The landmark trials established that rapid reperfusion can preserve function in selected patients: thrombolysis for eligible patients within a time window, and thrombectomy for some large-vessel blockages. Brain imaging and organized transfer systems made patient selection and timely treatment possible. These advances do not apply to every stroke, and delays or unequal access can determine who benefits [230, 231].

Timeline

1995 The NINDS trial establishes benefit of early intravenous tPA for selected ischemic strokes.

1996 The United States approves alteplase for acute ischemic stroke.

2015 Randomized trials establish mechanical thrombectomy for selected large-vessel strokes.

2018 Imaging-based studies extend thrombectomy to some patients treated later after onset.

Chapter 97

Intensive-Care Medicine

Intensive-care medicine supports people whose breathing, circulation, or other vital functions are failing. It brings continuous monitoring, organ-support technologies, and teams of trained clinicians together in a specialized setting. Intensive care emerged from the recognition that patients with acute organ failure could sometimes recover if support was immediate and coordinated.

97.1 The Copenhagen polio epidemic

During the 1952 polio epidemic in Copenhagen, many patients developed respiratory paralysis, but few ventilators were available. Anesthesiologist Bjørn Ibsen proposed tracheostomy and positive-pressure ventilation for patients with respiratory failure. Medical and dental students manually ventilated patients in shifts, and survival improved. In 1953, Ibsen helped establish an intensive-care unit in Copenhagen, concentrating staff and resources around critically ill patients [232, 233].

Other centers developed shock wards and intensive-care units during the 1950s and 1960s. Mechanical ventilation, cardiac monitoring, intravenous fluids, renal support, and laboratory testing expanded the ability to manage multiple forms of organ failure.

97.2 A specialty and a system

Intensive care became a specialty that depends on teamwork across medicine, nursing, respiratory therapy, pharmacy, and many other fields. Its technology can sustain life during severe illness, but decisions about treatment goals, recovery, and end-of-life care remain central to practice.

The ICU's history is a reminder that machines alone do not create critical care. Success depends on continuous observation, skilled staff, rapid response, infection control, and communication with patients and families.

Intensive care developed by bringing continuous observation, trained teams, respiratory support, and organ monitoring into a dedicated setting. The polio epidemics accelerated this model, but modern ICU care grew through advances in ventilation, infection control, and cardiovascular support. Intensive treatment can rescue people from reversible organ failure, while delirium, weakness, and burdensome treatment remain important outcomes [232, 233].

Timeline

1952 The Copenhagen polio epidemic prompts round-the-clock positive-pressure ventilation.

1953 An early dedicated intensive-care unit opens in Copenhagen.

1960s ICUs spread and add continuous monitoring and organ support.

Late twentieth century
Intensive-care medicine develops as a distinct clinical specialty.

Chapter 98

Pulse Oximetry

Pulse oximetry estimates the oxygen saturation of arterial blood through a sensor placed on a finger, toe, or ear. It does so without drawing blood, using red and infrared light to detect changes in absorption caused by pulsating arterial blood. Before pulse oximetry, clinicians often relied on visible signs and intermittent blood samples to assess oxygenation.

98.1 Aoyagi's insight

Japanese engineer Takuo Aoyagi developed the principle of pulse oximetry in 1974 while working on a noninvasive cardiac-output monitor. He realized that the pulsating component of light absorption could distinguish arterial blood from the surrounding tissue and provide an estimate of oxygen saturation. Early devices were difficult to use, and commercialization took further engineering work [235, 236].

In the late 1970s and early 1980s, commercial pulse oximeters became more practical. Their use spread through operating rooms and intensive-care units, where continuous readings allowed clinicians to detect oxygen desaturation more quickly. The device later became common in emergency departments, hospital wards, and outpatient settings.

98.2 A new vital sign

Pulse oximetry made oxygen saturation visible as a continuous vital sign. It reduced reliance on repeated arterial blood sampling for routine monitoring, although blood-gas tests remained important when precise measurements were needed. Like any measurement, the reading must be interpreted alongside the patient's condition and the instrument's limitations.

The device transformed patient monitoring through a simple combination of optics, signal

processing, and engineering. Its widespread use shows how a compact monitor can change clinical practice when it provides useful information continuously and noninvasively.

Noninvasive oxygen measurement changed bedside monitoring because clinicians could follow oxygenation continuously rather than rely only on intermittent blood samples or visible signs such as skin color. It became routine in anesthesia, intensive care, and emergency medicine. Readings can be affected by poor circulation, motion, nail products, and skin pigmentation, so the number must be interpreted alongside the patient and other measurements. In 2025, the FDA proposed updated performance-testing recommendations intended to improve accuracy across skin tones [235, 236, 234].

Timeline

1930s–1940s
: Early oximeters use light absorption to estimate blood oxygen.

1974 Takuo Aoyagi presents the pulse-oximetry principle.

Late 1970s–1980s
: Practical commercial devices become available.

1980s onward
: Pulse oximetry spreads into routine anesthesia and critical-care monitoring.

Chapter 99

Polymerase Chain Reaction

Before the 1980s, studying a particular stretch of DNA often required large amounts of biological material and laborious cloning. The polymerase chain reaction (PCR) changed that scale. By repeatedly copying a selected DNA segment in a test tube, it made small samples useful for research, diagnosis, forensics, and public-health investigations.

99.1 A method for amplifying DNA

In 1983, biochemist Kary Mullis conceived a way to use short DNA primers and a heat-stable DNA polymerase to copy a chosen region. The reaction cycles through heating to separate the strands, cooling so primers bind, and warming to let polymerase extend them. Each round can approximately double the target sequence, so repeated cycles produce millions of copies. Mullis received half of the 1993 Nobel Prize in Chemistry for inventing PCR [237].

Early PCR required researchers to add fresh enzyme after each high-temperature step. The isolation of heat-tolerant Taq polymerase from a bacterium living in hot springs made the process easier to automate. Thermal cyclers could control temperature shifts and carry out many cycles consistently, transforming PCR from a laboratory procedure into a broadly usable tool [237, 238].

99.2 From research tool to clinical test

PCR can detect a sequence even when it is present in tiny quantities. In infectious-disease testing, a test may first convert viral RNA into DNA and then amplify a selected target. Inherited variants can be examined in the same way, while forensic laboratories can amplify DNA from small or degraded samples. PCR also supports cloning and DNA sequencing by

producing material for further analysis.

The method's sensitivity is a strength and a source of error: contamination with stray DNA can produce a misleading positive result. A PCR result also answers only the question built into the test. It does not by itself show whether an organism is alive, whether a genetic variant causes disease, or how severe an illness will be. PCR made molecular detection faster and more accessible, while increasing the importance of careful controls and interpretation.

PCR's medical impact grew as thermally stable DNA polymerase made repeated cycles practical and automated instruments reduced the labor involved. The method later became central to infectious-disease testing, inherited-disease analysis, and forensic identification. Its speed is powerful, but a positive result detects genetic material and must be interpreted in context; it does not by itself prove active disease or infectiousness [238, 237].

Timeline

1983 Kary Mullis conceives the PCR method.

1985 The first published demonstration applies PCR to a human gene.

Late 1980s
Heat-stable Taq polymerase and thermal cyclers make routine amplification practical.

1993 Mullis receives the Nobel Prize in Chemistry for PCR.

Chapter 100

DNA Sequencing

Knowing that DNA carries hereditary information was not enough: researchers also needed ways to read its sequence. DNA sequencing identifies the order of the four chemical bases in a DNA molecule. It made it possible to find genes, compare organisms, identify disease-associated variants, and assemble whole genomes.

100.1 Two methods open the field

In the 1970s, Frederick Sanger developed a chain-termination method that copies DNA while occasionally incorporating modified bases that stop extension. The resulting fragments differ in length by one base; separating them reveals the sequence. Allan Maxam and Walter Gilbert developed a different chemical-cleavage method around the same time. Sanger and Gilbert shared the 1980 Nobel Prize in Chemistry for contributions to determining base sequences in nucleic acids [239].

Sanger sequencing became especially useful after researchers adapted it to fluorescent labels, capillary instruments, and computer analysis. These changes enabled automated reading of many DNA fragments and supported the large international effort to sequence the human genome. The method remained reliable for targeted regions and for checking results from newer platforms.

100.2 What a sequence can tell us

Sequencing can reveal a change in a gene, but interpretation requires biological and clinical evidence. Some variants are harmless; others alter a protein or gene regulation and contribute to disease. A person's sequence also contains information about relatives, ancestry, and future

health risks, so testing raises questions about consent, privacy, and how uncertain results should be explained.

Sequencing has moved from reading short fragments one at a time to analyzing millions of fragments in parallel. Each technology has trade-offs in cost, speed, accuracy, and the kinds of changes it detects. The central achievement is that hereditary information became experimentally readable, allowing genetics to become a molecular science and a clinical tool.

Sanger sequencing made it possible to read relatively short DNA fragments accurately and became a foundation for identifying disease-causing variants and assembling reference genomes. The later increase in speed and scale came from new platforms, not from replacing the importance of reliable interpretation. A sequence result can raise questions about uncertain variants and should be paired with clinical and family information [239].

Timeline

1970s Sanger and Maxam–Gilbert methods establish practical DNA sequencing.

1977 Sanger's group publishes the sequence of bacteriophage phi X174, an early complete genome.

1980 Sanger and Gilbert share the Nobel Prize in Chemistry.

1980s–1990s
Fluorescent labels and automation expand sequencing throughput.

Chapter 101

Human Genome Project

The Human Genome Project was an international effort to determine a reference sequence for human DNA and build the tools needed to analyze it. Its scale was unprecedented: the project set out to read billions of DNA bases, map genes, improve sequencing technology, and make data broadly available to researchers.

101.1 An international project

Planning began in the 1980s, and the coordinated project formally started in 1990. Publicly funded laboratories in the United States and other countries worked alongside a private effort led by Celera Genomics. In 2001, the public consortium and Celera announced draft sequences. The international consortium declared the project complete on 14 April 2003, ahead of its original schedule [240, 241].

The project did more than produce a sequence. It improved mapping, sequencing, computing, and data-sharing methods, and it supported research on the genomes of model organisms. The reference gave scientists a shared coordinate system for locating genes and comparing genetic variants. A reference genome is a research scaffold, however, not a perfect sequence for every person; early references left gaps and represented only a limited sample of human diversity.

101.2 A foundation for genomic medicine

Researchers used the reference to identify genes involved in inherited conditions, study cancer mutations, and design genome-wide association studies. The project's technology-development and data-sharing approach also influenced later large-scale biological research. Subsequent

projects have improved the reference and expanded representation of global genetic diversity.

The Human Genome Project did not reveal a simple genetic blueprint that predicts each person's future. Most common diseases reflect many genes, environment, and chance. Its lasting contribution was to make a common genomic reference and the tools for studying it available, enabling many follow-on discoveries rather than providing a final explanation of human biology [242].

The project produced a shared reference that accelerated gene discovery and made large-scale genomic research possible. It did not deliver a complete explanation of disease or an immediate cure: many conditions reflect multiple genes, environment, and chance. Its lasting patient impact has come through the diagnostic and research tools built on the reference, alongside continuing questions about privacy and representation [240, 242].

Timeline

1990 The international Human Genome Project begins.

2001 The public consortium and Celera publish draft human genome sequences.

2003 The consortium announces completion of the project.

2004 A detailed description of the finished sequence is published.

Chapter 102

Next-Generation Sequencing

Traditional Sanger sequencing reads DNA fragments one reaction at a time. Next-generation sequencing (NGS) changed the economics of the task by reading millions of fragments in parallel. The result was a dramatic fall in the cost per base and a rapid expansion of genetic studies and clinical testing.

102.1 Parallel sequencing

The first commercial next-generation platform appeared in 2005. Instead of tracking a single DNA chain, early systems attached many DNA fragments to a surface or beads and recorded signals as bases were added or detected. Different platforms used different chemistries, but their shared principle was massively parallel measurement [243].

A sequencing run produces short reads that computers align to a reference or assemble into longer sequences. Laboratories can examine selected genes, whole exomes (the protein-coding portions of genes), or nearly all genomic DNA. Parallel sequencing enabled large population studies, rapid analysis of tumors, and the sequencing of pathogens during outbreaks.

102.2 Speed, scale, and interpretation

More data do not automatically produce a diagnosis. Short reads can be hard to place in repetitive regions, and some structural changes require other methods. Laboratories must check quality, distinguish genuine variants from technical errors, and interpret findings using clinical context. Results can also reveal unexpected information about relatives or disease risks.

NGS made comprehensive sequencing feasible for many more laboratories and patients.

Its clinical value depends on the question being asked, the quality of the analysis, and access to genetic counseling and follow-up.

Massively parallel sequencing lowered the cost and time required to examine many genes or whole genomes, enabling applications such as tumor profiling and diagnosis of rare inherited disorders. More data do not always yield a clear answer: variants may be uncertain, incidental findings can emerge, and interpretation requires expertise. The breakthrough is therefore both a laboratory technology and a new clinical information challenge [243].

Timeline

2005 The first commercial next-generation sequencing system reaches the market.

Late 2000s
Sequencing costs fall and whole-genome studies become more practical.

2010s Clinical laboratories increasingly use panels, exomes, and genomes in diagnosis.

Chapter 103

Genetic Diagnosis

For many families, an inherited disorder once remained unnamed after years of symptoms and testing. Genetic diagnosis uses DNA analysis to identify a variant that explains a condition or changes a person's medical care. It can end a diagnostic search, guide treatment, and help relatives understand their own risks.

103.1 From single genes to exomes and genomes

Early molecular diagnosis focused on one suspected gene at a time. As sequencing became faster and less expensive, clinicians began using panels that examine many genes linked to a symptom, followed by exome or genome sequencing when narrower tests do not find an answer. Exome and genome sequencing can identify variants missed by single-gene testing, particularly when a condition has many possible genetic causes [244, 245].

The test must match the clinical question. A pathogenic variant can confirm a diagnosis, while a negative result does not rule out a genetic cause: the variant may lie outside the regions examined or remain beyond current knowledge. A variant of uncertain significance should not be treated as a confirmed cause. Family testing can help clarify whether a variant tracks with disease, but interpretation may change as evidence accumulates.

103.2 Clinical use and family implications

Genetic findings can support surveillance, treatment selection, reproductive planning, and cascade testing of relatives. Before testing, patients should understand what the test can and cannot answer and what kinds of secondary findings may be reported. A genetic result can carry implications for family members, so clinicians balance useful information with privacy

and the patient's preferences.

Sequencing has made diagnosis more precise, but it has not removed the need for clinical judgment. The result becomes useful when a laboratory finding is interpreted alongside a person's history, examination, and family pattern.

Genetic testing can end a long search for a diagnosis, guide treatment or surveillance, and clarify risks for relatives. The same result may affect family members who did not seek testing, and uncertain findings can create difficult decisions. Counseling, consent, data privacy, and access to confirmatory care are therefore part of the clinical advance, not optional additions [244, 245].

Timeline

1980s onward DNA tests begin identifying the molecular causes of inherited disorders.

2000s Microarrays and gene panels broaden diagnostic testing.

2010s Exome and genome sequencing enter clinical practice for selected patients.

Chapter 104

CRISPR-Cas Gene Editing

CRISPR-Cas systems began as part of bacterial defense against viruses. In bacteria, short stored DNA sequences help guide enzymes to matching genetic material. Researchers adapted this natural system into a programmable way to cut DNA at selected locations, creating a new approach to laboratory genetics and potential treatment.

104.1 From bacterial defense to genome editing

In 2012, Jennifer Doudna and Emmanuelle Charpentier showed that a CRISPR-Cas9 system could be programmed with a guide RNA to cut a chosen DNA sequence. Cells repair the cut, sometimes disabling the gene or using a supplied template to make a planned change. The method was easier to retarget than many earlier genome-editing tools. Doudna and Charpentier received the 2020 Nobel Prize in Chemistry for developing a method for genome editing [246].

CRISPR is a family of systems, and Cas9 is only one enzyme. Researchers have also developed base editors and other tools that can alter DNA without making the same double-strand cut. Each approach has distinct limits, including unintended edits, delivery into the right cells, and difficulty reaching enough cells in the body.

104.2 Promise and responsibility

Editing blood-forming cells outside the body can allow clinicians to test and select modified cells before returning them to a patient. Editing cells inside the body is more difficult because the editor must reach the right tissue safely. Changes to embryos could be inherited by future generations; that raises broad ethical questions and differs from treatment of a patient who

can consent.

CRISPR made targeted editing more accessible, but a precise design does not guarantee a safe or effective therapy. Medical use requires evidence about off-target changes, long-term effects, delivery, and benefit to patients.

In December 2023, the FDA approved Casgevy, the first CRISPR-Cas9 therapy approved in the United States, for selected people with sickle-cell disease. It edits a patient's blood-forming stem cells outside the body to increase fetal hemoglobin. The cells are returned after intensive conditioning chemotherapy. This is a somatic, ex-vivo treatment, not editing every cell in the body or changing inherited DNA. It demonstrates a real medical use while also showing how specialized delivery and conditioning still constrain reach; the treatment's history and patient outcomes are described in the sickle-cell gene-therapy chapter [249, 251].

Timeline

1980s Repeated CRISPR DNA sequences are described in bacteria.

2012 Researchers demonstrate programmable CRISPR-Cas9 genome editing.

2020 Charpentier and Doudna receive the Nobel Prize in Chemistry.

2023 The first CRISPR-based treatment is authorized by the US FDA for sickle-cell disease.

Chapter 105

Gene Therapy

Gene therapy aims to treat disease by adding, replacing, silencing, or editing genetic material in a patient's cells. The idea is powerful because some illnesses arise from a specific genetic defect, but the path from concept to dependable treatment required decades of work on delivery, safety, and clinical evidence.

105.1 Early trials and hard lessons

The first human gene-therapy trial began in 1990 for adenosine deaminase deficiency, a cause of severe combined immunodeficiency. Researchers modified a patient's own immune cells to carry a working ADA gene and returned them to the body. Early approaches showed that gene transfer was possible, though their effects could be limited or temporary [247, 248].

Some later studies exposed serious risks. Insertion of genetic material can disrupt normal cell controls, and immune reactions to delivery vectors can cause harm. A 1999 death in a gene-therapy trial intensified scrutiny and led to stricter attention to trial design, consent, monitoring, and reporting. Researchers developed improved viral vectors, nonviral delivery methods, and strategies that target long-lived stem cells.

105.2 Treating cells and tissues

In ex vivo therapy, clinicians remove cells, modify them in a laboratory, check them, and return them to the patient. In vivo therapy delivers genetic material directly into the body, often using a viral vector. Some treatments add a working gene; others edit a sequence or regulate gene activity. The approach depends on the disease, target tissue, and durability needed.

Gene therapies can be transformative for particular diseases, but they are complex products with demanding manufacturing and follow-up. Their history shows how a promising mechanism becomes a treatment only through careful clinical testing and long-term observation.

Clinical successes have been disease-specific: gene transfer has helped address particular inherited immune and blood disorders, while other conditions remain difficult to treat because the target cells are hard to reach or the effect may fade. Long-term follow-up is needed to establish durability and detect delayed harms [247, 248].

Timeline

1990 The first clinical gene-therapy trial begins for ADA-SCID.

1999 A trial-related death prompts renewed safety scrutiny.

2010s Several gene therapies reach regulatory approval in different countries.

2020s Genome editing joins gene addition as a clinical treatment strategy.

Chapter 106

Sickle-Cell Gene Therapy

Sickle-cell disease results from a change in the gene for hemoglobin. The abnormal hemoglobin can make red blood cells rigid and sickle-shaped, causing painful crises, anemia, and damage to organs. For generations, care focused on preventing complications and treating episodes. Gene-based therapies now offer some patients a way to alter their own blood-forming stem cells.

106.1 Two strategies for changing blood cells

In December 2023, the US Food and Drug Administration approved two cell-based gene therapies for eligible people aged 12 years and older. Casgevy uses CRISPR-Cas9 to edit a regulatory region of the BCL11A gene in a patient's blood stem cells. This increases fetal hemoglobin, which can reduce the sickling of red cells. Lyfgenia uses a lentiviral vector to add a modified beta-globin gene that produces an anti-sickling hemoglobin. Casgevy was the first FDA-approved therapy using CRISPR-Cas9; both were the first approved gene therapies for sickle-cell disease in the United States [249, 250].

For either treatment, stem cells are collected from the patient and modified outside the body. Before infusion, patients receive intensive chemotherapy to clear space in the bone marrow. The process requires specialized centers, hospital care, and careful follow-up. The conditioning treatment itself can cause serious complications, and long-term durability and late effects continue to be monitored.

106.2 Benefits and limits of the trial evidence

The pivotal studies found strong reductions in severe pain crises among participants who could be evaluated, but they were single-arm studies rather than head-to-head comparisons. In the FDA's 2023 Casgevy review, 29 of 31 evaluable participants (93.5%) had no severe vaso-occlusive crisis for at least 12 consecutive months within the 24-month follow-up period. For Lyfgenia, 28 of 32 participants (88%) had complete resolution of vaso-occlusive events during the specified six-to-18-month period. These outcomes use different definitions and follow-up windows, so the percentages should not be read as a direct comparison. They demonstrate substantial benefit for many trial participants, while longer follow-up is needed to establish durability and late effects [250, 249].

Access has begun to widen: in July 2026, the FDA expanded Casgevy's US approval to children aged 2 years and older with recurrent crises. Lyfgenia's US approval remains for eligible patients aged 12 years and older. The pediatric evidence included a small trial, and the younger-age indication was extended using product and clinical data; long-term outcomes still require follow-up [251].

106.3 What stands between a breakthrough and broad reach

Both approaches require collecting a patient's stem cells, modifying them in a specialized facility, and giving intensive chemotherapy before the cells are returned. Patients need hospital care and long-term monitoring; chemotherapy can cause serious complications and may affect fertility. These demands limit delivery to equipped centers and make treatment difficult to access for many people, especially where specialist services are scarce. A broader population benefit will depend on safer conditioning, more treatment capacity, practical financing, and durable results across diverse patients. Those are goals for future work, not benefits already established by approval.

The safety profiles also differ. Lyfgenia carries an FDA boxed warning because blood cancers have occurred after treatment; its prescribing information calls for long-term monitoring. Casgevy's FDA labeling warns about possible unintended genome edits as well as serious risks related to stem-cell engraftment and conditioning. These risks should be weighed alongside the potential reduction in pain crises during specialist counseling [252, 251].

Timeline

1950s The molecular basis of sickle hemoglobin is established.

2010s Gene-addition and gene-editing trials begin for sickle-cell disease.

2023 FDA approves Casgevy and Lyfgenia for eligible patients aged 12 and older in the United States.

2026 FDA expands Casgevy's US indication to children aged 2 and older.

Chapter 107

mRNA Vaccine Technology

Messenger RNA carries instructions that cells use to make proteins. A vaccine can deliver a short-lived mRNA instruction for a harmless piece of a pathogen, allowing the immune system to learn to recognize it. The idea was studied for decades before advances in chemistry, delivery, and manufacturing made practical vaccines possible.

107.1 Overcoming the inflammatory response

Early laboratory mRNA could trigger a strong innate immune response and be degraded quickly. Katalin Karikó and Drew Weissman found that modifying certain nucleosides in mRNA could reduce unwanted inflammation while improving protein production. Their discoveries helped enable mRNA vaccines, and they received the 2023 Nobel Prize in Physiology or Medicine [253].

Lipid nanoparticles protect mRNA and help it enter cells. Once inside, cells temporarily make the encoded antigen; the mRNA is then broken down. The instruction does not need to enter the cell nucleus or change a person's DNA. Researchers can alter the sequence to encode a different antigen while using a related production platform.

107.2 A platform with conditions

mRNA technology permits rapid design after a pathogen's genetic sequence is known, but each vaccine still needs testing for safety and effectiveness. Storage, manufacturing quality, immune response, and the choice of antigen remain important. The platform also supports research into vaccines for other infections and potential cancer treatments.

The platform depended on decades of work on RNA stability, delivery into cells, and

reducing unwanted inflammatory responses before it could be used in vaccines. COVID-19 demonstrated that mRNA could support rapid vaccine design once a pathogen's sequence was known, while also revealing the importance of cold-chain logistics and public confidence. The technology is promising, but each vaccine still requires its own testing for safety and effectiveness [253].

Timeline

1990s Early experiments demonstrate protein production after mRNA delivery.

2005 Karikó and Weissman publish work on modified nucleosides in mRNA.

2020 mRNA vaccines are authorized during the COVID-19 pandemic.

2023 Karikó and Weissman receive the Nobel Prize in Physiology or Medicine.

Chapter 108

COVID-19 Vaccines

The COVID-19 pandemic created an urgent need for vaccines against a newly identified coronavirus. Vaccine development moved unusually quickly because researchers could build on decades of work in virology, immunology, vaccine platforms, and clinical-trial methods. The speed came from overlapping research, testing, manufacturing, and review, rather than skipping the need to evaluate safety and effectiveness.

108.1 From viral sequence to vaccines

After SARS-CoV-2 was identified, its genetic sequence was shared in January 2020. Several approaches entered development, including mRNA, viral-vector, protein-based, and inactivated-virus vaccines. The first vaccines authorized in the United States included the Pfizer-BioNTech mRNA vaccine in December 2020, followed shortly by Moderna's mRNA vaccine. Large clinical trials showed substantial protection against symptomatic disease in the conditions studied at the time [254, 255].

The vaccines trained the immune system to recognize the coronavirus spike protein. As the virus evolved, protection against infection changed, and updated formulations were developed to better match circulating variants. Vaccination continued to provide important protection against severe disease, though the strength and duration of protection varied with age, health, previous infection, vaccine formulation, and time since a dose.

108.2 A public-health intervention at scale

Clinical trials established initial evidence, while ongoing surveillance identified uncommon adverse events and informed recommendations. Vaccine programs also depended on manufac-

turing, cold-chain logistics, prioritization, and public confidence. The pandemic showed how established platforms can accelerate response, while reminding health systems that access and clear communication shape who benefits.

Large randomized trials showed that the first vaccines substantially reduced symptomatic COVID-19, and subsequent evidence established strong protection against severe disease and death, especially before immune-evasive variants emerged. Protection against infection waned and changed as the virus evolved, leading to updated formulations and recommendations. Rapid development rested on decades of platform research, trial networks, manufacturing investment, and regulatory review [255, 254].

Timeline

January 2020

The SARS-CoV-2 genetic sequence is shared publicly.

December 2020

The first COVID-19 vaccines receive emergency authorization in the United States.

2021 onward

Vaccination expands globally and updated formulations address evolving variants.

Chapter 109

Proton-Pump Inhibitors

Acid in the stomach helps digest food, but excess acid can contribute to ulcers and reflux symptoms. Earlier medicines neutralized acid or blocked histamine signals that stimulated acid secretion. Proton-pump inhibitors (PPIs) act at the final step of acid production, providing a more powerful way to reduce stomach acid.

109.1 Blocking the acid pump

Parietal cells in the stomach lining secrete acid through the H^+/K^+-ATPase, often called the proton pump. PPIs are activated in the acidic environment near these cells and inhibit the pump. Omeprazole became the first widely used medicine in the class in the late 1980s, followed by related drugs [256].

PPIs transformed the treatment of gastroesophageal reflux disease and peptic-ulcer disorders. They also became part of combination treatment to eradicate *Helicobacter pylori*. By reducing acid, they can help ulcers heal while antibiotics address the infection. The medicine and duration are chosen according to the condition and clinical guidance.

109.2 Benefits and appropriate use

For many patients, acid suppression relieves symptoms and supports healing. Long-term or unnecessary use can expose people to avoidable medication risks, so clinicians periodically review whether continued treatment is needed. PPIs changed acid-related care by acting directly on the final mechanism of acid secretion, while their use still depends on an appropriate indication.

By suppressing the stomach's acid-producing proton pump, these medicines provided more

sustained acid reduction than earlier treatments and transformed care for ulcers and reflux disease. Their use expanded widely, which later prompted scrutiny of long-term prescribing and potential harms. The historical lesson is to distinguish clear indications and short-term benefit from routine continuation without review [256].

Timeline

1970s Researchers identify the gastric proton pump as the final pathway of acid secretion.

1980s Omeprazole is developed and enters clinical use.

1990s onward
PPIs become central treatments for reflux and ulcer disease.

Chapter 110

Helicobacter pylori

For much of the twentieth century, peptic ulcers were commonly attributed to stress, personality, or excess acid. In the early 1980s, Australian researchers Robin Warren and Barry Marshall identified a bacterium in the stomach lining of people with gastritis and ulcers. Their work showed that many ulcers were infections that could be treated with antibiotics.

110.1 A bacterium survives in the stomach

Warren observed curved bacteria in stomach-biopsy samples. Marshall joined the investigation, and the pair succeeded in culturing the slow-growing organism in 1982. They proposed that it caused gastritis and peptic-ulcer disease, challenging the prevailing view that bacteria could not persist in the acidic stomach. Marshall's self-experimentation and subsequent clinical studies added evidence that the bacterium could cause gastritis. Warren and Marshall received the 2005 Nobel Prize in Physiology or Medicine for the discovery of *Helicobacter pylori* and its role in gastritis and peptic ulcer disease [257, 258].

110.2 From acid treatment to eradication

Antacids and acid-reducing medicines could relieve symptoms and help an ulcer heal, but they did not remove the underlying infection. Studies showed that antibiotics combined with acid suppression could eradicate *H. pylori* and reduce ulcer recurrence. The discovery changed ulcer treatment from long-term symptom control or surgery toward testing for infection and, when indicated, a course of combination therapy.

H. pylori infection is also associated with gastric cancer and other stomach disorders, although infection does not mean that cancer will occur. Treatment choices depend on

diagnostic results, prior antibiotic exposure, and local resistance patterns. The history demonstrates how identifying a cause can change both the explanation of a disease and its treatment.

Timeline

1979 Warren begins documenting bacteria in stomach biopsies.

1982 Warren and Marshall culture the organism and propose its role in ulcers.

1994 A US NIH consensus statement recognizes eradication therapy for many ulcer patients.

2005 Warren and Marshall receive the Nobel Prize in Physiology or Medicine.

Chapter 111

Continuous Renal Replacement Therapy

Intermittent hemodialysis removes waste and excess fluid over a few hours. For critically ill patients whose blood pressure is unstable, that rapid change can be difficult to tolerate. Continuous renal replacement therapy (CRRT) filters blood gradually over many hours, allowing intensive-care teams to control fluid and solute removal more slowly.

111.1 A continuous approach

In 1977, German physician Peter Kramer described continuous arteriovenous hemofiltration for patients with acute kidney failure. Blood pressure provided the force to move blood through a filter. Later systems used pumps and venovenous access, making treatment more controllable and broadly applicable [259, 260].

CRRT can remove fluid and dissolved waste while clinicians replace selected electrolytes and fluids. It is commonly used in intensive care when acute kidney injury occurs alongside shock, severe fluid overload, or other critical illness. It does not cure the underlying cause of kidney failure; it supports the patient while clinicians treat that cause and allow time for recovery or plan further renal replacement.

111.2 A tool for intensive care

CRRT is an option when clinicians need gradual fluid and waste removal, such as in some patients with unstable blood pressure or raised intracranial pressure. Intermittent dialysis remains suitable for many patients who can tolerate faster treatment. Randomized compar-

isons have not established a survival advantage for CRRT over intermittent renal replacement therapy; the choice is guided by a patient's physiology, treatment goals, equipment, and staffing rather than a claim that one method is best for everyone [261, 260].

CRRT requires continuous monitoring, vascular access, trained staff, and careful management of anticoagulation and fluid balance. Circuit clotting can interrupt treatment; anticoagulation can cause bleeding, and patients can develop hypothermia or electrolyte imbalances. The method supports patients while clinicians treat the cause of acute kidney injury and assess whether kidney function will recover.

Timeline

1977 Kramer reports continuous arteriovenous hemofiltration.

1980s–1990s
Pump-driven continuous therapies develop for critical care.

Today CRRT is used in selected intensive-care patients with acute kidney injury.

Chapter 112

Hearing Devices

Hearing devices help people hear and communicate when part of the auditory system is damaged. Hearing aids amplify and shape sound for a person's remaining hearing. Cochlear implants use electrical signals to bypass damaged sensory cells and stimulate the auditory nerve. These technologies address different kinds and degrees of hearing loss; neither restores normal hearing for every user.

112.1 From acoustic aids to digital hearing aids

Before electronic amplification, ear trumpets and related acoustic devices collected sound and directed it into the ear. Telephone technology and the carbon microphone helped make electronic hearing aids possible in the early twentieth century. Vacuum tubes increased amplification but made devices bulky. Transistors in the 1950s allowed smaller, portable aids, while later digital signal processing made it possible to adjust amplification by frequency and develop features such as directional microphones and noise management. A commercially successful digital hearing aid appeared in 1996 [323].

112.2 How hearing aids help

Most hearing aids have a microphone, amplifier, and speaker. They make sounds louder and can shape amplification to a person's hearing profile. This can improve access to speech and everyday sounds when enough functioning sensory cells remain in the inner ear. Hearing aids do not restore normal hearing, and benefit depends on the type of hearing loss, fitting, adjustment, and use [324].

112.3 Cochlear implants bypass damaged sensory cells

In 1957, French researchers André Djourno and Charles Eyries reported that electrical stimulation of the auditory nerve could produce sound sensations. William House and other teams developed implantable devices and refined surgical approaches. Early single-channel devices established that electrical hearing was possible; multichannel implants conveyed more information and improved speech perception. The FDA approved the first commercial cochlear implant in the United States in 1984 [262, 263].

A microphone and external processor convert sound into a pattern of signals. A transmitter sends those signals to an implanted receiver, which activates electrodes in the cochlea. The brain must learn to interpret this input, so fitting, rehabilitation, and practice are important. Implants require surgery and ongoing support; results vary, especially in noisy settings. Their use also raises cultural and personal questions within Deaf communities about identity, language, and choices for children.

Hearing aids and cochlear implants expanded communication options through different routes: one makes sound more accessible to the ear's remaining sensory cells, while the other provides electrical stimulation beyond them. Access, maintenance, rehabilitation, and the user's goals shape their value.

Timeline

Early twentieth century
: Telephone-derived electronics make practical hearing aids possible.

1950s Transistors allow smaller, portable hearing aids.

1957 Djourno and Eyries demonstrate auditory sensations from electrical stimulation.

1960s–1970s
: House and others develop implantable single-channel systems.

1984 The FDA approves a commercial cochlear implant in the United States.

1996 The first commercially successful digital hearing aid is introduced.

Chapter 113

Advanced Prosthetic and Bionic Limbs

Artificial limbs have been made for thousands of years, but modern prosthetic devices increasingly combine lightweight materials, powered joints, electronic sensors, and signals from the user's muscles. They can restore some mobility or grasp after amputation, yet no device fully reproduces the sensation and movement of a natural limb.

113.1 From passive limbs to powered control

Traditional prostheses used body-powered cables or passive joints. Myoelectric systems detect electrical signals produced by residual muscles and use them to control a motorized hand or limb. Advances in batteries, processors, materials, and sensors made powered movement more practical. Microprocessor knees can adjust resistance during walking, while advanced hands offer multiple grips [264, 265].

Targeted muscle reinnervation, developed for people with amputations, moves remaining nerves to muscles that can serve as biological signal amplifiers. Those signals can improve control of powered prostheses and, in some systems, provide feedback. Research in osseointegration and neural interfaces seeks more stable attachment and closer communication between a device and the nervous system.

113.2 Function, fit, and access

The usefulness of a prosthesis depends on fit, comfort, training, repair, and the tasks a person wants to perform. Advanced devices can be expensive and may need regular servicing; simpler designs remain more appropriate for many users. The term bionic can suggest a complete replacement, but current prostheses restore selected functions and require a skilled team and

the wearer's own adaptation.

Modern devices combine lighter materials, powered joints, sensors, and user-controlled signals, but function depends as much on fit, comfort, repair, and rehabilitation as on sophisticated electronics. A prosthesis can improve mobility and independence, yet it does not recreate a biological limb and may be costly or unavailable. User experience and accessible design are essential measures of the breakthrough [264, 265].

Timeline

1960s Myoelectric control is developed for powered prosthetic limbs.

1980s onward

Microprocessors improve control of powered knees and hands.

2000s onward

Targeted muscle reinnervation and advanced interfaces expand control options.

Chapter 114

Biomaterials and Medical Plastics

An implant or medical device succeeds only if it works mechanically and can remain in contact with the body safely. Biomaterials are materials designed for such contact. Their history is not a single discovery: it is a continuing effort to make devices durable, cleanable or sterile, and compatible with blood and tissue. Metals, ceramics, polymers, and composites each solve different problems, and none is suitable for every use [328].

114.1 Plastics change care outside the operating room

Glass bottles, rubber tubing, and metal instruments once dominated hospital equipment. The growth of polymer chemistry and manufacturing in the mid-twentieth century made it possible to produce light, flexible, inexpensive devices in large quantities. Polyvinyl chloride (PVC) enabled flexible blood bags and IV tubing; silicone could be made into soft tubing and catheters; polyethylene and polypropylene became useful in containers, syringes, and other devices. Plastic blood bags were easier to handle than rigid glass bottles and supported the collection, storage, separation, and transport of blood components [329, 330].

Disposable syringes, tubing, and other single-use equipment also made it easier to provide sterile supplies at scale and reduced the need to clean and resterilize some complex devices between patients. These changes supported vaccination, transfusion, intravenous treatment, dialysis, and intensive care. They did not make infection control automatic: correct use, safe disposal, manufacturing quality, and access to sterile supplies still mattered.

114.2 Materials make implants possible

The same materials revolution reached devices intended to remain in the body. Titanium and some alloys can form stable interfaces with bone; polyethylene reduced friction in artificial joints; acrylic polymers made early intraocular lenses possible; and silicone has been used in selected implants and medical tubing. The hip-replacement, cataract-lens, stent, and prosthetic-limb chapters show how material choice shaped each device's practical success. In dentistry, Brånemark's studies of titanium and bone helped establish osseointegrated dental implants [331].

Biocompatibility is not a guarantee that a material is harmless in every use. Devices can wear, corrode, trigger inflammation, release particles or chemical components, or fail mechanically. Plastics can also interact with medicines and stored blood products, so materials must be tested for the particular device and duration of contact [329, 330]. The advance was the ability to match a material to a clinical task and then evaluate how it performed in real patients over time.

Timeline

1940s Flexible plastic blood bags begin to replace glass containers in some transfusion systems.

1950s–1960s
Polymer tubing, disposable syringes, and new implant materials spread into routine medical practice.

1960s onward
Titanium implants, polyethylene joint surfaces, and polymer intraocular lenses demonstrate distinct uses for biocompatible materials.

Today Material selection and safety testing remain central to device design and long-term patient care.

Chapter 115

Modern Dentistry

Tooth decay, gum disease, injury, and infection can cause pain, difficulty eating, missed school or work, and tooth loss. Modern dentistry changed this burden through prevention and procedures that preserve or replace teeth. Community prevention reached whole populations, while local anesthesia, improved instruments, restorative materials, and implant techniques expanded what could be treated in a dental clinic.

115.1 Preventing decay at population scale

In the early twentieth century, investigators noticed that naturally fluoridated water was associated with fewer cavities, though excessive fluoride could cause dental fluorosis. H. Trendley Dean and colleagues studied the balance between these effects. In 1945, Grand Rapids, Michigan, became the first city to adjust its water supply with fluoride as part of a long-term evaluation. The program found substantially less tooth decay among children, helping establish community water fluoridation as a preventive public-health measure [333, 334]. Fluoride toothpastes and professional fluoride treatments later added other ways to prevent or slow decay. Their reach depends on access, appropriate use, and local conditions.

115.2 Saving teeth with modern procedures

Local anesthetics made dental treatment more tolerable by numbing a limited area while the patient remained conscious; their development is part of the broader history of modern anesthesia in this book. Faster rotary handpieces, better lighting and imaging, and improved restorative materials made it practical to remove decay and repair teeth with fillings and crowns. Root canal treatment can preserve a tooth whose inner pulp is infected or damaged,

while periodontal treatment aims to control disease affecting the supporting gums and bone. These procedures can relieve pain and maintain chewing function, but their success depends on diagnosis, technique, follow-up, and the condition of the tooth and surrounding tissues.

115.3 Replacing missing teeth

Dentures and bridges restore some function after tooth loss, but they may be uncomfortable or require support from neighboring teeth. Research by Swedish orthopaedic surgeon Per-Ingvar Brånemark began with titanium chambers used to study bone. He observed that bone could become firmly attached to titanium, a process later called osseointegration. In 1965, his team placed titanium dental implants in a patient; later clinical studies helped establish implants as a way to support crowns, bridges, or full-arch prostheses [332, 331]. Implants can improve stability and chewing for selected patients, but they require surgery, healing, maintenance, and adequate bone and oral health. They are not suitable or accessible for everyone.

Modern dentistry has improved daily life by preventing some cavities, reducing pain, preserving natural teeth, and restoring function after tooth loss. These gains depend on routine prevention and affordable access as much as on advanced procedures. Dental implants also show how biomaterials can connect engineering to ordinary activities such as eating and speaking.

Timeline

1930s–1940s
: Epidemiological research clarifies the relationship between fluoride, fluorosis, and tooth decay.

1945 Grand Rapids begins a landmark community water-fluoridation study.

Mid-twentieth century
: Improved local anesthetics, rotary instruments, and restorative materials expand dental treatment.

1952 Brånemark observes close bone attachment to titanium during experimental research.

1965 His team places titanium dental implants in a patient.

1980s onward

Osseointegrated implants gain wider clinical acceptance.

Chapter 116

Treating Major Health Challenges Today

The breakthroughs in this book now work together in routine care. Vaccines, safer water, screening, laboratory tests, imaging, medicines, surgery, rehabilitation, and long-term monitoring address diseases that remain major causes of death and disability. Modern medicine often controls or slows a condition rather than curing it. Its benefits depend on early diagnosis, affordable treatment, reliable health services, and support that fits a person's life. The following overview connects current medical tools to 15 major health challenges.

116.1 Cardiovascular disease: prevent damage and restore blood flow

Coronary heart disease and stroke remain the world's leading causes of death. Modern care reduces risk through detection and treatment of high blood pressure, high cholesterol, diabetes, and tobacco exposure. When a heart attack occurs, antiplatelet medicines and rapid reperfusion with primary PCI or clot-dissolving medicines can restore blood flow; after an ischemic stroke, eligible patients may receive thrombolysis or mechanical thrombectomy. Stroke units, anticoagulation for atrial fibrillation, and rehabilitation can reduce further injury and help people regain function. These advances build on the book's chapters on statins, antihypertensive medicines, stents, heart attack care, and stroke treatment, but access to fast emergency care remains unequal [295, 296].

116.2 Cancer: prevention, early diagnosis, and treatment matched to the tumor

Cancer includes many diseases, so there is no single treatment or screening test for every cancer. Tobacco control, vaccination against HPV and hepatitis B, and screening for selected cancers can prevent cases or find them earlier. Pathology and molecular testing help identify the cancer type and, in some cases, the molecular changes that guide targeted medicines or immunotherapy. Surgery, radiotherapy, chemotherapy, hormone therapy, targeted drugs, and checkpoint immunotherapy are combined according to the cancer and its stage; palliative care can improve comfort at every stage. Survival gains remain uneven where diagnosis, radiotherapy, surgery, or medicines are hard to reach [297].

116.3 COPD and asthma: control airway disease and protect lung function

Inhaled medicines deliver treatment directly to the airways. In asthma, inhaled corticosteroids reduce airway inflammation and lower the risk of dangerous attacks; bronchodilators relieve airway narrowing. COPD care uses long-acting bronchodilators, smoking cessation, vaccination, pulmonary rehabilitation, and oxygen for selected people with severe low oxygen. These measures can ease breathlessness, reduce flare-ups, and support daily activity, but they do not reverse all lung damage. Correct inhaler use and access to clean air and ongoing care make a major difference [298].

116.4 Type 2 diabetes: protect the heart, kidneys, eyes, and nerves

Blood glucose monitoring and individualized treatment help control type 2 diabetes, while regular eye, kidney, foot, and cardiovascular checks can detect complications before they cause irreversible damage. Metformin, insulin, and other medicines remain important, but newer SGLT2 inhibitors and GLP-1 receptor agonists can also reduce cardiovascular and kidney risks for many people with diabetes and related conditions. Blood-pressure and cholesterol treatment, nutrition support, and access to retinal care help prevent blindness, kidney failure, heart attacks, stroke, and limb injury. The choice of medicine depends on a person's health, risks, preferences, and access; no one regimen suits everyone [299].

116.5 Chronic kidney disease: find it early and slow its progression

Simple blood tests for kidney filtration and urine tests for albumin can identify chronic kidney disease before symptoms appear. Treating its causes, especially diabetes and high blood pressure, and using kidney-protective medicines for eligible patients can delay progression and reduce heart complications. When kidney failure develops, dialysis can replace some kidney functions and transplantation can offer longer-term replacement for eligible patients. Conservative kidney care can also focus on symptoms and quality of life. The availability of screening, medicines, dialysis, and transplantation varies sharply between regions [300, 301].

116.6 Dementia, Parkinson disease, and epilepsy: diagnosis and tailored care

Dementia has several causes, and Alzheimer's disease is the most common. Clinical assessment, brain imaging, and newer fluid or blood biomarkers can help clarify diagnosis. For some people with early Alzheimer's disease, anti-amyloid medicines can modestly slow cognitive decline, but they do not restore lost abilities or stop the disease, and they require careful assessment and monitoring. Dementia care also includes managing other health conditions, supporting independence, and helping caregivers. Parkinson disease symptoms can be eased with levodopa, other medicines, exercise, and deep-brain stimulation for selected patients; antiseizure medicines control seizures for many people, while surgery or implanted devices can help some whose epilepsy resists medicines. These treatments improve symptoms or slow some losses; they are not general cures [302, 303, 304, 305].

116.7 Lower respiratory infections: prevent, diagnose, and support breathing

Pneumonia and other lower respiratory infections remain especially dangerous for young children, older adults, and people with chronic illness. Vaccines against pneumococcus, Hib, measles, pertussis, and influenza prevent some important causes. Rapid assessment can distinguish people who need urgent hospital care; antibiotics treat bacterial pneumonia but do not work against viruses. Medical oxygen, pulse oximetry, fluids, and respiratory support help patients whose breathing is impaired. The gains depend on timely access to vaccination, diagnosis, appropriate antibiotics, and reliable oxygen supplies [306].

116.8 Tuberculosis: rapid tests and shorter treatment courses

Molecular tests can detect tuberculosis and some drug resistance quickly, helping clinicians choose an effective regimen sooner. Drug-susceptible TB is usually curable with a combination of antibiotics; newer all-oral regimens have shortened treatment for eligible people with drug-resistant disease. Preventive treatment can stop TB infection from progressing to active disease in people at higher risk. Drug resistance, delayed diagnosis, incomplete treatment access, and the overlap with HIV continue to cause avoidable illness and death [307].

116.9 HIV/AIDS: sustained treatment preserves health and prevents transmission

Combination antiretroviral therapy suppresses HIV, restores immune function, and allows most people with sustained access to live long lives. When viral load remains undetectable on treatment, HIV is not sexually transmitted. Testing, treatment during pregnancy, pre-exposure prophylaxis, condoms, and safer blood and injection practices help prevent new infections. HIV is not yet cured by routine treatment, and millions still face barriers to diagnosis and continued care [308].

116.10 Malaria: combine medicines, prevention, and childhood vaccines

Rapid diagnostic tests help identify malaria, and artemisinin-based combination therapies treat the most dangerous common form. Insecticide- treated nets and other vector-control measures reduce exposure; preventive medicines are used for some groups and seasons. WHO-recommended malaria vaccines now add protection for children in endemic areas, especially in Africa, as part of a wider prevention program. Drug and insecticide resistance, mosquito behavior, and gaps in health services mean these tools must be adapted and sustained [309, 277].

116.11 Diarrheal disease: rehydrate, nourish, vaccinate, and prevent infection

Oral rehydration solution replaces the water and salts lost during diarrhea; zinc treatment can shorten childhood episodes. Continued feeding and breastfeeding protect nutrition, while rotavirus vaccination prevents a significant cause of severe childhood diarrhea. Safe water, sanitation, handwashing, and food safety reduce exposure. Blood in stool, severe dehydration, or persistent illness requires clinical assessment because rehydration alone does not treat every cause. These low-cost measures save lives, yet many children still lack timely access to them [310].

116.12 Hepatitis B and C: prevent chronic liver disease and liver cancer

The hepatitis B vaccine, especially a timely birth dose followed by the recommended series, prevents infection that can become chronic and later cause cirrhosis or liver cancer. Antiviral medicines can suppress chronic hepatitis B and lower the risk of liver damage, though treatment is often long-term. Direct-acting antivirals cure hepatitis C in most treated people, usually with a short course of oral medicine; there is no hepatitis C vaccine. Testing, safe blood and injection practices, and access to treatment are essential to reduce deaths from both infections [311, 312].

116.13 Mental-health disorders: effective care combines medicines and support

Depression, anxiety, bipolar disorder, and schizophrenia can cause major disability, but evidence-based care helps many people recover or manage symptoms. Depending on the condition and the person's needs, treatment may include psychotherapy, antidepressants, mood stabilizers, antipsychotic medicines, psychosocial rehabilitation, and support from family or peers. Shared decisions and monitoring help balance benefits with side effects. Stigma, shortages of trained providers, cost, and limited community services still keep many people from effective care [313].

116.14 Musculoskeletal disease: relieve pain and preserve movement

Exercise, physical therapy, assistive devices, and rehabilitation help people with osteoarthritis and persistent low-back pain stay active and maintain function. Anti-inflammatory medicines and other pain treatments may help some people when chosen with their health risks in mind. For rheumatoid arthritis, disease-modifying medicines, including biologic drugs, can reduce inflammation and prevent joint damage; joint replacement can restore function when severe arthritis has destroyed a joint. Treatment aims to preserve movement and participation, not simply to remove pain at any cost [314, 315, 318].

116.15 Obesity and metabolic disease: treat a chronic condition at several levels

Obesity increases the risk of type 2 diabetes, cardiovascular disease, and some cancers. Modern care recognizes it as a chronic condition influenced by biology, environment, medicines, stress, and access to food and activity—not simply a failure of willpower. Structured nutrition and activity support, behavioral care, anti-obesity medicines such as GLP-1 therapies, and metabolic surgery can help some people achieve and maintain clinically meaningful weight loss. These treatments require individualized decisions and often long-term follow-up; cost and access remain major barriers, and population policies that make healthy choices more available are also necessary [316, 317].

116.16 What progress means

Medical science helps through different routes: preventing infection, lowering risk, finding disease earlier, restoring blood flow, suppressing a pathogen, slowing organ damage, relieving symptoms, and supporting recovery. For some conditions it offers cure; for others it makes long-term life safer and more independent. The continuing challenge is to deliver proven care fairly, preserve trust, measure outcomes, and keep improving treatments as new evidence emerges.

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